Operator Manual

Screw Compressor

AIRCENTER SX Tri-Voltage

901837 46 USE

Read this manual before using this product.

Failure to follow the instructions and safety precautions in this manual can result in serious injury or death.

Manufacturer:

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1 Regarding this Document

1.1 Using this document

The operating manual is a component of the product. It describes the machine as it was at the time of first delivery after manufacture.

➤ Keep the operating manual in a safe place throughout the life of the machine.
➤ Supply any successive owner or user with this operating manual.
➤ Please insert any amendment or revision of the operating manual sent to you.
➤ Enter details from the machine nameplate and individual items of equipment in the table in chapter 2.

1.2 Further documents

Further documents included with this operator manual are:

- Certificate of acceptance / operating instructions for the pressure vessel
- User manual for SIGMA CONTROL 2

Missing documents can be requested from KAESER.

➤ Make sure all documents are complete and observe the instructions contained in them.
➤ Make sure you provide the data from the nameplate when ordering documents.

1.3 Copyright

This service manual is copyright protected. Queries regarding use or duplication of the documentation should be referred to KAESER. Correct use of information will be fully supported.

1.4 Symbols and labels

➤ Please note the symbols and labels used in this document.

1.4.1 Warnings

Warning notices indicate dangers that may result in injury when disregarded.

Warning notices indicate three levels of danger identified by the corresponding signal word:

Signal term	Meaning	Consequences of disregard
DANGER	Warns of an imminent danger	Will very likely result in death or severe injury
WARNING	Warns of a potentially imminent danger	May result in death or severe injury
CAUTION	Warns of a potentially dangerous situation	May result in a moderate physical injury

Tab. 1 Danger levels and their definition (personal injury)

Warning notices preceding a chapter apply to the entire chapter, including all sub-sections.

Example:

⚠ DANGER

The type and source of the imminent danger is shown here!
The possible consequences of ignoring a warning are shown here.
If you ignore the warning notice, the "DANGER" signal word indicates a lethal or severe injury will occur very likely.

➤ *The measures required to protect yourself from danger are shown here.*

Warning notes referring to a sub-section or the subsequent action are integrated into the procedure and numbered as an action.

Example:

1. **⚠ WARNING** *The type and source of the imminent danger is shown here!*
   The possible consequences of ignoring a warning are shown here.
   If you ignore the warning notice, the "WARNING" signal word indicates that a lethal or severe injury may occur.
   ➤ *The measures required to protect yourself from danger are shown here.*
2. Always read and comply with warning instructions.

1.4.2 Potential damage warnings

Contrary to the warnings shown above, damage warnings do not indicate a potential personal injury.

Warning notices for damages are identified by their signal term.

Signal term	Meaning	Consequences of disregard
NOTE	Warns of a potentially dangerous situation	Damage to property is possible

Tab. 2 Danger levels and their definition (damage to property)

Example:

NOTICE

The type and source of the imminent danger is shown here!
Potential effects when ignoring the warning are indicated here.

➤ *The protective measures against the damages are shown here.*

➤ Carefully read and fully comply with warnings against damages.

1.4.3 Other alerts and their symbols

This symbol identifies particularly important information.

Material — Here you will find details on special tools, operating materials or spare parts.

Precondition — Here you will find conditional requirements necessary to carry out the task.

The conditions relevant to safety shown here will help you to avoid dangerous situations.

Option H1

➤ This symbol denotes lists of actions comprising one stage of a task.
Operating instructions with several steps are numbered in the sequence of the operating steps.
Information relating to one option only are marked with an option code (e.g., H1 indicates that this section applies only to machines with machine mountings). Option codes used in this operator manual are explained in chapter 2.2.

Information referring to potential problems are identified by a question mark.

The cause is named in the help text ...

➤ ... as is a solution.

This symbol identifies important information or measures regarding the protection of the environment.

Further information — Further subjects are introduced here.

2 Technical Data

2.1 Nameplate

The machine's nameplate provides the model designation and important technical information.

The nameplate is located on the outside of the machine:

- above the cooler
- on the rear of the machine

➤ Enter here the nameplate data as a reference:

Characteristic	Value
Rotary screw compressor	
Material No.	
Serial No.	
Ambient temperature	
Rated power	
Maximum working pressure PS	
Rated motor speed	
Volume flow rate	
Phases	
Voltage	
Full load current	
Full load current drive motor	
Short circuit current	
Supply fuse	
Class	
Electrical wiring Diagram	

Tab. 3 Nameplate

2.2 Options

The table contains a list of possible options. The options for this machine are shown near the nameplate.

➤ Enter options here as a reference:

Option	Option code	Available?
MODULATINGcontrol	C1	
SIGMA CONTROL 2 (Connection to control technology available)	C3	—
Provided: ✓ Not available: —		

Option	Option code	Available?
SIGMA CONTROL 2 (Connection to control technology not provided)	C48	✓
KAESER FILTER KE	F1	
Bolt-down machine mounts	H1	
Air-cooling	K1	✓
Transformer power supply for refrigerated dryer	T2	
Provided: ✓ Not available: —		

Tab. 4 Options

2.3 Weight

The values shown are maximum values. The actual weight of individual machines depends on equipment fitted.

	SX 3	SX 4	SX 5	SX 7.5
Weight [lb]	628	628	639	661

Tab. 5 Weight

2.4 Temperature

	SX 3	SX 4	SX 5	SX 7.5
Minimum cut-in temperature [°F]	40	40	40	40
Typical airend discharge temperature during operation [°F]	150 – 200	150 – 200	150 – 200	150 – 200
Maximum airend discharge temperature (automatic safety shutdown) [°F]	230	230	230	230

Tab. 6 Temperature

2.5 Ambient conditions

	SX 3	SX 4	SX 5	SX 7.5
Maximum altitude AMSL* [ft]	3000	3000	3000	3000
Permissible ambient temperature [°F]	40 – 115	40 – 115	40 – 115	40 – 115
* Higher altitudes are permissible only after consultation with the manufacturer.				

	SX 3	SX 4	SX 5	SX 7.5
Cooling air temperature [°F]	40 – 115	40 – 115	40 – 115	40 – 115
Inlet air temperature [°F]	40 – 115	40 – 115	40 – 115	40 – 115
* Higher altitudes are permissible only after consultation with the manufacturer.				

Tab. 7 Ambient conditions

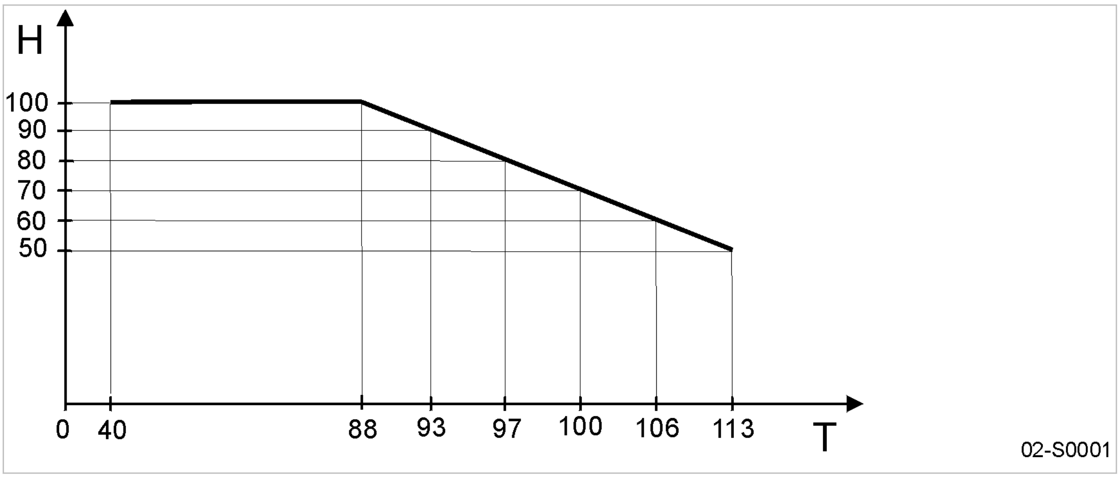

Fig. 1 Maximum relative humidity of inlet air

T Inlet air temperature [°F]
H Maximum relative humidity of inlet air [%]

2.6 Ventilation

The values given are minimum guide values.

	SX 3	SX 4	SX 5	SX 7.5
Inlet aperture Z see figure 14 [sq.ft.] (free area)	2.2	2.2	2.2	2.2
Extractor for forced ventilation: Flow rate [cfm] at 0.4 in. wc	883	1001	1236	1530

Tab. 8 Ventilation

2.7 Pressure

Maximum working pressure: see nameplate

Max. gauge working pressure [psig]	SX 3	SX 4	SX 5	SX 7.5
90	—	—	—	—
125	155	155	155	155

Max. gauge working pressure [psig]	SX 3	SX 4	SX 5	SX 7.5
160	230	230	230	230
217	—	232	232	232

Tab. 9 Safety relief valve activating pressure

2.8 Air receiver

Volume

	SX 3	SX 4	SX 5	SX 7.5
Volume [gal]	53	53	53	53

Tab. 10 Air receiver: Volume

Safety relief valve activating pressure [psi]

Maximum working pressure: see air receiver nameplate

Max. working pressure [psi]	SX 3	SX 4	SX 5	SX 7.5
160	160	160	160	160
217	—	220	220	220

Tab. 11 Safety relief valve: Activating pressure

2.9 Flow rate (constant delivery volume relative to intake conditions)

Flow rate [cfm]:

Max. working pressure [psig]	SX 3	SX 4	SX 5	SX 7.5
90	—	—	—	—
125	12	16	21	28
160	9	13	17	24
217	—	9	13	19
Flow rate as per ISO 1217:2009 (Annex C)				

Tab. 12 Flow rate

2.10 Cooling Oil Recommendation

A sticker showing the type of oil filled is located near the oil separator filler.

Information for ordering cooling oil can be found in chapter 11.

2.10.1 Basic Information

Lubrication of an air compressor is essential to reliable operation. Carbon and varnish can form in compressor cooling oils. These deposits block the flow of lubricant and cause excessive wear and failure of moving parts. Contamination of the lubricant can allow the formation of acids, causing extensive internal corrosion. Water may be condensed decreasing the lubricity.

Lubricants in rotary compressors do much more than lubricate. During the compression process, it acts as a sealant in the airend which is important for maximum efficiency. The lubricant also absorbs much of the heat of compression to cool the airend and reduce the temperature of the compressed air. It’s not enough that a compressor cooling oil lubricates well, it must stand up to the heat, pressure and contaminants that are present in every air compressor.

2.10.2 KAESER Lubricants

KAESER synthetic lubricants should be stored in a protected location to prevent contamination. Do not re-use drums; flush and send to reconditioner.

Although the KAESER synthetic is not highly flammable, it will burn. While KAESER synthetic compressor cooling oil is less flammable than equal viscosity mineral oils, it cannot be classified as a fire-resistant fluid. It has a flash point above 460 °F. Since the user has total control over the conditions of the compressor lubricant, he assumes total responsibility for its safe usage.

Material Safety Data Sheets are available for each lubricant from your authorized KAESER SERVICE representative.

Regardless of the lubricant selected, the KAESER SIGMA lubricants will separate readily from water. If condensate occurs it can easily be removed. Let the compressor sit so that any water can drain back to the separator tank and separate to the bottom. See chapter 10.15 proper draining procedure.

KAESER has several lubricants available that are specially formulated to match these demands. They feature excellent lubricity, outstanding demulsibility (ability to separate from water), and long life.

M-SERIES

- M-Series SIGMA compressor cooling oils are **semi-synthetic** lubricants.
- M-Series SIGMA compressor cooling oils are the highest quality petroleum lubricants. M–460 is specially blended to provide reliable performance in KAESER screw compressors.

S-SERIES

- S-Series SIGMA compressor cooling oils are **synthetic** lubricants.
- S-Series SIGMA compressor cooling oils are formulated from the most advanced synthetic lubricants. These ”synthetic” lubricants begin as high quality petroleum feed stock. They are then refined, processed and purified into fluids with very consistent molecular structure. These oils are carefully blended to produce extremely consistent lubricants with superior properties. SIGMA synthetic lubricants feature all the advantages of both PAO and diester fluids.
- S-460 lubricant is recommended for compressors operating in ambient temperatures between 40 °F and 105 °F.

Specialty KAESER LUBRICANTS

- S-680 lubricant may be used when ambient temperatures are always between 70 °F and 105 °F.
- FG-460 synthetic hydrocarbon based food grade lubricant is designed for use in rotary screw compressors in the application where incidental food contact may occur with the discharge air. This lubricant meets the requirements of the FDA Regulation 21 CFR §178.3570 and is USDA H-1approved and NSF certified. FG-460 is approved for canning, food packing, meat and poultry processing and other applications where incidental food contact may occur.

2.11 Cooling oil charge quantity

	SX 3	SX 4	SX 5	SX 7.5
Total charge [qt]	3.0	3.0	3.0	3.0
Topping up volume [qt] (minimum–maximum)	0.13	0.13	0.13	0.13

Tab. 13 Cooling oil charge quantity

2.12 Motors and power

2.12.1 Compressor motor

	SX 3	SX 4	SX 5	SX 7.5
Rated power [hp]	3	4	5	7.5
Rated speed [rpm]	3490	3520	3520	3540
Protection enclosure	TEFC	TEFC	TEFC	TEFC

Tab. 14 Compressor motor

2.13 Sound emission

	SX 3	SX 4	SX 5	SX 7.5
Sound emission [dB(A)]	61	62	63	66
Sound pressure level in operation at maximum working gauge pressure as per ISO 2151 and the basic standard ISO 9614-2, tolerance: ±3 dB(A)				

Tab. 15 Sound emission

2.14 Power Supply

Basic requirements

The machine is designed for an electrical supply according to National Electric Code (NEC), edition 2020, particularly article 670 and NFPA 79, edition 2021, particularly section 4.4. In the absence of any user-specified alternatives, the limits given in these standards must be adhered to. Consult manufacturer for any other specific power supply.

The incoming line within the control panel should be as short as possible.

If external sensors or communication lines are to be connected to the machine, use shielded cables and insert the same through EMC fittings into the control panel.

Three-phase

Do **NOT** operate package on any unsymmetrical power supply. Also do **NOT** operate package on power supplies such as a three-phase WYE system with the center point not solidly grounded or three-phase (open) delta.

The machine requires a symmetrical three-phase power supply transformer with a WYE configuration output as shown in Figure 2 and Figure 3. In a symmetrical three phase supply the phase angles and voltages are all the same.

Other power supplies are not suitable.

Fig. 2 Three-phase star (wye system); 4 wire; center point solidly grounded

Fig. 3 Three-phase star (wye system); 3 wire; grounded center point solidly grounded

Further information

Please contact an authorized KAESER service representative for options.

The electrical diagram 13.4 contains further specifications for electrical connection.

2.15 Power Supply specifications

The following multi-strand copper core wires are given according to 2020 NEC 310.14, 310.15, 310.16 and table 310.16 adjusted for 40 °C ambient temperature.

If other local conditions prevail, like for example high temperature, the cross section should be checked and adjusted according to 2020 NEC 110.14(C), 220.3, 310.14, 310.15, 310.16, table 310.15(B)(1), table 310.15(C)(1), 430.6, 430.22, 430.24, 670.4(A) and other local codes.

Dual element time delay fuses should be selected according to 2020 NEC 240.6, 430.52 and tables 430.52, 430.248 and 430.250.

It is recommended to use a ground conductor the same size as the current carrying conductors, if local codes allow. Neither the minimum ground wire size as pointed out in 2020 NEC table 250.122 nor using conduit as the sole ground connection is recommended.

Rated power supply 208V, 3-ph, 60Hz

	SX 3	SX 4	SX 5	SX 7.5
Pre-fuse [A]	15	15	25	30
Supply per phase and ground	AWG14 1)	AWG14 1)	AWG10 1)	AWG10 1)
1) 75 °C				

	SX 3	SX 4	SX 5	SX 7.5
Consumption [A]	11	13	18	23
1) 75 °C				

Tab. 16 Supply 208V/3/60Hz

Rated power supply 230V, 3-ph, 60Hz

	SX 3	SX 4	SX 5	SX 7.5
Pre-fuse [A]	15	15	25	30
Supply per phase and ground	AWG14[1)]	AWG14[1)]	AWG10[1)]	AWG10[1)]
Consumption [A]	10	13	18	22
1) 75 °C				

Tab. 17 Supply 230V/3/60Hz

Rated power supply 460V, 3-ph, 60Hz

	SX 3	SX 4	SX 5	SX 7.5
Pre-fuse [A]	10	10	15	15
Supply per phase and ground	AWG14[1)]	AWG14[1)]	AWG14[1)]	AWG14[1)]
Consumption [A]	5.1	6.3	9.0	11
1) 75 °C				

Tab. 18 Supply 460V/3/60Hz

2.16 Refrigerated dryer

Model

	SX 3	SX 4	SX 5	SX 7.5
Model*				
* Read off the dryer model from the dryer nameplate and enter it in the table.				

Tab. 19 Refrigerated dryer: Model

Compressed air system

	ABT 4	ABT 8
Pressure drop [psi] (referred to 100 psig working pressure)	1.6	1.6
Maximum permitted working pressure [psig]	232	232

Tab. 20 Refrigerated dryer: Compressed air system

Refrigerant system

The refrigerated dryer contains a refrigerant that is classified as a fluoridated global warming gas. This refrigerant is required for the function.

The refrigeration system is hermetically sealed.

	ABT 4	ABT 8
Refrigerant	R-513A	R-513A
Global warming potential (GWP)	631	631
Charge quantity[1] [lb]	0.37	0.44
Charge quantity as CO_2 equivalent [t]	0.1	0.1
Maximum permitted working pressure [psig] (high pressure side)	305	305
Maximum permitted working pressure [psig] (low pressure side)	232	232
Safety pressure switch: Cut-out pressure [psig]	305	305

[1] Volume of fluoridated global warming gases for which the refrigerant system was designed

Tab. 21 Refrigerated dryer: Refrigerant system

3 Safety and Responsibility

3.1 Basic instructions

The machine is manufactured to the latest engineering standards and acknowledged safety regulations. Nevertheless, dangers can arise through its operation:

- danger to life and limb of the operator or third parties,
- damages to the machine and other material assets.

Disregard of warning or safety instructions can cause serious injuries!

➤ Use this machine only if it is in a technically perfect condition and only for the purpose for which it is intended; observe all safety measures and the instructions in the operator manual.

➤ Immediately rectify (have rectified) any faults that could be detrimental to safety!

3.2 Specified use

The machine is intended solely for generating compressed air for industrial use. Any other use is considered incorrect. The manufacturer is not liable for any damages that may result from incorrect use. The user alone is liable for any risks incurred.

➤ Keep to the specifications listed in this service manual.

➤ Operate the machine only within its performance limits and under the permitted ambient conditions.

➤ Do not use compressed air for breathing purposes unless it is specifically treated.

➤ Do not use compressed air for any application that will bring it into direct contact with food products unless it is specifically treated.

3.3 Improper use

Improper usage can cause damage to property and/or (severe) injuries.

➤ Only use the machine as intended.

➤ Never direct compressed air at persons or animals.

➤ Use hot cooling air for heating purposes only if there is no risk to the health of humans or animals. If necessary, hot cooling air should be treated by suitable means.

➤ Do not allow the machine to take in toxic, acidic, flammable of explosive gases or vapors.

➤ Do not operate the machine in areas in which specific requirements with regard to explosion protection are in force.

3.4 User's Responsibilities

3.4.1 Observe statutory and universally accepted regulations

This is, for example, nationally applied European directives and/or valid national legislation, safety and accident prevention regulations.

➤ Observe relevant statutory and accepted regulations during installation, operation and maintenance of the machine.

3.4.2 Qualified personnel

These are people who, by virtue of their training, knowledge and experience as well as their knowledge of relevant regulations can assess the work to be done and recognize the possible dangers involved.

Authorized operators possess the following qualifications:

- are of legal age,
- are conversant with and adhere to the safety instructions and sections of the operator manual relevant to operation,
- have received adequate training and authorization to operate electrical and compressed air devices.
- Additional qualifications for compressors with refrigerated dryers:
  - Adequate training and authorization on refrigeration devices.

Authorized installation and maintenance personnel have the following qualifications:

- are of legal age,
- have read, are conversant with and adhere to the safety instructions and sections of the operator manual applicable to installation and maintenance,
- are fully conversant with the safety concepts and regulations of electrical and compressed air engineering,
- are able to recognize the possible dangers of electrical and compressed air devices and take appropriate measures to safeguard persons and property,
- have received adequate training and authorization for the safe installation and maintenance on this equipment.
- Additional qualifications for compressors with refrigerated dryers:
  - fully conversant with the safety concepts and regulations concerning refrigeration devices,
  - must be able to recognize the possible dangers of refrigeration devices and take appropriate measures to safeguard persons and property.

➤ Ensure that operating, installation and maintenance personnel are qualified and authorized to carry out their tasks.

3.4.3 Adherence to inspection schedules and accident prevention regulations

The machine is subject to local inspection schedules.

➤ Ensure that local inspection schedules are adhered to.

3.5 Dangers

Basic instructions

The following describes the various forms of danger that can occur during machine operation.

Basic safety instructions are found in this operator manual at the beginning of each chapter in the section entitled “Safety”.

Warning instructions are found before a potentially dangerous task.

3.5.1 Safely dealing with sources of danger

The following describes the various forms of danger that can occur during machine operation.

Electricity

Touching voltage carrying components can result in electric shocks, burns or death.

- ➤ Allow only qualified and authorized electricians or trained personnel under the supervision of a qualified and authorized electrician to carry out work on electrical equipment according to electrical engineering regulations.
- ➤ Before commissioning or re-commissioning the machine, the user must ensure adequate protection against electric shock from direct or indirect contact.
- ➤ Before starting any work on electrical equipment:
  Switch off and lock out the power supply disconnecting device and verify the absence of any voltage.
- ➤ Switch off any external power sources.
  These could be connections to floating relay contacts or the electrical machine heating, for example.
- ➤ Use fuses corresponding to machine power.
- ➤ Check regularly that all electrical connections are tight and in proper condition.

Forces of compression

Compressed air is contained energy. Uncontrolled release of this energy can cause serious injury or death. The following information concerns work on components that could be under pressure.

- ➤ Close shut-off valves or otherwise isolate the machine from the distribution network to ensure that no compressed air can flow back into the machine.
- ➤ Depressurize all pressurized components and enclosures.
- ➤ Do not carry out welding, heat treatment or mechanical modifications on pressurized components (e.g. pipes and vessels) as this influences the component's resistance to pressure.
  The safety of the machine is then no longer ensured.

Compressed air quality

The composition of the compressed air must be suitable for the actual application in order to preclude health and life-threatening dangers.

- ➤ Use appropriate systems for air treatment before using the compressed air from this machine as breathing air and/or for the processing of food products.
- ➤ Use food-grade cooling oil whenever compressed air is to come into contact with food products.

Spring forces

Springs under tension or compression store energy. Uncontrolled release of this energy can cause serious injury or death.
Minimum pressure / check valves, safety relief valves and inlet valves are powerfully spring-loaded.

- ➤ Do not open or dismantle any valves.

Rotating components

Touching the fan wheel, the coupling or the belt drive while the machine is switched on can result in serious injury.

➤ Do not open the enclosure while the machine is activated.
➤ Switch off and lock out the power supply disconnecting device and verify the absence of any voltage.
➤ Wear close-fitting clothes and a hair net if necessary.
➤ Make sure all covers and safety guards are in place and secured before re-starting.

Temperature

High temperatures are generated during compression. Touching hot components may cause injuries.

➤ Avoid contact with hot components.
These include, for example, compressor airends or blocks, oil and compressed air lines, coolers, oil separator tanks, motors and machine heaters.
➤ Wear protective clothing.
➤ If welding is carried out on or near the machine, take adequate measures to prevent sparks or heat from igniting oil vapors or parts of the machine.

Noise

The enclosure absorbs the machine noise to a tolerable level. This function will be effective only if the enclosure is closed.

➤ Operate the machine only with intact sound insulation.
➤ Wear hearing protection if necessary.
The blowing-off of the safety relief valve can be particularly loud.

Operating fluids/materials

The used operating fluids and materials can cause adverse health effects. Suitable safety measures must be taken in order to prevent injuries.

➤ Strictly forbid fire, open flame and smoking.
➤ Follow safety regulations when dealing with oils, lubricants and chemical substances.
➤ Avoid contact with skin and eyes.
➤ Do not inhale oil mist or vapor.
➤ Do not eat or drink while handling cooling and lubricating fluids.
➤ Keep suitable fire extinguishing agents ready for use.
➤ Use only KAESER approved operating materials.

Unsuitable spare parts

Unsuitable spare parts compromise the safety of the machine.

➤ Use only spare parts approved by the manufacturer for use in this machine.
➤ Use only genuine KAESER replacement parts on pressure bearing parts.

Conversion or modification of the machine

Modifications, additions to and conversions of the machine or the controller can result in unpredictable dangers.

- ➤ Do not convert or modify the machine!
- ➤ Obtain written approval by the manufacturer prior to any technical modification or expansion of the machine, the controller, or the control programs.

Extending or modifying the compressor station

If dimensioned appropriately, safety relief valves reliably prevent an impermissible rise in pressure. New dangers may arise if you modify or extend the compressed air station.

- ➤ When extending or modifying the compressed air system:
  Check the blow-off capacity of safety relief valves on air receivers and compressed air lines before installing a new machine.
- ➤ If the blow-off capacity is insufficient:
  Install safety relief valves with larger blow-off capacity.

3.5.2 Safe machine operation

The following is information supporting you in the safe handling of the machine during individual product life phases.

Personal protective equipment

When working on the machine you may be exposed to dangers that can result in accidents with severe adverse health effects.

- ➤ Wear protective clothing as necessary.

Suitable protective clothing (examples):

- Safety workwear
- Protective gloves
- Safety boots
- Eye protection
- Ear protection

Transporting

The weight and size of the machine require safety measures during its transport to prevent accidents.

- ➤ Use suitable lifting gear that conforms to local safety regulations.
- ➤ Allow transportation only by personnel trained in the safe movement of loads.
- ➤ Attach lifting gear only to suitable lifting points.
- ➤ Be aware of the center of gravity to avoid tipping.
- ➤ Make sure the danger zone is clear of personnel.
- ➤ Do not step onto machine components to climb up the machine.

Assembly

- ➤ Only use only electrical cables that are suitable and approved for the surroundings and electrical loads applied.

- Never dismantle compressed air pipes until they are fully vented.
- Only use pressure lines that are suitable and approved for the maximum working pressure and the intended medium.
- Do not allow connection pipes to be placed under mechanical stress.
- Do not induce any forces into the machine via the connections, so that the compressive forces must be balanced by bracing.

Positioning

A suitable installation location for the machine prevents accidents and faults.

- Install the machine in a suitable compressor room.
- Ensure sufficient and suitable lighting such that the display can be read and work carried out comfortably and safely.
- Ensure accessibility so that all work on the machine can be carried out without danger or hindrance.
- If installed outdoors, the machine must be protected from frost, direct sunlight, dust, rain and splashing water.
- Do not operate in areas in which specific requirements regarding explosion protection are in force.
- Ensure adequate ventilation.
- Place the machine in such a manner that the working conditions in its environment are not impaired.
- Comply with limit values for ambient temperature and humidity.
- The intake air must not contain any damaging contaminants,
  Damaging contaminants are for instance: explosive or chemically instable gases and vapors, acid or base forming substances such as ammonia, chlorine or hydrogen sulfide.
- Do not position the machine in warm cooling outlet air from other machines.
- Keep suitable fire extinguishing agents ready for use.

Commissioning, operation and maintenance

During commissioning, operation and maintenance you may be exposed to dangers resulting from, e.g., electricity, pressure and temperature. Careless actions can cause accidents with severe adverse effects for your health.

- Allow maintenance work to be carried out only by authorized personnel.
- Wear close-fitting, flame-resistant clothing. Wear protective clothing as necessary.
- Switch off and lock out the power supply isolating device and verify the absence of voltage.
- Check that there is no voltage on floating relay contacts.
- Close shut-off valves or otherwise isolate the machine from the compressed air network to ensure that no compressed air can flow back into the machine.
- De-pressurize all pressurized components and enclosures.
- Allow the machine to cool down.
- Do not open the cabinet while the machine is switched on.
- Do not open or dismantle any valves.
- Use only spare parts approved by KAESER for use in this machine.

- ➢ Carry out regular inspections:
  for visible damages,
  of safety installations,
  of the EMERGENCY STOP push button,
  of any components requiring monitoring.
- ➢ Pay particular attention to cleanliness during all maintenance and repair work. Cover components and openings with clean cloths, paper or tape to keep them clean.
- ➢ Do not leave any loose components, tools or cleaning rags on or in the machine.
- ➢ Components removed from the machine can still be dangerous.
  Do not attempt to open or destroy any components taken from the machine.

De-commissioning, storage and disposal

Improper handling of old operating fluids and components represent a danger for the environment.

- ➢ Drain off fluids and dispose of them according to environmental regulations.
  These include, for example, compressor oil and cooling water.
- ➢ Have refrigerant disposed of by authorized bodies only.
- ➢ Dispose of the machine in accordance with local environmental regulations.

3.5.3 Organizational Measures

- ➢ Designate personnel and their responsibilities.
- ➢ Give clear instructions on reporting faults and damage to the machine.
- ➢ Give instructions on fire reporting and fire-fighting measures.

3.5.4 Danger Areas

The table gives information on the areas dangerous to personnel.

Only authorized personnel may enter these areas.

Activity	Danger area	Authorized personnel
Transport	Within a 10 ft radius of the machine.	Installation personnel for transport preparation. No personnel during transport.
	Beneath the lifted machine.	No personnel!
Installation	Within the machine. Within 3 ft radius of the machine and its supply cables.	Installation personnel
Operation	Within a 3 ft radius of the machine.	Operating personnel
Maintenance	Within the machine. Within a 3 ft radius of the machine.	Maintenance personnel

Tab. 22 Danger Areas

3.6 Safety devices

Various safety devices ensure safe working with the machine.

- ➢ Do not change, bypass or disable safety devices.

- ➤ Regularly check safety devices for their correct function.
- ➤ Do not remove or obliterate labels and notices.
- ➤ Ensure that labels and notices are clearly legible.

Further information More information on safety devices is contained in chapter 4, section 4.9.

3.7 Working life of safety functions

The safety-relevant components of the safety functions are designed for a working life of 20 years. The working life starts with original machine commissioning, and is not extended by times during which the machine is not in use.

The following components are affected:

- Resistance thermometer (Pt100 sensor for measuring the compression discharge temperature)
- EMERGENCY STOP push button
- Main contactor
- Door interlock switch

1. The components of the safety functions must be replaced by an authorized KAESER service representative after a working life of 20 years.
2. Have an authorized KAESER service representative check the reliability of the safety functions.

3.8 Safety signs

The figure shows the position of the safety signs on the machine. The table lists the various safety signs used and their meanings.

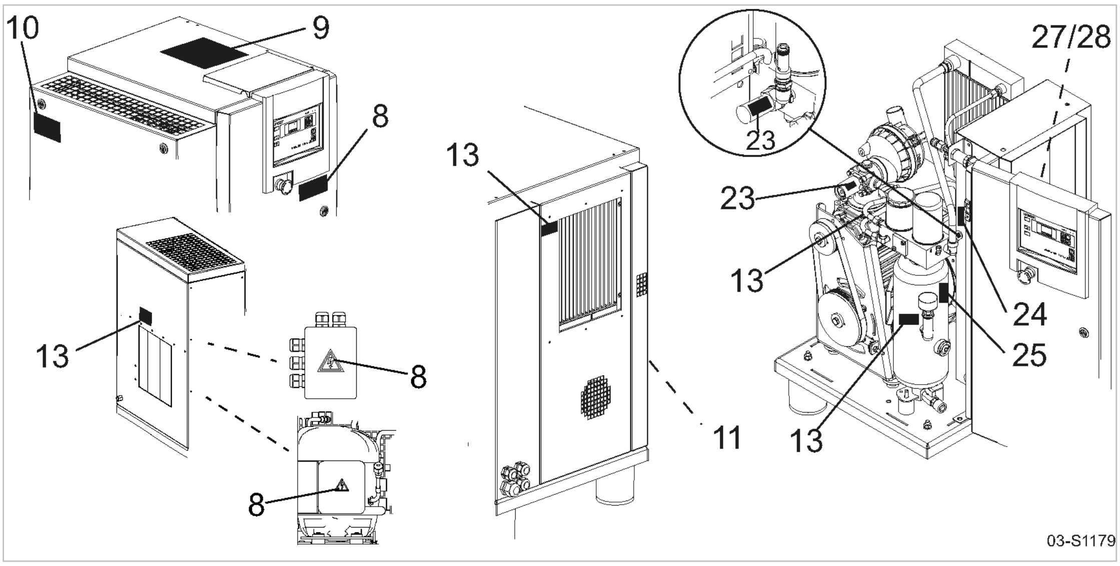

Fig. 4 Location of safety signs

Item	Symbol	Meaning
8		Danger of fatal injury from touching electrically live components! ➤ Switch off and lock out the power supply disconnecting device and check that no voltage is present.
9		Personal injury or damage to the machine by incorrect operation! ➤ Read and understand the service manual and all safety signs before switching on this machine.
		Machine starts automatically! Severe injury could result from rotating components, electrical voltage and air pressure. ➤ Switch off and lock out the power supply disconnecting device and check that no voltage is present before opening any machine enclosure or guard.
10		Rotating parts! Severe injury could result from touching the v-belt drive while it is rotating. ➤ Never switch the machine on without guard in place. ➤ Switch off and lock out the power supply disconnecting device and check that no voltage is present.
11		Injury and/or contamination can result from breathing compressed air! Contamination of food can result from using untreated compressed air for food processing! ➤ Never breathe untreated compressed air. ➤ Air from this compressor must meet OSHA 29CFR1910.134 and FDA 21CFR178.3570 standards, if used for breathing or food processing. Use proper compressed air treatment.
13		Hot surface can cause burns! ➤ Let the machine cool down. ➤ Wear long-sleeved garments (not synthetics such as polyester) and protective gloves.
23		Serious injury or death can result from loosening or opening component that is under pressure and heavily spring loaded! ➤ Do not open or dismantle the valve. ➤ Call an authorized KAESER service representative if a fault occurs.
24		Serious injury or death can result from loosening or opening component under pressure! ➤ De-pressurize all pressurized components and enclosures. ➤ Secure so that machine stays de-pressurized. ➤ Check that machine is de-pressurized.
25		Ear damage and burns can result from loud noise and/or oil mist when the safety relief valve opens! ➤ Wear ear protection and protective clothing. ➤ Close all access doors and cover panels.
27		Risk of electric shock! If the interrupter has tripped, current-carrying components of the controller should be examined and replaced if damaged to reduce the risk of fire or electric shock.

Item	Symbol	Meaning
28		Risk of electric shock! To maintain overcurrent short-circuit, and ground-fault protection, the manufacturer's instructions for setting the interrupter must be followed to reduce the risk of fire or electric shock.

Tab. 23 Safety signs

3.9 Emergency situations

3.9.1 Correct fire fighting

Suitable measures

Calm and prudent action can safe lives in the event of a fire.

➤ Keep calm.

➤ Give the alarm.

➤ Shut off supply lines if possible.
Main disconnecting device (all poles)
Cooling water (if present)
Heat recovery (if present)

➤ Warn and move endangered personnel to safety.

➤ Help incapacitated persons.

➤ Close the doors.

➤ When trained accordingly: Attempt to extinguish the fire.

Extinguishing substances

➤ Suitable extinguishing media:
Foam
Carbon dioxide
Sand or soil

➤ Unsuitable extinguishing media:
Strong jet of water

3.9.2 Treating injuries from handling cooling oil

Eye contact

Cooling oil can cause irritation.

➤ Rinse open eyes thoroughly for a few minutes under running water.

➤ Seek medical help if irritation persists.

Skin contact

Cooling oil may irritate after prolonged contact.

➤ Wash thoroughly with skin cleaner, then with soap and water.

➤ Contaminated clothing should be dry-cleaned before reuse.

Inhalation

Cooling oil mist may make breathing difficult.

➤ Clear air passages of oil mist.
➤ Seek medical help if difficulty with respiration continues.

Ingestion

➤ Wash out the mouth immediately.
➤ Do not induce vomiting.
➤ Seek medical aid.

3.9.3 Injury from Handling Refrigerant

Eye contact:

Severe eye irritation, watering, reddening and swelling of the eyelids.
Risk of caustic burns and frostbite.

➤ Open eyelids wide to allow product to evaporate.
➤ Hold the eyelid wide and rinse the eye with running water.
➤ Consult an ophthalmologist if you experience lasting pains.

Skin contact:

Initially a sensation of chill, skin may redden subsequently.
Risk of frostbite.

➤ Allow the product to evaporate.
➤ Rinse with lukewarm water.
➤ Consult a physician if experiencing lasting pain or reddened skin.

Inhalation:

At high concentrations, risk of cardiac irregularity (arrhythmia).
At very high concentration, risk of asphyxia caused by oxygen deficiency.

➤ Remove victim to the fresh air.
➤ If necessary Respiration with respirator or administration of oxygen.
➤ Consult a physician if experiencing breathing or nerve complaints.

3.10 Warranty

This operator manual contains no independent warranty commitment. Our general terms and conditions of business apply with regard to warranty.

A condition of our warranty is that the machine is used for the purpose for which it is intended under the conditions specified.
Due to the multitude applications for which the machine is suitable the obligation lies with the user to determine its suitability for his specific application.

In addition, we accept no warranty obligation for:

- the use of unsuitable parts or operating materials,
- unauthorized modifications,

- incorrect maintenance,
- incorrect repair.

Correct maintenance and repair includes the use of original spare parts and operating materials.

➤ Obtain confirmation from KAESER that your specific operating conditions are suitable.

3.11 Environmental protection

The operation of this machine may cause dangers for the environment.

➤ Do not allow cooling oil to escape to the environment or into the sewage system.

➤ Store and dispose of operating materials and replaced parts in accordance with local environmental protection regulations.

➤ Observe national regulations.
This applies particularly to parts contaminated with compressor cooling oil.

4 Design and Function

4.1 Enclosure

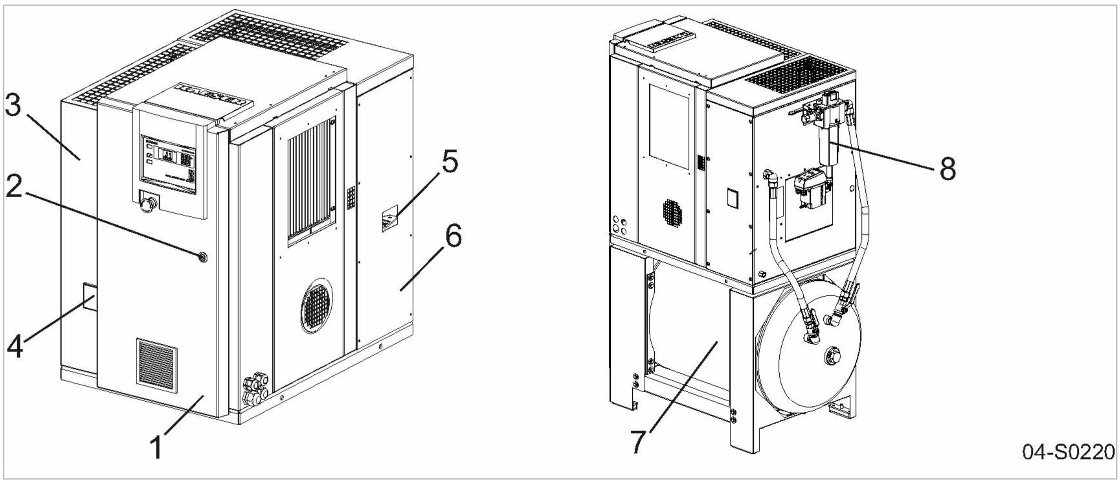

Fig. 5 Enclosure overview

1	Control cabinet door	5	Sight glass: V-belt tension
2	Latch	6	Add-on cabinet: Refrigerated dryer
3	Panel (removable)	7	Air receiver
4	Sight glass: Oil level indicator	8	Compressed air filter (optionally)

When closed, the enclosure serves various functions:

- Sound insulation
- Protection against contact with components
- Cooling air flow

The enclosure is not suitable for the following uses:

- Walking on, standing or sitting on.
- As resting place or storage of any kind of load.

Safe and reliable operation is only assured with the enclosure closed.

Access doors are hinged to swing open and removable panels can be lifted off. The latches should be opened using the supplied key.

4.2 Machine function

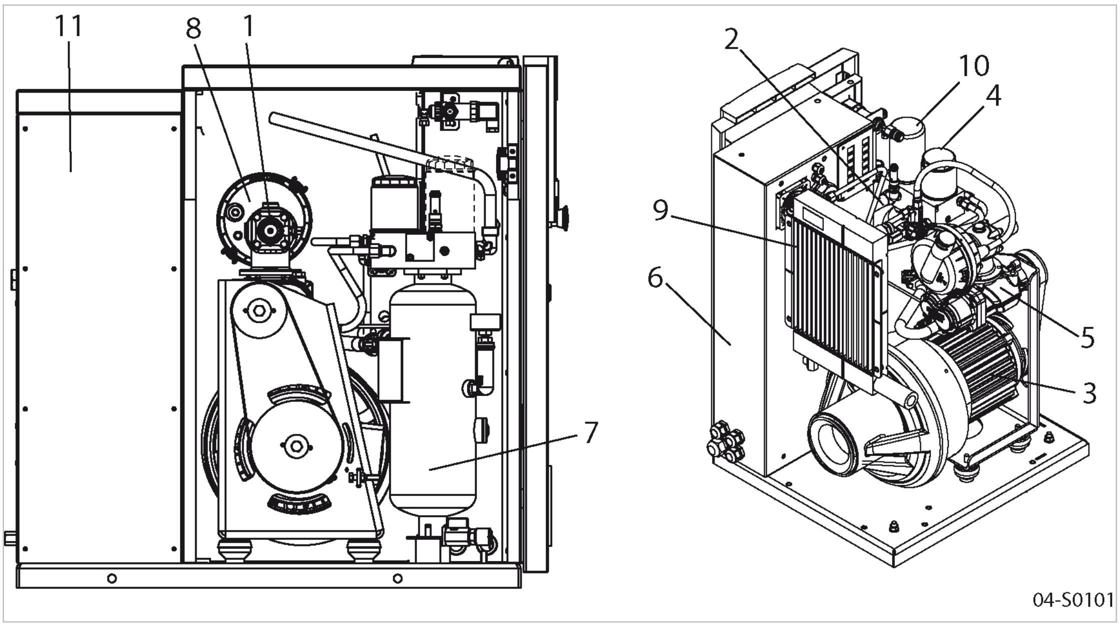

Fig. 6 Machine layout

1	Inlet valve	7	Oil separator tank
2	Minimum pressure / check valve	8	Air filter
3	Compressor motor	9	Oil cooler and compressed air cooler
4	Oil filter	10	Oil separator cartridge
5	Airend	11	Add-on cabinet for refrigerated dryer
6	Control cabinet		

Ambient air is cleaned as it is drawn in through the filter 8.

The air is then compressed in the airend 5.

The airend is driven by an electric motor 3.

Cooling oil is injected into the airend. It lubricates moving parts and forms a seal between the rotors themselves and between them and the airend casing. This direct cooling in the compression chamber ensures a very low airend discharge temperature.

Cooling oil recovered from the compressed air in the oil separator tank 7 and separator cartridge 10 gives up its heat in the oil cooler 9. The oil then flows through the oil filter 4 and back to the point of injection. Pressure within the machine keeps the oil circulating. A separate pump is not necessary. A thermostatic valve maintains optimum cooling oil temperature.

Compressed air passes through the minimum pressure / check valve 2 into the compressed air cooler 9 . The minimum pressure check valve ensures that there is always a minimum internal pressure sufficient to maintain cooling oil circulation in the machine.

The compressed air cooler brings down the compressed air temperature to 9 °F to 18 °F above the ambient temperature. Most of the moisture carried in the air is removed in this way.

4.3 Refrigerated dryer

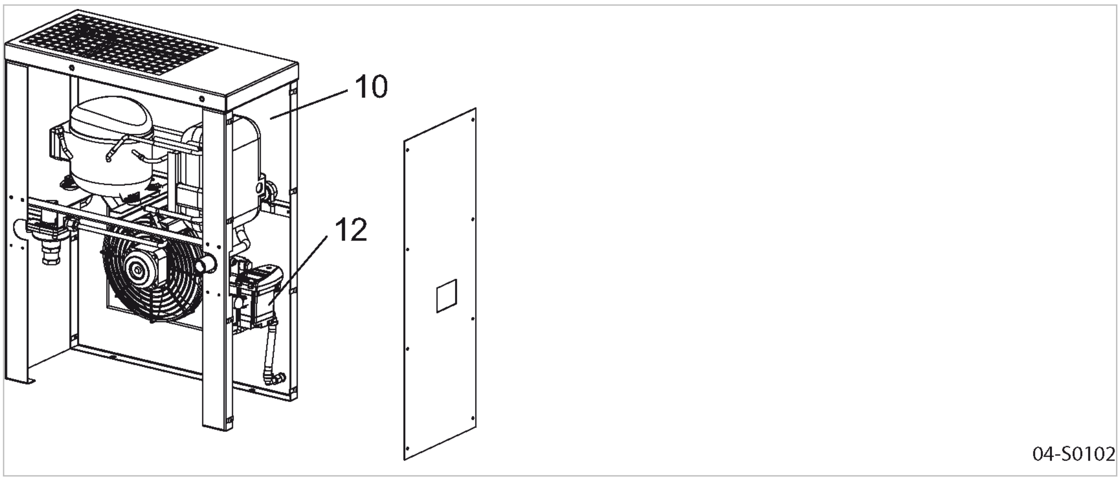

Fig. 7 Refrigerated dryer

10 Refrigerated dryer
12 Condensate drain

The downstream refrigerated dryer removes moisture from the compressed air.

The condensate drain ejects the precipitate.

4.4 Air receiver

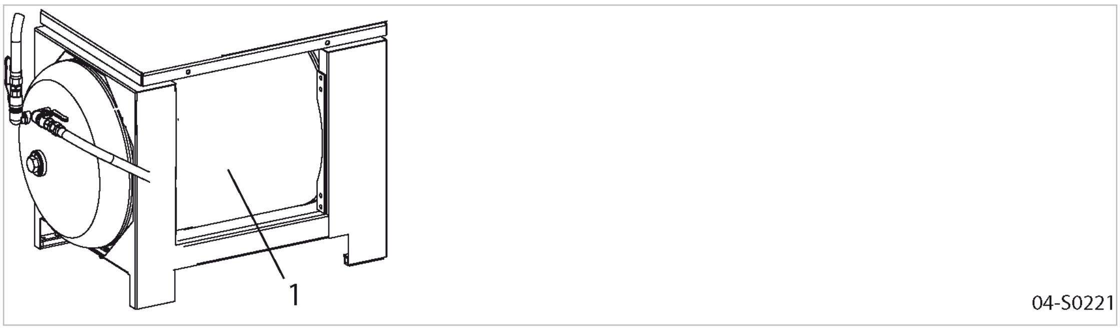

Fig. 8 Air receiver

1 Air receiver

The compressed air stored in the air receiver offsets peaks in air consumption.

Use a manual condensate drain to drain the condensate.

4.5 Floating relay contacts

Floating relay contacts are provided for the transfer of signals and messages.
Information on location, loading capacity and type of message or signal is found in the electrical diagram.

If the floating relay contacts are connected to an external voltage source, voltage may be present even when the machine is isolated from the power supply.

4.6 Options

The options available for your machine are described below.

4.6.1 Option C3/C48
SIGMA CONTROL 2 : Connection to control technology

Connection to various control technology systems is possible with the C3 option.

Option C3	Option C48
Main Control System (MCS): ■ Slot for a communication module to connect to a control technology system.	Main Control System Input Output (MCSIO): ■ Without slot for a communication module to connect to a control technology system. ■ Digital and analog inputs and outputs integrated.
Input-Output-Module (IOM): ■ Module with digital and analog inputs and outputs.	

Tab. 24 Components

4.6.2 Option H1
Machine mountings

These mountings allow the machine to be anchored firmly to the floor.

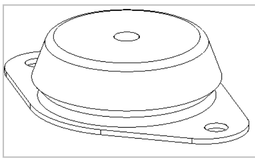

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Fig. 9 Machine mountings

4.6.3 Option F1
KAESER FILTER KE

➤ Comply with the specifications provided in the operating manual for the compressed air filter (see chapter 13.5).

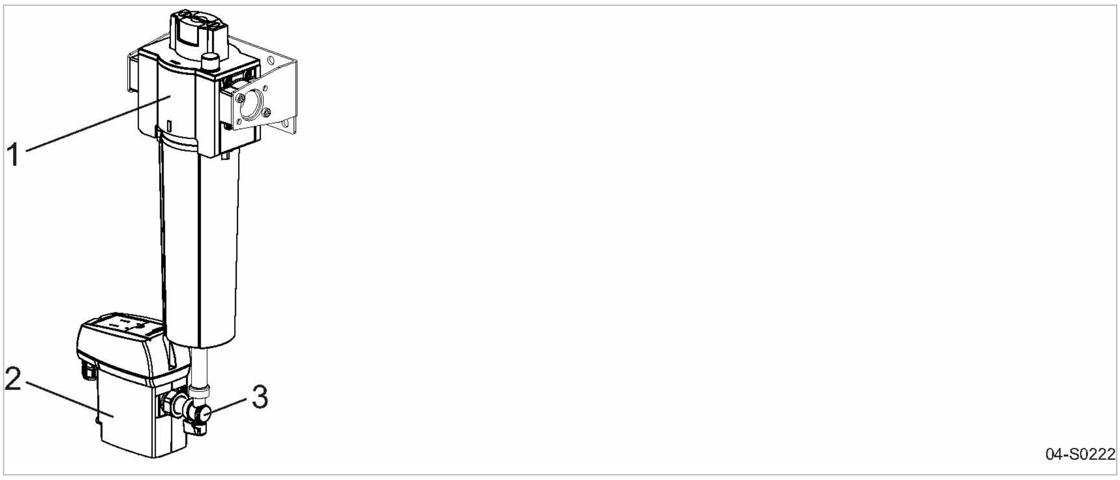


Fig. 10 KAESER FILTER KE

1 KAESER FILTER KE
2 Electronic condensate drain
3 Shut-off valve

KAESER FILTER KE removes aerosols and solid particles from the compressed air.

The condensate is removed by an electronic condensate drain. A hose coupling is provided for the connection of a flexible condensate line.

Further information The information provided in chapter 6.4 is to be used additionally for the condensate drain.

4.7 Operating modes and control modes

4.7.1 Machine operating modes

STOP

The machine is connected to the power supply.
The *voltage applied to controller* LED lights green.
The machine is switched off. The *ON* LED is extinguished.

READY

The machine has been activated with «ON»:

- The *ON* LED lights green.
- The drive motor is stopped.
- The inlet valve is closed.
- The minimum pressure/check valve isolates the oil separator tank from the air system.
- The venting valve is open.

The compressor motor starts as soon as pressure is demanded.

In addition, timer and/or remote control may affect the start of the motor.

LOAD

The compressor motor runs under load.
- The inlet valve is open.
- The compressor block delivers compressed air to the system.

IDLE

The compressor motor runs unloaded with low power consumption.
- The inlet valve is closed.
- The minimum pressure/check valve isolates the oil separator tank from the air system.
- The venting valve is open.

A small volume of air circulates through the bypass bore in the inlet valve, through the compressor block and back to the inlet valve via the venting line.

4.7.2 Control modes

Using the selected control mode, the controller switches the machine between its various operational states in order to maintain the gauge working pressure between the set minimum and maximum values, regardless of the drawn compressed air volume. The control mode also rules the degree of energy efficiency of the machine.

The shortest possible times for the various parameters is preset by the factory to ensure that the compressor motor earlier and more frequently is at standstill. If you want to change these parameters, select the shortest possible times in order for the machine working energy-efficiently.

The machine-dependant venting time between the LOAD and READY operating modes ensures load changes at minimum material stresses.

The following control modes can be set:
- DUAL
- QUADRO
- VARIO

Supplementary mechanical flow rate regulation:

Option C1
- MODULATING control

Energy-efficient control modes for various applications:

Application	Recommended control mode
Compressed air station with one machine or several machines supplying similar volumes.	VARIO
Machine for peak load in a compressed air station	DUAL
Machine for intermediate load in a compressed air station	VARIO
Machine for basic load in a compressed air station	QUADRO

Tab. 25 Energy-saving control modes

The QUADRO control mode is preset by the factory, unless a different agreement has been made with the manufacturer.

DUAL

In the DUAL control mode, the machine is switched back and forth between LOAD and IDLE to maintain the machine working pressure between the preset minimum and maximum values. When maximum pressure is reached, the machine switches to IDLE. When the preset *idling time* has elapsed, the machine switches to READY.

QUADRO

Unlike DUAL control mode, in QUADRO the machine will switch between LOAD to READY after short periods of being in IDLE.

Following extended times in the various operating modes, the machine switches from LOAD to READY.

In this event, the controller considers the time in READY mode as *standstill time*. The time in LOAD and IDLE operating modes are taken as *minimum runtime*.

VARIO

The VARIO mode is based on the DUAL control mode. The difference to DUAL is that the *idling time* is automatically lengthened or shortened to compensate for higher or lower machine starting frequencies.

Option C1

MODULATING control

The MODULATING control is an additional mechanical regulation. It continuously changes the flow rate within the machine's control range.

A control valve, the proportional controller, changes the degree of opening of the inlet valve when the machine transports compressed air into the air network (LOAD)

The load and power consumption of the drive motor rises and falls with the air demand.

4.8 Refrigerated Dryer Control Modes

The controller can operate in the following modes:

- CONTINUOUS
- TIMER

CONTINUOUS

The refrigerated dryer will remain activated even when the machine is in standby.

This mode of control is set up at the factory.

TIMER

The refrigerated dryer is switched on and off by a timer when the machine is in standby. In this mode, the operating temperature is held within tight limits.

Which control mode is the most practical, and when?

Control mode	Advantage	Disadvantage
CONTINUOUS	Constant dew point.	Higher power consumption when the machine is in standby mode.

Control mode	Advantage	Disadvantage
TIMER	Lower power consumption when the machine is in standby mode.	Brief increase in dew point when the compressor re-starts

Tab. 26 Refrigerated dryer control modes

4.9 Safety devices

The following safety devices are provided and must not be modified or disabled.

- EMERGENCY STOP push button:
  Stops the machine immediately in an emergency situation. The motor is stopped. The pressure system is vented.
- Safety relief valve:
  The safety relief valve protects the system against excessive pressure. It is factory set.
- Airend discharge temperature sensor:
  Monitoring the airend discharge temperature protects the compressed air system from impermissible temperature rises.
- Door interlock switch:
  The machine will stop automatically if a safety interlocked door or panel is opened or removed.
- Safety pressure monitor (Machine with refrigerated dryer):
  The safety pressure monitor protects the refrigerant circuit of the refrigerated dryer against excessive pressure. It cannot be set.
- Enclosures and guards for moving parts and electrical connections:
  Protect against accidental contact.

4.10 Operating panel SIGMA CONTROL 2

Keys

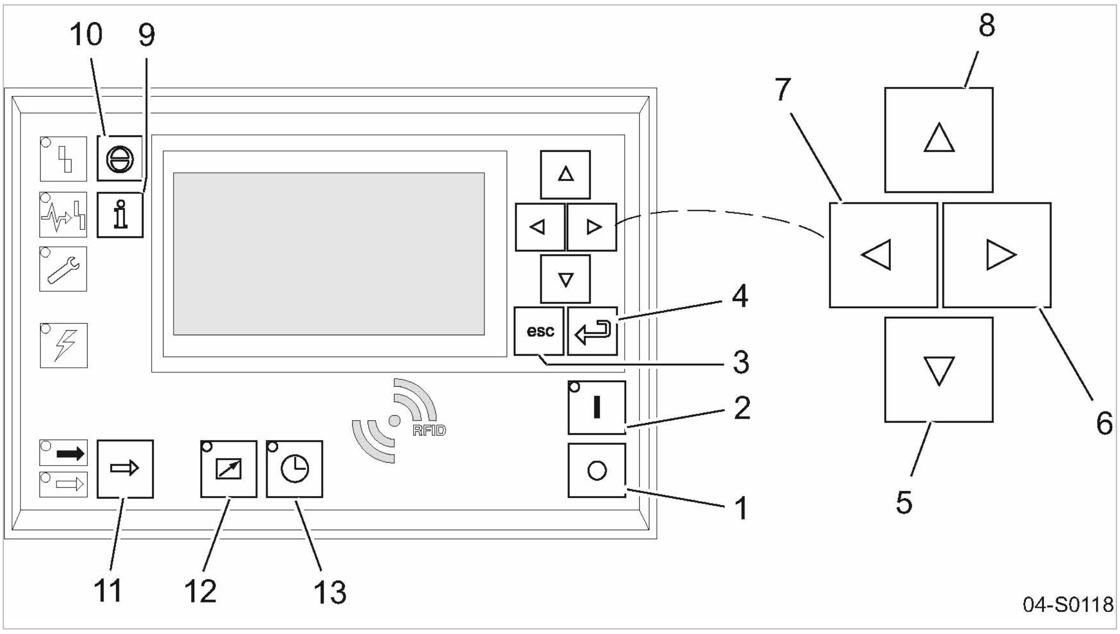


Fig. 11 Keys - overview

Position	Name	Function
1	«OFF»	Switches the machine off.
2	«ON»	Switches the machine on.
3	«Escape»	Returns to the next higher menu option level. Exits the edit mode without saving.
4	«Enter»	Jumps to the selected menu option. Exits the edit mode and saves.
5	«Down»	Scrolls down the menu options. Reduces a parameter value.
6	«Right»	Jumps to the right. Moves the cursor position to the next right field.
7	«Left»	Jumps to the left. Moves the cursor position to the next left field.
8	«Up»	Scrolls up the menu options. Increases a parameter value.
9	«Events and information»	Operating mode: Displays the event memory.
10	«Acknowledgement»	Acknowledges alarms and warning messages. If permissible: Resets the fault counter (RESET).
11	«LOAD/IDLE»	Toggles between the LOAD and IDLE operating modes.
12	«Remote control»	Switches the remote control on and off.
13	«Timer control»	Switches clock control (timer) on and off.

Tab. 27 Keys

Indicators

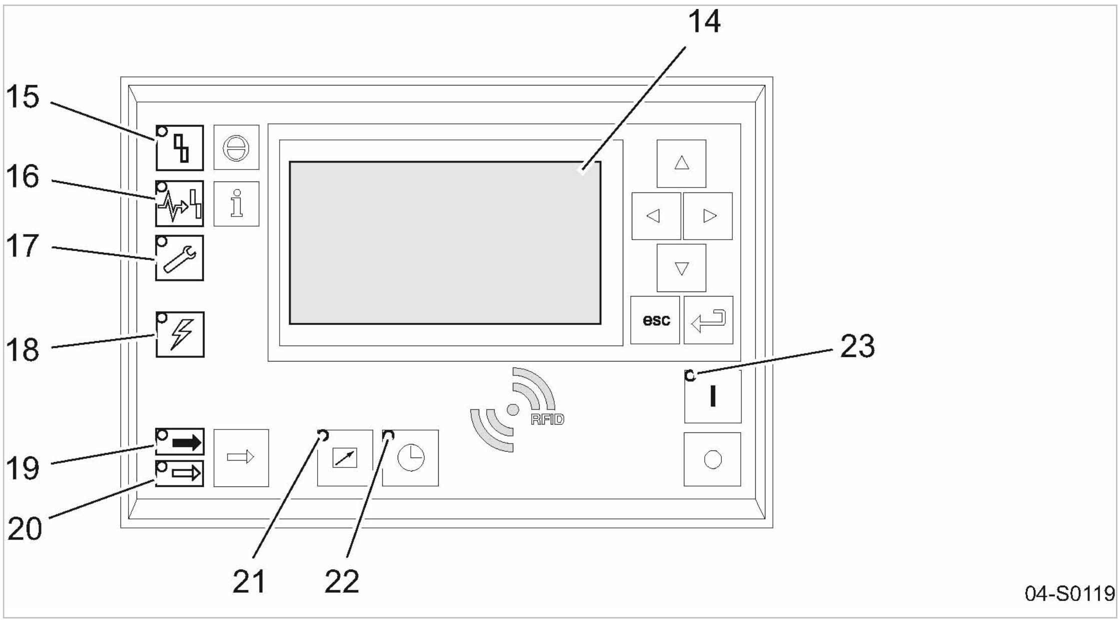


Fig. 12 Indicators – overview

Position	Name	Function
14	Indicator field or display	Graphic display with 8 lines, 30 characters each.
15	*Fault*	Flashes red when an alarm occurs. Lights continuously when acknowledged.
16	*Communications*	Continuous red illumination if a communication connection (Ethernet, USS, COM modules) has a fault.
17	*Warning*	Flashes yellow for the following events: ■ Maintenance work due ■ Warning message Lights yellow continuously when acknowledged.
18	*Controller voltage*	Lights green when the power supply is switched on.
19	*LOAD*	Lights green when the compressor is running under LOAD.
20	*IDLE*	Lights green when the compressor is running in IDLE. Flashes when the «LOAD/IDLE» toggle key is pressed.
21	*Remote control*	The LED lights when the machine is in remote control.
22	*Shift control*	The LED lights when the machine is in clock control (timer).
23	*Machine ON*	Lights green when the machine switched on.

Tab. 28 Indicators

RFID sensor field

RFID is the abbreviation for “Radio Frequency Identification” and enables the identification of persons or objects.
Placing a suitable transponder in front of the RFID sensor field of the controller will automatically activate the communication between transponder and SIGMA CONTROL 2.

A suitable transponder is the KAESER RFID EQUIPMENT CARD. Two of them have been provided with the machine.

Typical application:

- Users log on to the machine.
  (no manual input of the password required.)

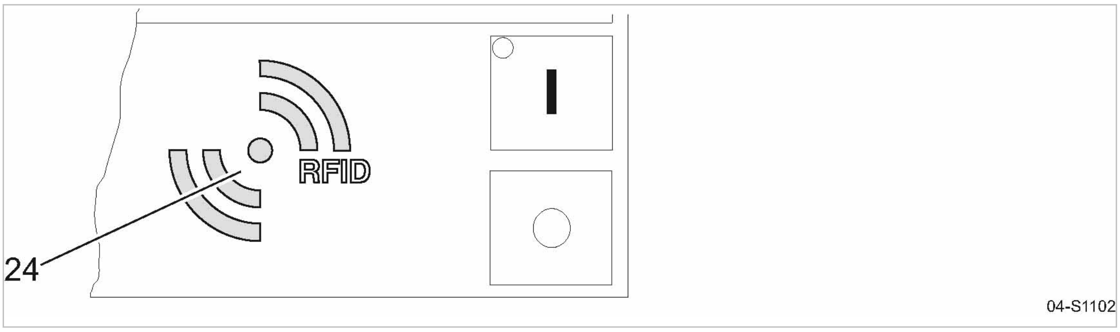


Fig. 13 RFID sensor field

Position	Name	Function
24	RFID	RFID sensor field for the communication with a suitable RFID transponder

Tab. 29 RFID sensor field

Further information

More information about the use of RFID technology is provided in the SIGMA CONTROL 2 operating manual.

5 Installation and Operating Conditions

5.1 Ensuring safety

The conditions in which the machine is installed and operated have a decisive effect on safety.

Warning instructions are located before a potentially dangerous task.

Disregard of warning instructions can cause serious injuries!

Complying with safety warnings

Disregard of safety warnings can cause unforeseeable dangers!

➤ Strictly forbid fire, open flame and smoking.
➤ If welding is carried out on or near the machine, take adequate measures to prevent sparks or heat from igniting oil vapors or parts of the machine.
➤ Do not store inflammable material in the vicinity of the machine.
➤ The machine is not explosion-proof!
Do not operate in areas in which specific requirements with regard to explosion protection are in force.
➤ Ensure sufficient and suitable lighting such that the display can be read and work carried out comfortably and safely.
➤ Keep suitable fire extinguishing agents ready for use.
➤ Ensure that required ambient conditions are maintained.

Required ambient conditions may be:

- Maintain ambient temperature and humidity
- Ensure the appropriate composition of the air within the machine room:
  - Clean with no damaging contaminants (e.g., dust, fibers, fine sand).
  - Free of explosive or chemically unstable gases or vapors.
  - Free of acid/alkaline forming substances, particularly ammonia, chlorine or hydrogen sulfide.

5.2 Installation conditions

5.2.1 Determining location and clearances

The machine is intended for installation in an appropriate machine room. Information on distances from walls and ventilation is given below.

The distances quoted are recommended distances and ensure unhindered access to all machine parts.

➤ Please consult KAESER if they cannot be adhered to.

Precondition The floor must be level, firm, and capable of bearing the weight of the machine.

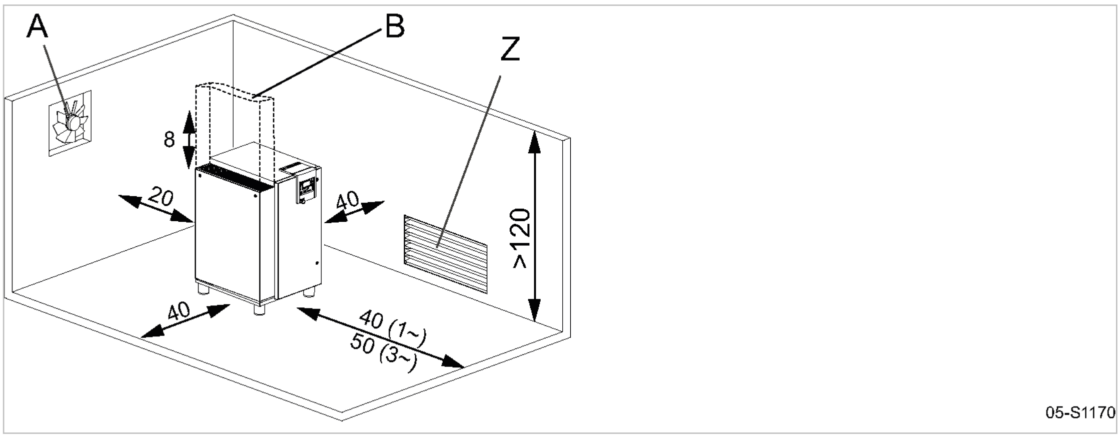


Fig. 14 Recommended machine placement and dimensions [in.]

- A Exhaust fan
- B Exhaust air duct
- Z Air inlet aperture

1. **NOTICE** *Ambient temperature too low.*
   Frozen condensate and highly viscous cooling oil can cause damage when starting the machine.
   - ➤ *Make sure that the temperature of the machine is at least +37 °F before starting.*
   - ➤ *Heat the machine room adequately or install an auxiliary heater.*
2. Ensure adequate lighting so that all work on the machine can be carried out without danger or hindrance.
3. Ensure that the indicators can be read without glare and that the controller display cannot be damaged by direct sunlight (UV radiation).
4. Ensure that all intake and exhaust apertures of the enclosure remain opened.
5. If installed outdoors, the machine must be protected from frost, direct sunlight, dust, and rain.

5.2.2 Ensuring the machine room ventilation

Adequate ventilation of the machine room has several tasks:

- It prevents subatmospheric pressure in the machine room.
- It evacuates the exhaust heat of the machine and thus ensures the required operating conditions.

➤ Consult with KAESER if you cannot ensure the conditions for an adequate ventilation of the machine room.

1. Ensure that the flow volume of fresh air is at least the same as the volume taken by the machine and exhaust fan from the machine space.
2. Make sure that the machine and exhaust fan can only operate when the inlet aperture is actually open.
3. Keep the inlet and exhaust apertures free of obstructions so that the cooling air can flow freely through the room.
4. Ensure clean air in order to support the proper functioning of the machine.

5.2.3 Exhaust duct design

At the cooling air inlet and exhaust, the machine can only overcome the air resistance resulting from the duct design. Any additional air resistance will reduce airflow and deteriorate machine cooling.

Maximum heat capacity available:

	SX 3	SX 4	SX 5	SX 7.5
Heat capacity [BTU/hr]	9213	11601	15013	20473

Tab. 30 Heat capacity (cooling air)

➤ Consult an authorized KAESER service representative before deciding on:
- Design of the exhaust air ducting
- Transition between the machine and the exhaust air duct
- Length of the ducting
- Number of duct bends
- Design of flaps or shutters

Further information

Further information on the design of exhaust air ducts can be found in chapter 13.3.

5.3 Operating the machine in a compressed air system

When the machine is connected to a compressed air system, the system operating pressure must not exceed 232 psig.

Initial charging of a fully vented air system creates a very high rate of airflow through the air treatment devices. These conditions are detrimental to correct air treatment. Air quality suffers. To ensure the desired air quality when charging a vented air system, we recommend the installation of an air main charging system.

➤ Consult an authorized KAESER service representative for advice on this subject.

6 Installation

6.1 Ensuring safety

Follow the instructions below for safe installation.

Warning instructions are located before a potentially dangerous task.

Disregard of warning instructions can cause serious injuries!

Complying with safety notes

Disregard of safety notes can cause unforeseeable dangers!

➤ Follow the instructions in chapter 3 'Safety and Responsibility'.
➤ Installation work may only be carried out by authorized personnel.
➤ Make sure that no one is working on the machine.
➤ Ensure that all service doors and panels are locked.

When working on live components

Touching voltage carrying components can result in electric shocks, burns or death.

➤ Work on electrical equipment may only be carried out by authorized electricians.
➤ Switch off and lock out the power supply isolating device and verify the absence of voltage.
➤ Check that there is no voltage on floating relay contacts.

When working on the compressed air system

Compressed air is contained energy. Uncontrolled release of this energy can cause serious injury or death. The following safety concerns relate to any work on components that could be under pressure.

➤ Close shut-off valves or otherwise isolate the machine from the compressed air network to ensure that no compressed air can flow back into the machine.
➤ Depressurize all pressurized components and enclosures.
➤ Check all hose couplings in the compressed air system with a hand-held pressure gauge to ensure that they all read 0 psig.
➤ Do not open or dismantle any valves.

When working on the drive system

Touching voltage carrying components can result in electric shocks, burns or death.
Touching the fan wheel, the coupling or the belt drive while the machine is switched on can result in serious injury.

➤ Switch off and lock out the power supply isolating device and verify the absence of voltage.
➤ Do not open the cabinet while the machine is switched on.

Further information

Details of authorized personnel are found in chapter 3.4.2.

Details of dangers and their avoidance are found in chapter 3.5.

6.2 Reporting Transport Damage

1. Check the machine for visible and hidden transport damage.
2. Inform the carrier and the manufacturer in writing of any damage without delay.

6.3 Connecting the machine with the compressed air network

On the AIRCENTER the user's shut-off valve is already fitted to the machine.

Condensate in the compressed air network can damage the pipework:

➤ Install only corrosion-resistant pipes.
➤ Use fluoroelastomers as gaskets.
➤ Be aware of galvanic or dissimilar metal corrosion when using other metal piping materials.
➤ Consult with KAESER for suitable materials for the compressed air network.

Precondition: The compressed air system is vented completely to atmospheric pressure.

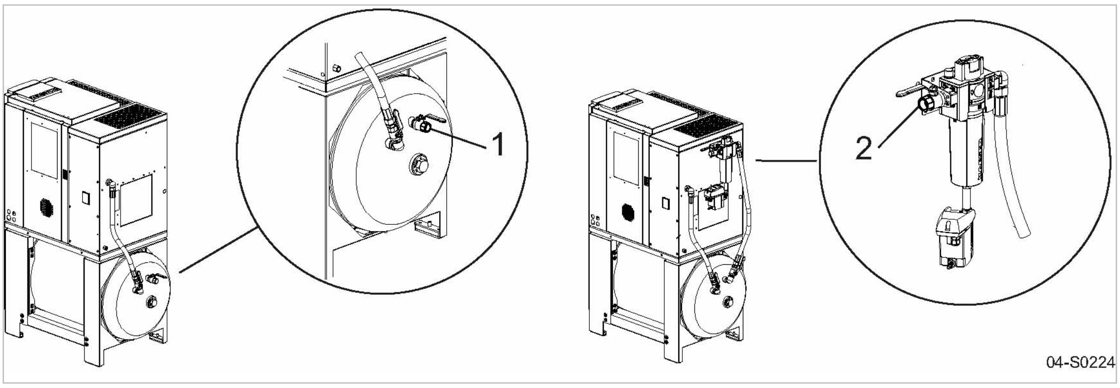


Fig. 15 Compressed air connection

1 Compressed air connection
2 Compressed air connection (Option F1)

1. **⚠ WARNING** *Serious injury or death can result from loosening or opening components under pressure!*
   ➤ *Fully vent all pressurized components and enclosures.*
2. Connect a flexible pressure hose.

Further information: The dimensional drawing in chapter 13.3 provides the size and position of the compressed air connection.

6.4 Connecting the condensate drain

A threaded hose connection is provided to attach a condensate drain hose.

The condensate must be able to drain freely.

➤ Only machines with 232 psig maximum permissible working pressure may be connected to the condensate collecting line.

Fig. 16 illustrates the recommended installation.
Condensate flows downward in the collecting line. This prevents condensate flowing back to the compressor.

If condensate flows at several points into the condensate collecting line, you must install shut-off valves in the condensate lines to shut the condensate line off before commencing maintenance work.

Condensate line

Feature	Value
Max. length[1] [ft]	50
Max. delivery head [ft]	16
Material (pressure-resistant, corrosion-proof)	Copper Stainless steel Plastics Hose line
[1] For longer lengths, please contact KAESER before installation.	

Tab. 31 Condensate line

Condensate collecting line

Feature	Value
Gradient [%]	≥3
Max. length[1] [ft]	65
Material (pressure-resistant, corrosion-proof)	Copper Stainless steel Plastics Hose line
[1] For longer lengths, please contact KAESER before installation.	

Tab. 32 Condensate collecting line

Compressed air flow rate[1] [cfm]	Line diameter ["]
<350	3/4
350 – 730	1
731 – 1410	1 1/2
>1410	2
[1] Compressed air flow rate as guide for the condensate volume to be expected	

Tab. 33 Condensate collecting line: Line diameter

Connecting the condensate drain

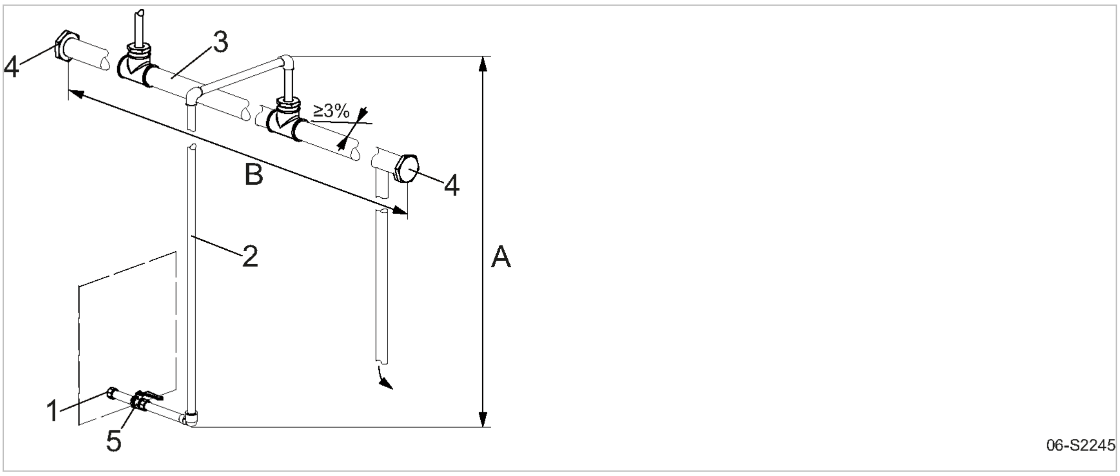


Fig. 16 Connecting the condensate drain

1	Threaded connection	5	Shut-off valve
2	Condensate line	A	Delivery head
3	Condensate collecting line	B	Length of the condensate collecting line
4	Screw plug		

Depending on the machine model, you may have several condensate drains.

➤ Directly connect every condensate drain to the condensate collecting line.

➤ Collect the condensate in a suitable container and dispose of in accordance with environmental regulations.

Further information The dimensional drawing in chapter 13.3 provides the size and position of the connection port.

6.5 Connecting the external pressure transducer

Material Retrofit kit: "External pressure transducer SIGMA CONTROL 2"
Use suitable shielded, copper-core cable (e.g.: LIYCY 2 x0.75 mm^2 for ambient temperature up to 30°C and laying method C).

Precondition The power supply disconnecting device is switched off,
the device is locked off,
the absence of any voltage has been verified.

Cable length between the machine and the pressure transducer: < 98 ft.

By means of a pressure transducer, the pressure in the compressed air network can be measured at any selected location and this signal used to regulate the compressor.

This ensures optimum compressor regulation with regard to the network pressure at the selected location.

Safety monitoring of the machine's internal pressure is unaffected.

An authorized KAESER SERVICE representative will be glad to provide support on planning and executing an individual solution.

1. Install the pressure transducer at the selected location in the compressed air network.

2. Using a suitable cable, connect the pressure transducer to a spare analog input.
   ➤ Connect as large an area of the shielding as possible to the mounting plate in the control cabinet or use an EMC fitting to make contact.
3. When commissioning the machine with a SIGMA CONTROL 2, select the *<Network actual pressure>* setting in the *<AI/>* menu.
4. Select and activate the used analog input (AII).

Further information
The electrical diagram in chapter 13.4 contains further details of the pressure transducer connection.

6.6 Option T2 Refrigerated dryer: Connecting the transformer according to the power supply.

The refrigerated dryer transformer has tappings for various power supply voltages.

1. Check that the correct connections are made for the supply voltage provided for the machine.
2. If necessary, re-connect the transformer to match the power supply voltage.

Further information
The electrical diagram in chapter 13.4 contains further details of the power supply connection.

6.7 Making the Power Supply Connection

Contact an authorized KAESER service representative prior to making any changes to the electrical connections!

The machine is **not** wired ready for operation!

This is a Tri-Voltage machine.

The machine can be set up to one of the following supply voltages:

- 208V
- 230V
- 460V

Precondition
The power supply disconnecting device is switched off.
The device is locked out and tagged out.
The absence of any voltage has been verified.

1. Have the electrical connections carried out by authorized personnel only.
2. Carry out protection measures as stipulated in relevant regulations (OSHA, NFPA, NEC for example) and in national accident prevention regulations. In addition, the regulations of the local electricity supplier or municipality must be observed.
3. Test the overcurrent protection cut-out to ensure that the time it takes to disconnect in response to a fault is within the permitted limit.
4. Use supply conductors and fuses in accordance with local regulations.
5. The user must provide the machine with a lockable power supply disconnecting device.
   This could be, for example, a switch-disconnector with fuses. If a circuit breaker is used it must be suitable for the motor starting characteristics.

6. Check that the correct taps on the control voltage transformer are connected according to the supply voltage.
   If this is not correct, change the connection to suit the power supply voltage.
7. **⚠ DANGER** *Danger of fatal injury from electric shock!*
   ➢ *Switch off, lock out and tag out the power supply disconnecting device and verify the absence of any voltage.*
8. If necessary: Set up the machine for the correct power supply voltage as described in chapter 6.7.1.
9. Connect the power supply.

Further information: The electrical diagram 13.4 contains further specifications for electrical connection.

6.7.1 Changing main voltage connections

Machine set up for [V]	208	230	460
Machine may be modified [V]	230	208	208
	460	460	230

Tab. 34 Voltage selection

The following parts have to be considered for making the change:

- Motor connection terminals in the control cabinet
- Drive motor overload protection relay
- Control transformer
- Transformer for the refrigerated dryer (only if present).

Material: If necessary, an alternative overload protection relay is provided in the control cabinet.

Precondition: The absence of any voltage has been verified.

Changing the drive motor connection

➢ Open the control cabinet and connect the motor in accordance with the electrical diagram.

Adjusting the overload protection cutout

➢ Check the overload protection relay setting.

	SX 3	SX 4	SX 5	SX 7.5
208V, 3-ph, 60Hz [A]	4.6[1)] / 5.2[2)]	6.2[1)] / 6.9[2)]	8.3[1)] /9.3[2)]	10.9[1)] / 12.2[2)]
230V, 3-ph, 60Hz [A]	4.3[1)] / 4.8[2)]	5.8[1)] / 6.5[2)]	8.3[1)] / 9.3[2)]	10.4[1)] / 11.6[2)]
460V, 3-ph, 60Hz [A]	2.1[1)] / 2.4[2)]	2.9[1)] / 3.3[2)]	4.2[1)] / 4.7[2)]	5.2[1)] / 5.8[2)]

[1)] NEC/CEC setting

[2)] NEC430.32(C) incremental setting

Tab. 35 Overload protection relay settings.

Connecting the control transformer

The pimary winding of the control transformer is not connected. The machine will not run without connecting the control transformer according to the power supply.

➤ Open the control cabinet and connect the control transformer in accordance with the electrical diagram.

6.8 Options

6.8.1 Option H1 Anchoring the machine

➤ Use appropriate fixing bolts to anchor the machine.

Further information: Details of the fixing holes are contained in the dimensional drawing in chapter 13.3.

6.8.2 Option W1 Connecting the external heat recovery system

An unsuitable heat exchanger or incorrect installation may influence the cooling oil circuit within the compressor. Damage to the machine will follow.

➤ Consult KAESER on a suitable heat exchanger and have an authorized KAESER service representative do the installation.

Further information: The dimensional drawing in chapter 13.3 gives the flow direction, size, and position of the connection ports.

6.8.3 Option F1 Connecting the condensate drain

➤ Connecting the condensate hose, as described in the KAESER FILTER assembly and operating manual.

Further information: The operating instructions for the KAESER FILTER are supplied in chapter 13.5.

7 Initial Start-up

7.1 Ensuring safety

Here you will find instructions for a safe commissioning of the machine.

Warning instructions are located before a potentially dangerous task.

⚠ Disregard of warning instructions can cause serious injuries!

Complying with safety notes

Disregard of safety notes can cause unforeseeable dangers!

- ➤ Follow the instructions in chapter 3 'Safety and Responsibility'.
- ➤ Commissioning tasks may only be carried out by authorized personnel!
- ➤ Make sure that no one is working on the machine.
- ➤ Ensure that all service doors and panels are locked.

When working on live components

Touching voltage carrying components can result in electric shocks, burns or death.

- ➤ Work on electrical equipment may only be carried out by authorized electricians.
- ➤ Switch off and lock out the power supply isolating device and verify the absence of voltage.
- ➤ Check that there is no voltage on floating relay contacts.

When working on the compressed air system

Compressed air is contained energy. Uncontrolled release of this energy can cause serious injury or death. The following safety concerns relate to any work on components that could be under pressure.

- ➤ Close shut-off valves or otherwise isolate the machine from the compressed air network to ensure that no compressed air can flow back into the machine.
- ➤ De-pressurize all pressurized components and enclosures.
- ➤ Check all hose couplings in the compressed air system with a hand-held pressure gauge to ensure that they all read 0 psig.
- ➤ Do not open or dismantle any valves.

When working on the drive system

Touching voltage carrying components can result in electric shocks, burns or death.
Touching the fan wheel, the coupling or the belt drive while the machine is switched on can result in serious injury.

- ➤ Switch off and lock out the power supply isolating device and verify the absence of voltage.
- ➤ Do not open the cabinet while the machine is switched on.

Further information

Details of authorized personnel are found in chapter 3.4.2.

Details of dangers and their avoidance are found in chapter 3.5.

7.2 Instructions to be observed before commissioning

Incorrect or improper commissioning can cause injury to persons and damage to the machine.

➤ Commissioning may be carried out only by authorized installation and service personnel who have been trained on this machine.

Special measures for recommissioning after storage/standstill

Storage period/ standstill period longer than	Measure
3 months	➤ Manually fill the airend with cooling oil.
12 months	➤ Change the oil filter. ➤ Change the oil separator cartridge. ➤ Change the cooling oil. ➤ Manually fill the airend with cooling oil.
36 months	➤ Have the overall technical condition checked by an authorized KAESER service representative.

Tab. 36 Re-commissioning after storage/standstill

7.3 Checking installation and operating conditions

➤ Check and confirm all the items in the check list before first start-up of the machine.

Check:	see chapter	Confirmed?
➤ Are the operators completely familiar with safety regulations?	–	
➤ Have all the positioning conditions been fulfilled?	5	
➤ Is a user-supplied lockable power supply disconnecting device installed?	6.7	
➤ Are the tolerance limits of the power supply within the permissible tolerance limits of the rated machine voltage? (see nameplate in the control cabinet)	13.4	
➤ Are the power supply cable cross sections and fuse ratings adequate?	2.15	
➤ Drive motor overload protection switch set according to the power supply voltage?	7.4	
➤ Option T2: Check that the correct connections are made for the supply voltage provided for the machine.	6.6	
➤ Have all electrical connections been checked for tightness?	–	
➤ Has the direction of rotation been checked?	7.7	
➤ Has the connection to the air system been made with a shut-off valve and a flexible hose?	6.3	
➤ Is the condensate drain connected?	6.4	
➤ Is there sufficient cooling oil in the separator tank?	10.12	

Check:	see chapter	Confirmed?
➤ Sufficient cooling oil in the airend?	7.5	
➤ Is the machine firmly anchored to the floor? (Option H1)	6.8.1	
➤ Are door interlock switches aligned and the function checked?	7.10	
➤ Are all access doors closed and latched and removable panels in place and secured?	–	

Tab. 37 Installation conditions checklist

7.4 Setting the overload protection relay

Electrical diagram 13.4 gives the location of the overload protection relay.

With star-delta starting, the phase current is fed via the overload protection relay. This phase current is 0.58-times the rated motor current.

To prevent the overload protection relay from being triggered by voltage fluctuations, temperature influences or component tolerances, the setting can be higher than the arithmetical phase current.

➤ Check the overload protection relay setting.

?

The overload protection relay shuts the machine down despite being correctly set?

➤ Contact an authorized KAESER service representative.

7.5 Pouring cooling oil into the airend

Before starting the compressor for the very first time and before re-starting after a shutdown period of more than 3 months it is necessary to manually add a quantity of cooling oil into the airend. In order to avoid that the cooling oil exceeds the permissible level, drain the required quantity from the de-pressurized oil separator tank.
Chapter 10.15 provides detailed information on how to drain cooling oil from the oil separator tank.

Material 0.26qt. cooling oil

Precondition The supply disconnecting device is switched off,
the device is locked off,
the absence of voltage has been verified.

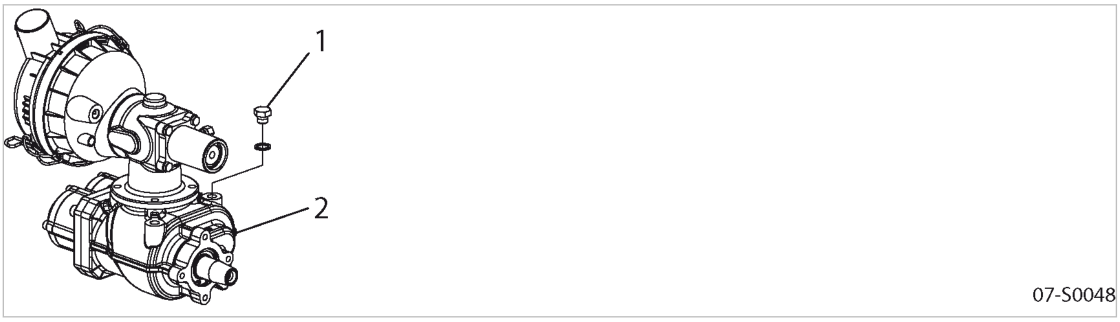


Fig. 17 Inlet valve filling port

1 Filler plug
2 Airend

1. Unscrew and remove the filler plug.
2. Pour in the cooling oil and retighten the filler plug.
3. Turn the airend manually by means of the belt pulley to distribute the cooling oil.

7.6 Option C1 Activating and deactivating the MODULATING control

Use a shut-off valve to activate and deactivate the MODULATING control. If the MODULATING control is deactivated, the machine always delivers the maximum possible compressed air quantity in LOAD mode.

MODULATING control	Shut-off valve
Switch on	open
Switch off	close

Tab. 38 MODULATING control: Setting the shut-off valve

Precondition The power supply disconnecting device is switched off,
the device is locked off,
the absence of any voltage has been verified.

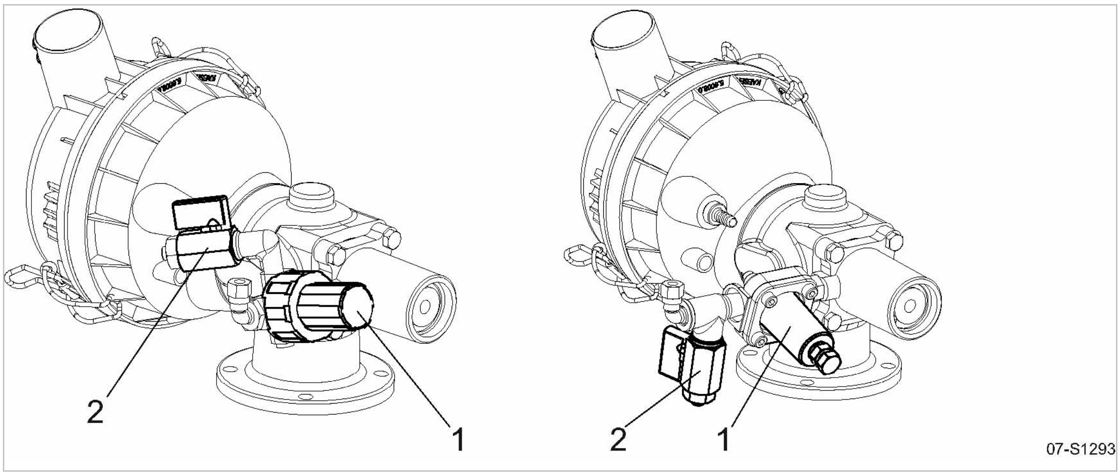


Fig. 18 MODULATING control: Setting the shut-off valve

1 Control valve (proportional control)
2 Shut-off valve

➤ Open or close the control valve, depending on the required control mode.

The regulating valve is factory set. The setting should not be changed without consultation with an authorized KAESER service representative.

7.7 Checking the direction of rotation

The machine is designed for a clockwise phase sequence.

Material Phase sequence meter

1. Check the direction of rotation using a phase sequence meter on the machine supply cables.
2. If the direction of rotation is incorrect (i.e. a counterclockwise phase sequence), exchange machine supply cabels L1 and L2.

If you do not have access to a phase sequence meter:

➤ Have the phase sequence to be checked by an authorized KAESER service representative.

7.8 Starting the machine for the first time

Machines of this type are generally only lightly utilized. Condensate can build up in lightly utilized machines with detrimental effects to the cooling oil and the machine itself. You can determine if your machine falls into this category by regular sampling of the cooling oil during the first week of operation.

Precondition No personnel are working on the machine.
All access doors are closed.
All removable panels in place and secured.

1. Open the user's shut-off valve to the air network.
2. Switch on the power supply disconnecting device.

   After the controller has carried out a self-test, the green *Control voltage* LED is lit continuously.
3. If required:
   Change the display language as described in chapter 7.11.
4. Press the «ON» key.

   The compressor motor runs up and after a short time the machine switches to LOAD and delivers compressed air.

➤ Watch for any faults occurring in the first hour of operation.

➤ After the first 50 operating hours, check all electrical connections and tighten where necessary.

Does the machine stop when the compressor motor rotates in the wrong direction?

➤ Switch off and lock out the power supply disconnecting device and verify the absence of voltage.

➤ Changeover phase lines L1 and L2.

➤ Acknowledge any existing alarm messages and switches the machine on again.

Recognizing damaging condensate

If the utilization conditions exceed these reference values, the probability of permanent condensate formation is low.

	SX 3	SX 4	SX 5	SX 7.5
Compressed air demand[1)] [cfm] (average value)	4.3	4.3	4.2	4.2
Compressed air demand[1)] in 7 hours [cfm]	1800	1800	1770	1770
Minimum running period[2)] [%]	36	27	20	15

	SX 3	SX 4	SX 5	SX 7.5

[1)] Relative to 115 psi mains set-point pressure
[2)] Time portions in operating mode LOAD, relative to a utilization time of 7 hours per day
. Prerequisite: >59 °F ambient temperature

Tab. 39 Preventing condensate

Make sure you are familiar with the procedure before draining off any oil.

1. A small quantity of cooling oil should be drained into a transparent, oil-proof vessel at least once a week during the first 4 weeks of operation.
2. Let the cooling oil sit for one hour approximately.

Result Any condensate will settle on the bottom of the vessel.

Condensate in the cooling oil?
➤ Contact an authorized KAESER service representative.

➤ Dispose of used oil in accordance with environment protection regulations.

Further information Information on draining off cooling oil is given in chapter 10.15.

7.9 Setting the set point pressure

The system pressure pA is factory set to the highest possible value.

Adjustment is necessary for individual operating conditions.

Do not set the set point pressure of the machine higher than the maximum working pressure of the compressed air system.

The machine may not toggle more than twice per minute between LOAD and IDLE.

To reduce the cycling (toggling) frequency:
➤ Increase the difference between cut-in and cut-out pressure.
➤ Add a larger air receiver downstream to increase buffer capacity.

➤ Set the set point pressure as described in the SIGMA CONTROL 2 operating manual.

7.10 Checking the Door Interlock Switch

The interlock switch stops the machine as soon as a door or access panel is opened.
Check the interlock switch function on commissioning.

The door interlock switch is an important safety device.
The machine may only be operated with a correctly functioning interlock switch.

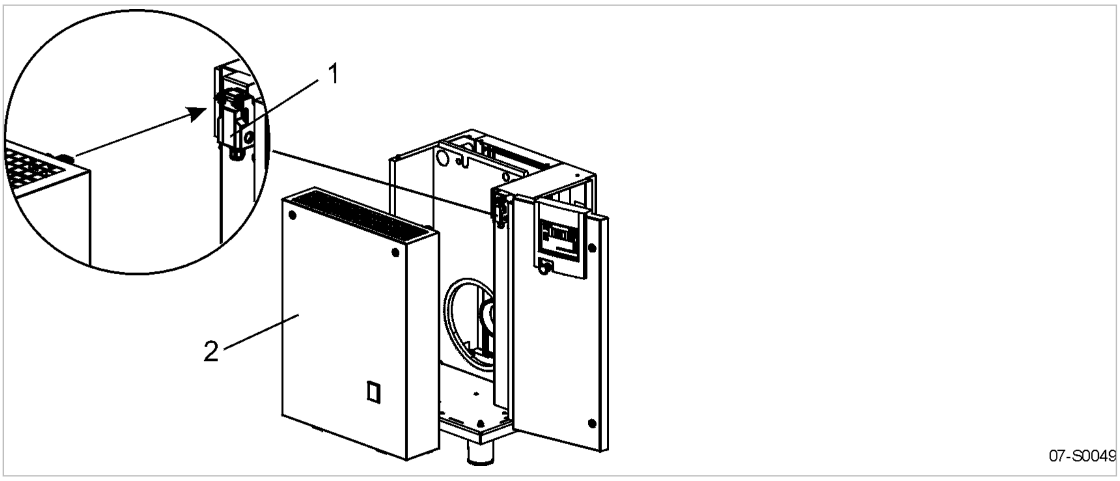


Fig. 19 Interlock switch position

1 Door interlock switch
2 Panel

1. Open the access panel 2 while the machine is running.
   The machine switches off automatically. The controller displays an alarm message.
2. Close the panel and acknowledge the alarm.

The machine does not switch off automatically?.

➤ Have the interlock switch checked by an authorized KAESER service representative agent.

7.11 Setting the display language

The controller can display text messages in several languages.

You can set the language for texts on the display. This setting will be retained even when the machine is switched off.

1. In operating mode, switch to the main menu with the «Return» key.
2. Press the «UP» or «DOWN» keys until the current language is shown as active line (inverse):

88 psig 176 °F	
——— de_DE Deutsch ———	Current language (active line)
▸1 xxxxxxxxxx	Submenu
▸2 xxxxxxxxxx	Submenu
▸3 xxxxxxxxxx	Submenu
▸4 xxxxxxxxxx	Submenu
▸5 xxxxxxxxxx	Submenu
▸6 xxxxxxxxxx	Submenu

3. Use the «Return» key to switch to setting mode.
   The language display flashes.
4. Move to the required language with «UP »or «DOWN».
5. Confirm the setting with the «Enter» key.

Result The display texts are now in the selected language.

Further information Detailed information can be found in the SIGMA CONTROL 2 operating manual.

8 Operation

8.1 Switching on and off

Always switch the machine on with the «ON» key and off with the «OFF» key.

A power supply disconnecting device should be installed by the user.

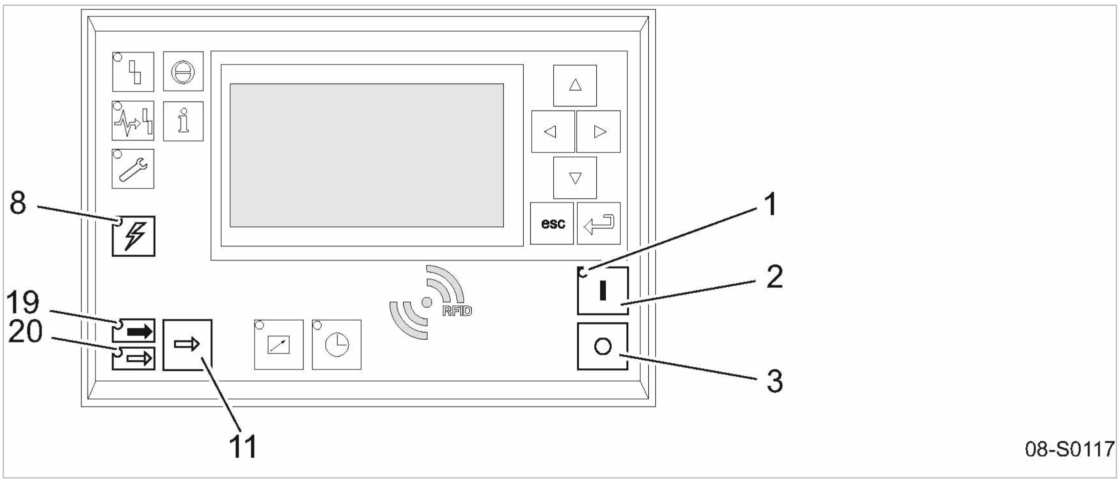


Fig. 20 Switching on and off

1	*Machine ON* LED	11	«LOAD/IDLE» toggle key
2	«ON» key	19	*LOAD* LED
3	«OFF» key	20	*IDLE* LED
8	*Controller voltage* LED		

8.1.1 Switching on

Precondition No personnel are working on the machine.

All access doors and panels are closed and secure.

1. Switch on the power supply disconnecting device.

   The *Controller voltage* LED lights green.
2. Press the «ON» key.

   The *ON* LED lights green.

If a power failure occurs, the machine is **not** prevented from restarting automatically when power is resumed.
It can restart automatically as soon as power is restored.

Result The compressor motor starts as soon as system pressure is lower than the setpoint pressure (cut-off pressure).

8.1.2 Switching off

1. Press the «OFF» key.

   The machine switches to IDLE and the *IDLE* LED flashes. The SIGMA CONTROL 2 displays *Stopping*. The *ON* LED extinguishes as soon as the automatic shut-off action is completed.
2. Switch off and lock out the power supply disconnecting device.

Result The *Controller voltage* LED extinguishes. The machine is switched off and disconnected from the power supply.

In rare cases, if you need to shut the machine down immediately and cannot wait until the automatic shut-down process is finished:

➤ Press «OFF» once again.

8.2 Switching off in an emergency and switching on again

The EMERGENCY STOP push button is located below the control panel.

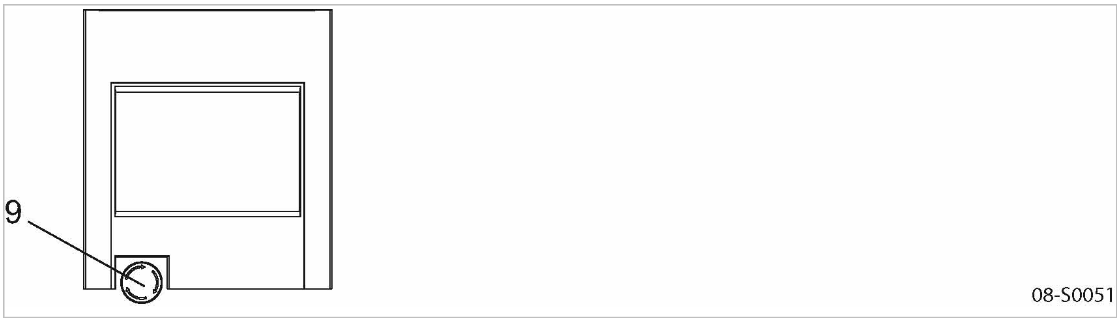


Fig. 21 Switching off in an emergency

9 EMERGENCY STOP push button

Switching off

➤ Press the EMERGENCY STOP push button.

Result The EMERGENCY STOP push button remains latched after actuation.

The compressor's pressure system is vented and the machine is prevented from automatically re-starting.

Switching on

Precondition The fault has been rectified

1. Turn the EMERGENCY STOP push button in the direction of the arrow to unlatch it.
2. Acknowledge any existing alarm messages.

Result The machine can now be started again.

8.3 Switching on and off from a remote control center

Precondition A link to the remote control center exists.

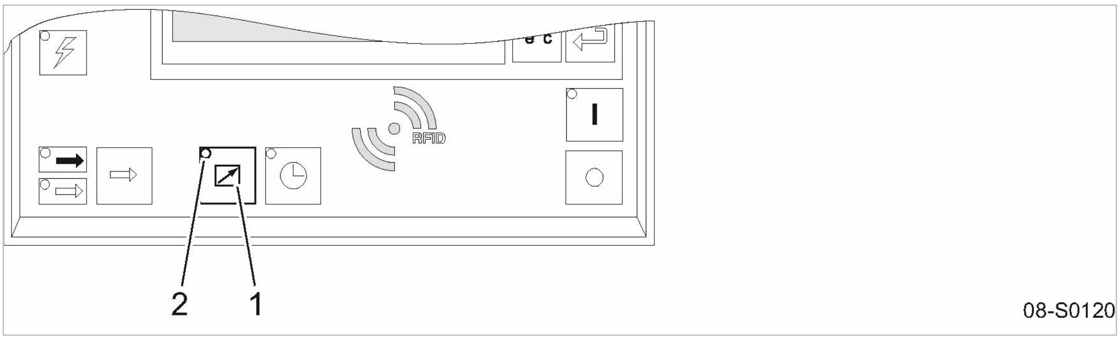


Fig. 22 Switching on and off from a remote control center

1 «Remote control» key
2 *Remote control* LED

1. Attach an easily seen notice to the machine that warns of remote operation.

⚠ WARNING

Remote control: Risk of injury caused by unexpected starting!

➤ Make sure that the power supply disconnecting device is switched off before commencing any work on the machine.

Tab. 40 Machine identification

2. Label the starting device in the remote control center as follows:

⚠ WARNING

Remote control: Risk of injury caused by unexpected starting!

➤ Before starting, make sure that no one is working on the machine and that it can be safely started.

Tab. 41 Remote control center identification

3. Press the «Remote control» key.
   The *Remote control* LED lights. The machine can be remotely controlled.

8.4 Using the timer for switching on and off

Precondition The clock is programmed.

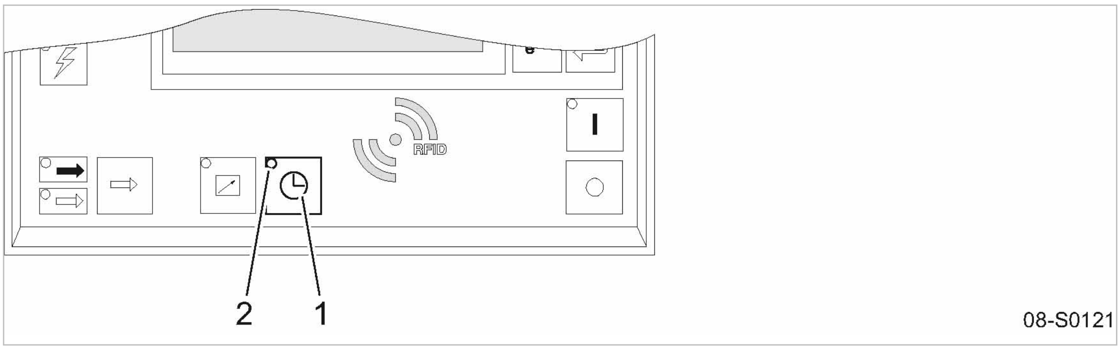


Fig. 23 Using the timer for switching on and off

1 «Timer» key
2 *Timer* LED

1. Attach an easily seen notice to the machine that warns of time-controlled operation:

⚠ WARNING
Timer control: Risk of injury caused by unexpected starting!
➢ Make sure that the power supply disconnecting device is switched off before commencing any work on the machine.

Tab. 42 Warning notice for timer control

2. Press the «Timer» key.
   The *Timer* LED lights. The timer switches the machine on and off.

8.5 Interpreting operation messages

The controller will automatically display operation messages informing you about the current operational state of the machine.

Operating messages are identified with the letter O.

Further information Detailed information can be found in the SIGMA CONTROL 2 operating manual.

8.6 Acknowledging alarm and warning messages

Messages are displayed on the "new value" principle:

- Message coming: LED flashes
- Message acknowledged: LED illuminates
- Message going: LED off

or

- Message coming: LED flashes
- Message going: LED flashes
- Message acknowledged: LED off

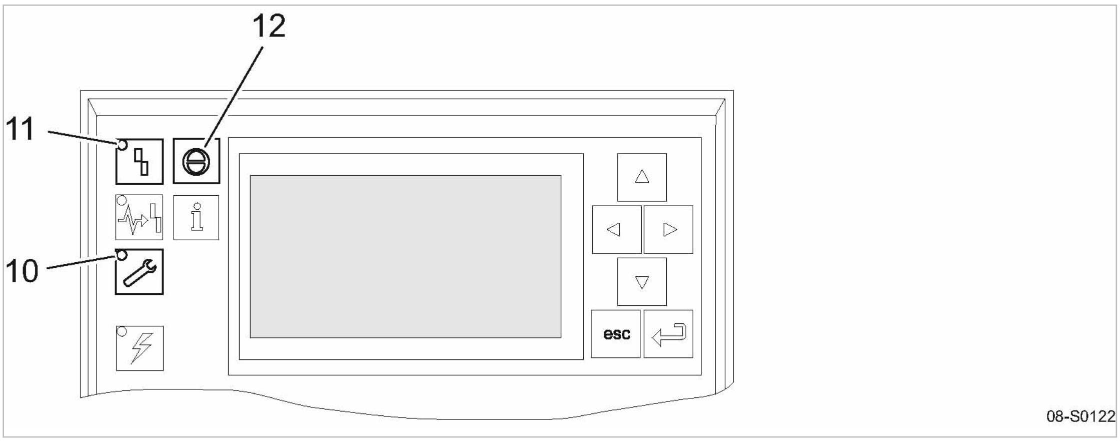


Fig. 24 Acknowledging messages

10 LED *Warning* (yellow)
11 LED *Alarm* (red)
12 Key «Acknowledge»

Alarm message

An alarm shuts the machine down automatically. The red *alarm* LED flashes.

The system displays the appropriate message.

Precondition The fault has been rectified.

➤ Acknowledge the message with the «acknowledge» key.
The *Fault* LED extinguishes.
The machine is again ready for operation.

If the machine was switched off with the EMERGENCY STOP button:

➤ Unlatch the EMERGENCY STOP button (turn in direction of the arrow) before acknowledging the alarm message.

Further information Please refer to the SIGMA CONTROL 2 operating manual for a list of possible fault messages during operation.

Warning message

If maintenance work is to be carried out or if a warning is displayed before an alarm, the yellow *warning* LED flashes.

The system displays the appropriate message.

Precondition The danger of an alarm is passed,
maintenance has been carried out.

➤ Acknowledge the message with the «Acknowledge» key.
The *warning* LED extinguishes.

Further information A list of possible alarm messages occurring during operation can be found in the service manual SIGMA CONTROL 2 .

9 Fault Recognition and Rectification

9.1 Basic instructions

There are 3 types of fault:

- Warning:
  - Warning messages *W*
- Fault (with indication):
  - Alarm messages *A*
  - System messages *Y*
  - Diagnostic messages *D*
- Other faults (without indication): See chapter 9.2

The messages valid for your machine are dependent on how the individual machine is equipped.

1. Do not attempt fault rectification measures other than those given in this manual!
2. In all other cases:
   Have the fault rectified by an authorized KAESER SERVICE representative.

Further information: Detailed information for the various messages can be found in the user manual SIGMA CONTROL 2.

9.2 Other Faults

Fault	Possible cause	Remedy
Machine does not run.	Control transformer not wired.	Connect control transformer in accordance with the electrical diagram. Machine with refrigeration dryer: Connect dryer transformer in accordance with the electrical diagram.
Machine runs but produces no compressed air.	Inlet valve not opening or only opening partially.	Call authorized KAESER Service representative.
	Venting valve not closing.	Call authorized KAESER Service representative.
	Leaks in the pressure system.	Check piping and connections for leaks and tighten any loose fittings.
	Air consumption is greater than the capacity of the compressor.	Check the air system for leaks. Shut down the consumer(s).
	Hose coupling or maintenance hose still plugged into the quick-release coupling on the oil separator tank.	Remove coupling or maintenance hose.

Fault	Possible cause	Remedy
Cooling oil runs out of the air filter.	Oil level in the oil separator tank too high.	Drain off oil until the correct level is reached.
	Inlet valve defective.	Call authorized KAESER Service representative.
Compressor switches between LOAD and IDLE more than twice per minute.	Air receiver too small.	Increase size of air receiver.
	Airflow into the compressed air network restricted.	Increase air pipe diameters. Check filter elements.
	The differential between cut-in and cut-out pressure too is small.	Check switching differential.
Cooling oil leaking into the floor pan.	Hose coupling or maintenance hose still plugged into the quick-release coupling on the oil separator tank.	Remove coupling or maintenance hose.
	Oil cooler leaking.	Call authorized KAESER Service representative.
	Leaking joints.	Tighten joints. Replace seals.
Cooling oil consumption too high.	Unsuitable oil is being used.	Use SIGMA FLUID cooling oil.
	Oil separator cartridge split.	Change the oil separator cartridge.
	Oil level in the oil separator tank too high.	Drain off oil until the correct level is reached.
	Oil return line clogged.	Check dirt trap in the return line.

Tab. 43 Other faults and actions

10 Maintenance

10.1 Ensuring safety

Follow the instructions below to ensure safe machine maintenance.

Warning instructions are located before a potentially dangerous task.

Disregard of warning instructions can cause serious injuries!

Complying with safety notes

Disregard of safety notes can cause unforeseeable dangers!

➤ Follow the instructions in chapter 3 “Safety and Responsibility”.

➤ Maintenance work may only be carried out by authorized personnel.

➤ Use the safety sign below to advise others that the machine is currently being serviced:

Sign	Meaning
	**WARNING** Serious injury or death can result from activating the machine during service! ➤ Do **not** activate the machine.

Tab. 44 Warn others that the machine is being serviced.

➤ Before switching on, make sure that nobody is working on the machine and all access doors and panels are closed and locked.

When working on live components

Touching voltage-carrying components can result in electric shocks, burns, or death.

➤ Work on electrical equipment may only be carried out by authorized electricians.

➤ Switch off and lock out the power supply disconnecting device and verify the absence of voltage.

➤ Check that there is no voltage on floating relay contacts.

When working on the compressed air system

Compressed air is contained energy. Uncontrolled release of this energy can cause serious injury or death. The following safety concerns relate to any work on components that could be under pressure.

➤ Close shut-off valves or otherwise isolate the machine from the compressed air network to ensure that no compressed air can flow back into the machine.

➤ Depressurize all pressurized components and enclosures.

➤ Check all hose couplings in the compressed air system with a hand-held pressure gauge to ensure that they all read 0 psig.

➤ Do not open or dismantle any valves.

When working on the drive system

Touching voltage-carrying components can result in electric shocks, burns, or death.

Touching the fan wheel, the coupling, or the drive while the machine is switched on can result in serious injury.

➤ Switch off and lock out the power supply disconnecting device and verify the absence of voltage.
➤ Do not open the cabinet while the machine is switched on.

Further information

Details of authorized personnel are found in chapter 3.4.2.

Details of dangers and their avoidance are found in chapter 3.5.

10.2 Following the maintenance plan

10.2.1 Logging maintenance work

The maintenance intervals given are those recommended for KAESER original components with average operating conditions.

➤ In adverse conditions, perform maintenance work at shorter intervals.

Adverse conditions are, e.g.:

- high temperatures
- much dust
- high number of load changes
- low load

➤ Adjust the maintenance intervals with regard to local installation and operating conditions.

➤ Document all maintenance and service work.

This enables the frequency of individual maintenance tasks and deviations from our recommendations to be determined.

Further information

A prepared list is provided in chapter 10.20.

10.2.2 Resetting maintenance interval counters

According to the way a machine is equipped, sensors and/or maintenance interval counters monitor the operational state of important functional devices. Required maintenance work is shown on SIGMA CONTROL 2.

Precondition

Maintenance performed and
maintenance message acknowledged.

➤ Reset the maintenance interval counter as described in the SIGMA CONTROL 2 operating manual.

10.2.3 Regular maintenance tasks

The table below lists the required maintenance tasks.

The refrigeration circuit is a closed system. Repairs may only be carried out by certified personnel.

➤ Take note of the controller's service messages and carry out tasks in a timely manner, taking ambient and operating conditions into account:

Interval	Maintenance task	See chapter
Weekly	Check the cooling oil level.	10.12
	Cooler: Check the filter mat.	10.3
	Control cabinet: Check the filter mat.	10.4
	Check the condensate drain.	10.18.2
	Air receiver: Drain condensate manually.	10.19
Up to 1000 h	Clean the cooler.	10.5
	Check the air filter.	10.6
	Cooler: Clean the filter mat.	10.3
	Control cabinet: Clean the filter mat.	10.4
	Clean the refrigerant condenser.	10.18.1
Up to 3000 h	Cooler: Change the filter mat.	10.3
	Control cabinet: Change the filter mat.	10.4
Up to 6000 h At least every 2 years	Condensate drain: Change the service unit.	10.18.2.2
Display: SIGMA CONTROL 2	Maintain the drive belt.	10.8
	Change the air filter.	10.6
Display: SIGMA CONTROL 2 At least every 2 years	Change the oil filter.	10.16
Display: SIGMA CONTROL 2 At least every 2 years	Change the oil separator cartridge.	10.17
Variable, see table 46	Change the cooling oil.	10.15
Variable	Service compressed air filter. (Option: F1)	13.5
Up to 6000 h	Replace the drive belt.	10.8
h = operating hours		

Interval	Maintenance task	See chapter
At least every 1 years	Machine: Check the safety relief valve.	10.9
	Air receiver: Check the safety relief valve.	10.19.2
	Check the function: Safety shut-down due to excessive airend discharge temperature.	10.10
	Check the EMERGENCY STOP push button.	10.11
	Check the function: Safety shut-down when the machine is opened.	7.10
	Have the pressure monitor checked by an authorized KAESER service representative.	—
	Check the cooler for leaks.	10.5
	Check that all electrical connections are tight.	—
h = operating hours		

Tab. 45 Regular maintenance tasks

10.2.4 Cooling oil changing interval

Machine utilization and ambient conditions are important criteria for the number and length of the change intervals.

An authorized KAESER SERVICE representative will support you in determining appropriate intervals and provide information on the possibilities of oil analysis.

➤ Please observe national regulations regarding the use of cooling oil in oil-injected screw compressors.

➤ Check operating conditions and adjust intervals as necessary.

KAESER LUBRICANTS			
SIGMA Lubricant	Description	Maximum Recommended Change Interval	
		First oil change	Subsequent oil change
M-460	ISO 46 Semi-Synthetic Lubricant	2000 Hours	3000 Hours
S-460	ISO 46 Synthetic Lubricant	6000 Hours	8000 Hours
S-680	ISO 68 Synthetic Lubricant	6000 Hours	8000 Hours
FG-460	ISO 46 Food Grade Synthetic Fluid	2000 Hours	3000 Hours

Tab. 46 Oil change intervals lubricants

10.2.5 Regular service tasks

The table below lists service tasks required.

➤ Only an authorized KAESER service representative should carry out service work.

➤ Have service tasks carried out punctually, taking ambient and operating conditions into account.

Interval	Service task
Display: SIGMA CONTROL 2	Service valves.
	Replace the compressor drive motor bearings.
Up to 36000 h Every 6 years at the latest.	Have plastic pipes and hose lines replaced.
Up to 36000 h	Replace the control cabinet fan.
After 20 years at the latest	Replace safety-relevant components of the safety devices.
h = operating hours	

Tab. 47 Regular service tasks

10.3 Cooler: Cleaning or Renewing the Filter Mats

The filter mats help to keep the cooler clean. If the filter mats are clogged, adequate cooling of the components is no longer ensured.

Material: Filter mats:
Warm water and household detergent
Spare parts (as required)

Precondition: The machine is switched off.

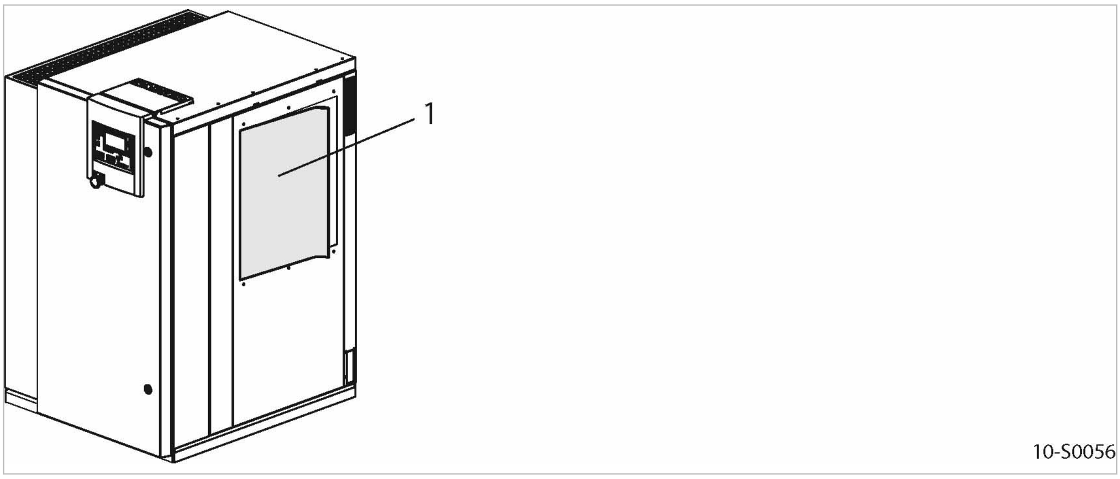


Fig. 25 Filter mat for the air and oil cooler

1 Filter mat

No tools are needed to remove the filter mat.

1. Carefully remove the filter mat from the retaining frame.
2. Beat the mat or use a vacuum cleaner to remove loose dirt. If necessary, wash with lukewarm water and household detergent.
3. Change the filter mat if cleaning is not possible or if the change interval has expired.
4. Carefully insert the filter mat in the retaining frame.

10.4 Control cabinet: Clean or renew the filter mat

A filter mat is placed behind every ventilation grill. Filter mats protect the control cabinet from ingress of dirt. If the filter mats are clogged, adequate cooling of the components is no longer ensured. In such a case, clean or replace the filter mats.

Material: Warm water and household detergent
Spare parts (as required)

Precondition: The power supply isolating device is switched off,
the device is locked off,
the absence of any voltage has been verified.
The machine has cooled down.

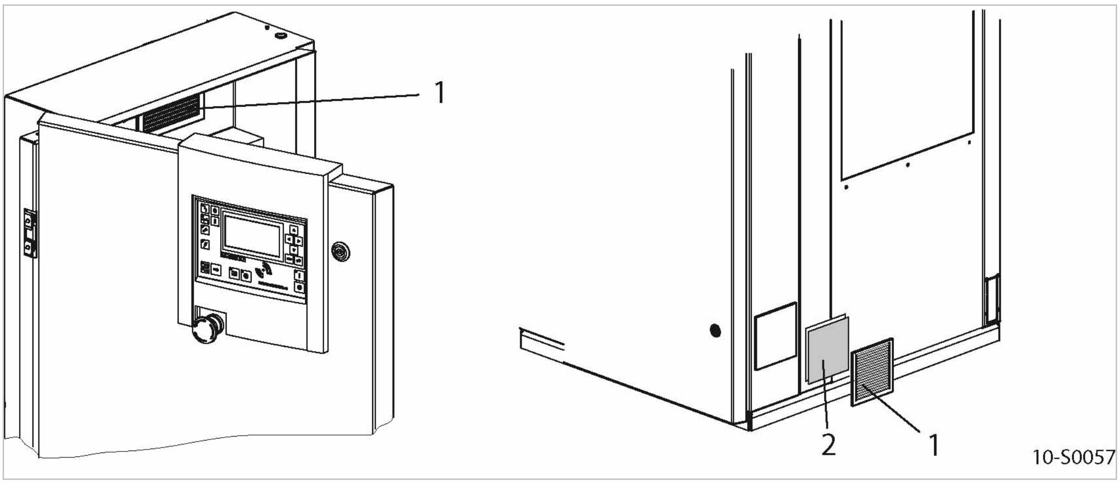


Fig. 26 Switching cabinet ventilation

1 Ventilation grill
2 Filter mat

1. Carefully remove the ventilation grill and take out the filter mat.
2. Beat the mat or use a vacuum cleaner to remove loose dirt. If necessary, wash with lukewarm water and household detergent.
3. Change the filter mat if cleaning is not possible or if the change interval has expired.
4. Insert the filter mat in the frame and latch in the ventilation grill.

10.5 Cooler maintenance

Regular cleaning of the cooler ensures reliable cooling of the machine and the compressed air. The frequency is mainly dependent on local operating conditions.

A leaking cooler results in loss of cooling oil and compressed air.

Clogged coolers are indicative of unfavorable ambient conditions. Such ambient conditions clog the cooling air ducts in the machine's interior and the motors resulting in increased wear and tear.

➤ Have the authorized KAESER service representative clean the cooling air ducts.

Material: Brush and vacuum cleaner
Face mask (as required)

Precondition: The power supply isolating device is switched off,
the device is locked off,
the absence of any voltage has been verified.
The machine has cooled down.

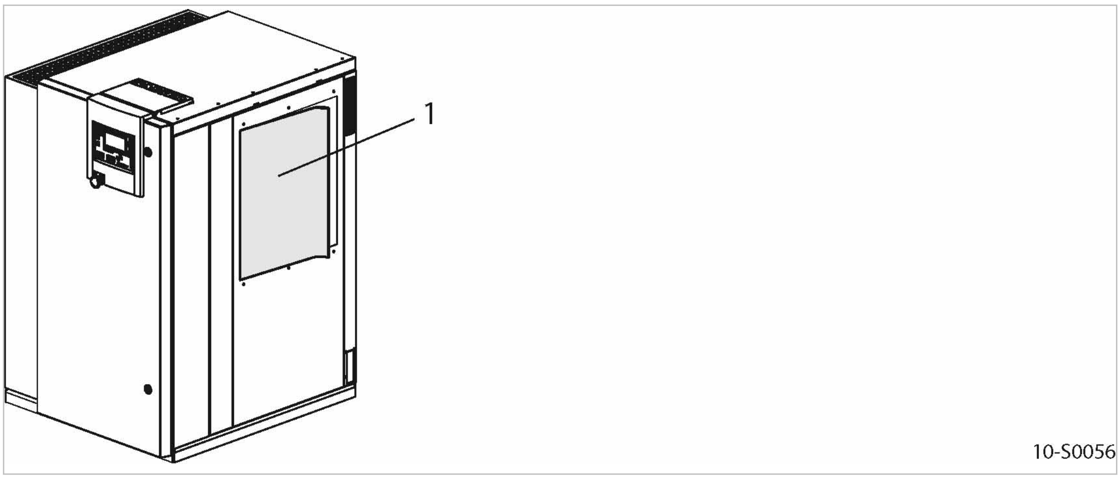


Fig. 27 Filter mat for the air and oil cooler

1 Filter mat

Cleaning the cooler

A filter mat helps to keep the cooler clean. Despite this fact, the cooler will clog over a period of time.

Do not use sharp objects to clean the cooler. It could be damaged.

Avoid creating clouds of dust.

1. Carefully remove the filter mat from the retaining frame.
2. Dry brush the oil and air coolers and use a vacuum cleaner to suck up the dirt.
3. Carefully insert the filter mat in the retaining frame.

The air and oil coolers can no longer be properly cleaned?

➤ Have severe clogging removed by an authorized KAESER service representative.

Checking the cooler for leaks

➤ Visual inspection: Did cooling oil escape?

Is a cooler leaking?

➤ Have the defective cooler repaired immediately by an authorized KAESER service representative.

10.6 Change the air filter

All sealing surfaces have been designed to match each other. The use of an unsuitable air filter can permit dirt to enter the pressure system, thus causing damage to the machine.

The air filter cannot be cleaned.

Material: Spare part

Precondition: The power supply disconnecting device is switched off,
lock out and tag out the device,
the absence of any voltage has been verified.
The machine has cooled down.

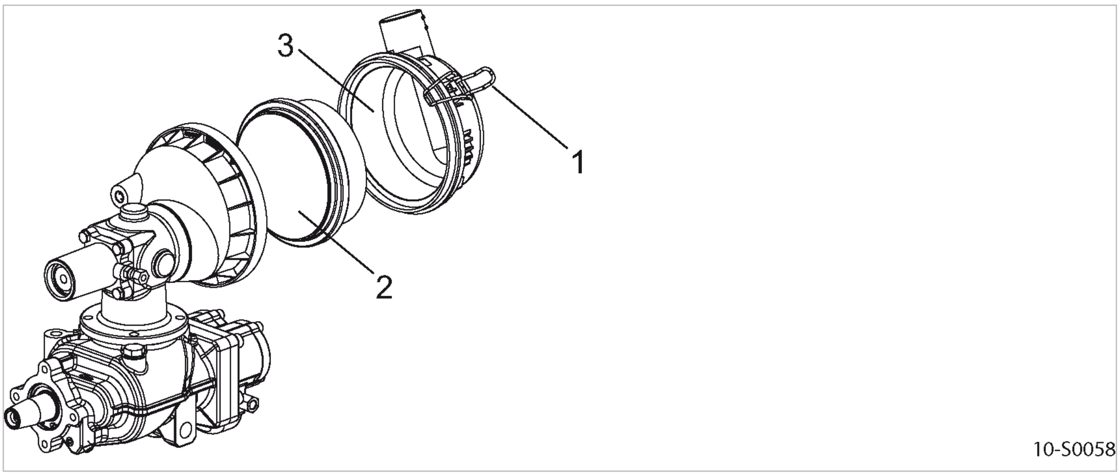


Fig. 28 Change the air filter

1 Fastener
2 Air filter
3 Air filter housing

1. Release the fasteners at the air filter housing and remove the air filter.
2. Clean all parts and sealing surfaces.
3. Insert the new air filter into the housing.
4. Close the air filter housing with the fasteners.

10.7 Compressor motor maintenance

The motor bearings of the compressor motor are permanently greased. Re-greasing is not necessary.

➤ Have the motor bearings checked by an authorized KAESER service representative during the course of a visit.

10.8 Maintaining the drive belt

Material: Spare parts (if required)

Precondition: The power supply disconnecting device is switched off,
lockout and tagout the device,
verify the absence of any voltage.
The machine has cooled down.

⚠ WARNING

Touching moving drive belt may result in severe bruising or even loss of limb or extremities!

➤ *Switch off and lockout / tagout the power supply disconnecting device and check that no voltage is present.*

Visually checking for damage.

1. Turn the pulley by hand so that all of the belt can be inspected for damage.
2. Change the belts immediately if any damage is found.

Replacing the drive belt

The drive motor must be moved in its bracket to change the belt. Use the appropriate tool and support the motor during belt changing.

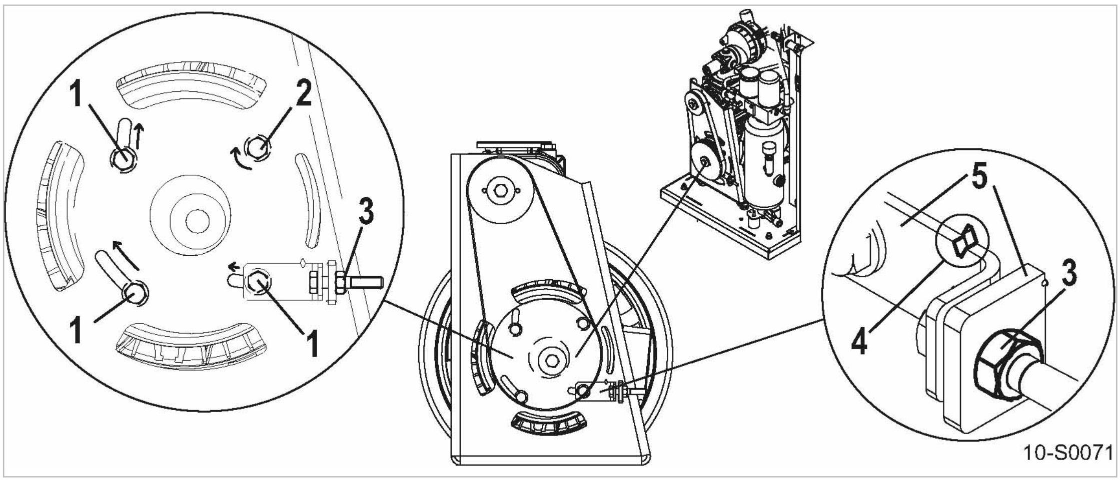


Fig. 29 Changing the belt

1	Fixing screw	4	Markings
2	Pivot point fixing screw	5	Tensioning device
3	Tensioning nut		

1. Loosen the tensioning nut 3 by approximately 0.4 in.
2. Remove the fixing screws 1.
3. Loosen the pivot point fixing screw 2 only sufficiently to allow the motor to shift to the side.
4. Move the motor to the side and fix it in position with a screw 1.
5. Place the new belt over the pulleys.
6. Ensure that all fixing screws 1 are loosened.
7. Tighten the belts by means of the tensioning nut 3 until the markings 4 coincide.
8. Tighten one fixing screw 1 to hold the motor in place then tighten the rest 1 and 2.

Result The drive belt is sufficiently tensioned.

It is not necessary to re-tension the belt.

10.9 Testing the safety relief valve

In order to check the safety relief valve, the machine's working pressure is raised above the blowoff pressure of the valve.

Blow off protection and air system pressure monitoring are switched off during the test. In normal operation, the blow-off protection will switch off the machine before the safety relief valve responds. During the inspection, the blow-off protection will switch off the machine only when the activating pressure of the safety relief valve has been exceeded by 14.5 psig

- ➤ Follow the detailed description of this procedure in theSIGMA CONTROL 2 operating manual
- ➤ Never operate the machine without a correctly functioning safety relief valve.
- ➤ Have a defective safety relief valve replaced immediately.

⚠ WARNING

Excessive noise is caused when the safety relief valve blows off!

- ➤ *Close all access doors, replace and secure all removable panels.*
- ➤ *Wear hearing protection.*

Precondition: The machine is switched off.

1. Close the user's shut-off valve between the machine and the air receiver.
2. Read off the activating pressure on the valve.
(the activating pressure is usually to be found at the end of the part identification)
3. Log on to SIGMA CONTROL 2 with access level 2.
4. Observe the display of pressure on SIGMA CONTROL 2 and call up the test function.
5. **⚠ WARNING** *Risk of burns due to released cooling oil and compressed air when blowing off the safety relief valve!*
   - ➤ *Close all access doors, replace and secure all removable panels.*
   - ➤ *Wear eye protection.*
6. End the test as soon as the safety relief valve blows off or working pressure exceeds the activating pressure of the pressure relief valve by nearly 14.5 psig.
7. If necessary, vent the machine and replace the defective safety relief valve.
8. Deactivate the test function
9. Open the user's shut-off valve between the machine and the air receiver.

10.10 Checking the overheating safety shutdown function

The machine should shut down if the airend discharge temperature reaches a maximum of 230 °F.

- ➤ Check the safety shutdown function as described in the SIGMA CONTROL 2 operating manual.

The machine does not shut down?

- ➤ Have the safety shutdown function checked by an authorized KAESER SERVICE representative.

10.11 Testing the EMERGENCY STOP push button

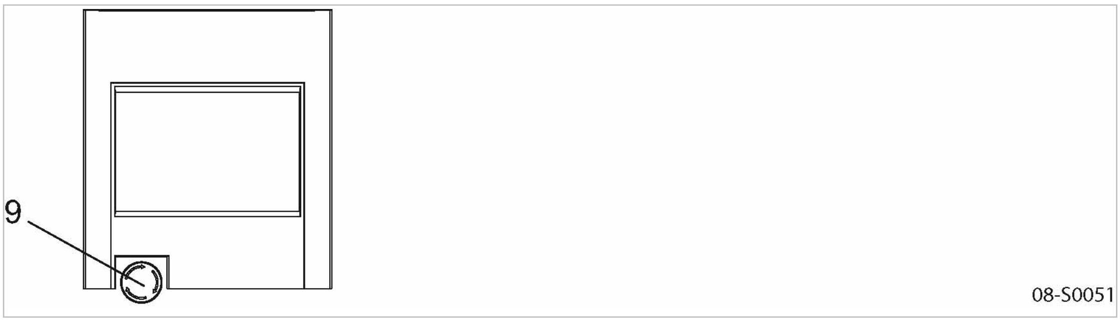


Fig. 30 Testing the EMERGENCY STOP push button

9 EMERGENCY STOP push button

Precondition The compressor motor is running.

1. Press the EMERGENCY STOP push button.

   The compressor motor stops, the pressure system is vented, and the machine is prevented from automatically restarting.

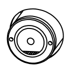
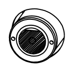
> The compressor motor does not stop?
>
> The safety function of the EMERGENCY STOP push button is no longer ensured.
>
> ➤ Shut down the machine immediately and call an authorized KAESER SERVICE representative.

2. Turn the EMERGENCY STOP push button in the direction of the arrow to unlatch it.
3. Acknowledge the alarm message.

10.12 Checking the cooling oil level

The sight glass allows a risk-free reading of the cooling oil level. The oil indicator should be fully filled with oil when the machine is at standstill. The correct oil level cannot be seen.

The ideal situation is with the oil level around the optimum mark when the machine is running.

Operating state	Minimum oil level	Maximum oil level
LOAD		

Tab. 48 Permissible cooling oil level under LOAD

Precondition The machine has been running at least 5 minutes under LOAD.

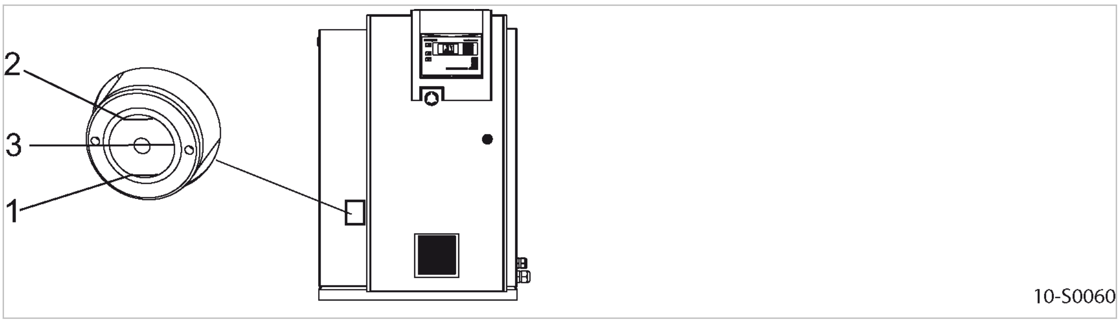


Fig. 31 Checking the cooling oil level

1 Minimum oil level
2 Maximum oil level
3 Optimum oil level

➤ Check the cooling oil level with machine running under LOAD.

Result As soon as the minimum level is reached: Replenish the cooling oil.

10.13 Venting the machine (de-pressurizing)

Venting takes place in three stages:

- Isolate the compressor from the air system.
- Vent the oil separator tank.
- Manually vent the air cooler.

⚠ The machine must be isolated from the compressed air network and completely vented before undertaking any work on the pressure system.

Material The maintenance hose with hose coupling and shut-off valve needed for venting is stowed beneath the oil separator tank.

Precondition The power supply disconnecting device is switched off,
The device is locked off,
A check has been made that no voltage is present.

⚠ CAUTION

Escaping oil mist is damaging to health.

➤ *Do not direct the maintenance hose at persons while venting.*

➤ *Do not inhale the oil mist.*

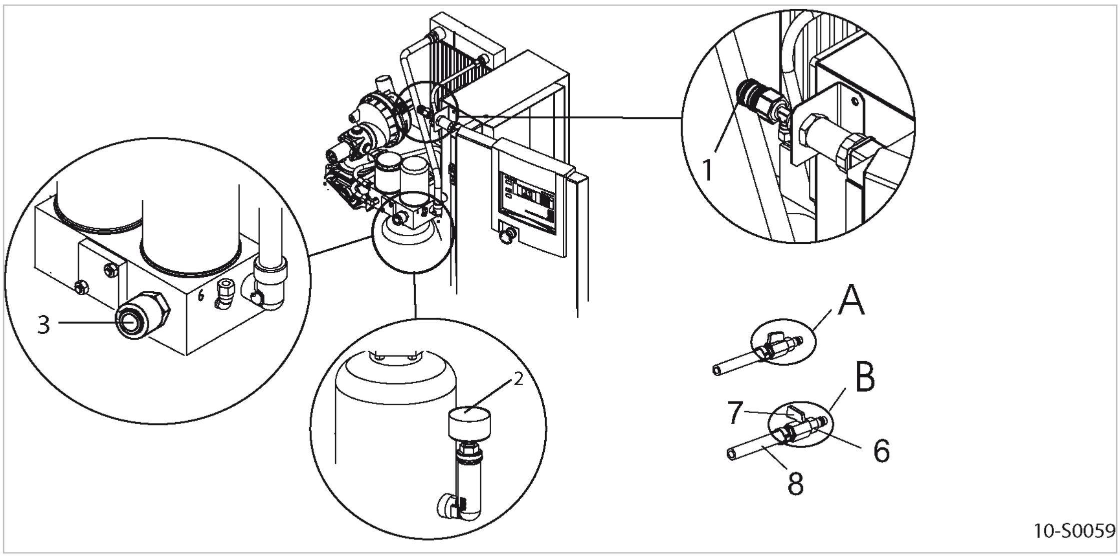


Fig. 32 Venting the machine

1	Hose coupling (air cooler venting)	7	Shut-off valve
2	Pressure gauge	A	Shut-off valve open
3	Hose coupling (oil separator tank venting)	B	Shut-off valve closed
6	Male hose fitting	8	Maintenance hose

Isolating the machine from the air system

➤ Close the user's shut-off valve between the machine and the air distribution network.

If no shut-off valve is provided by the user, the complete air distribution network must be vented.

Venting the oil separator tank

The oil circulation vents automatically as soon as the machine is stopped.

➤ Check that the oil separator tank pressure gauge reads 0 psig.

The pressure gauge does not read 0 psig after automatic venting?

➤ Make sure that the user's shut-off valve is closed or that the complete air system is vented.

➤ With the shut-off valve closed, insert the male hose fitting 6 into the hose coupling 3.

➤ Slowly open the shut-off valve 7 to release pressure.

➤ Disconnect the male hose fitting 6 and close the shut-off valve 7.

➤ If manual venting does **not** bring the oil separator tank pressure gauge to zero: Contact an authorized KAESER service representative.

Manually venting the air cooler

After shutting down the compressor and venting the oil separator tank, the machine is still under pressure from the air system or the section from the shut-off valve to the minimum pressure/check valve.

1. With the shut-off valve closed, insert the male hose fitting 6 into the hose coupling 1.
2. Slowly open the shut-off valve 7 to release pressure.
3. Disconnect the male hose fitting 6 and close the shut-off valve 7.

10.14 Replenishing the cooling oil

> The machine must be isolated from the compressed air network and completely vented before undertaking any work on the pressure system.

Material — The maintenance hose with hose coupling and shut-off valve needed for venting is stowed beneath the oil separator tank.

Precondition — The power supply disconnecting device is switched off,
The device is locked off,
A check has been made that no voltage is present.

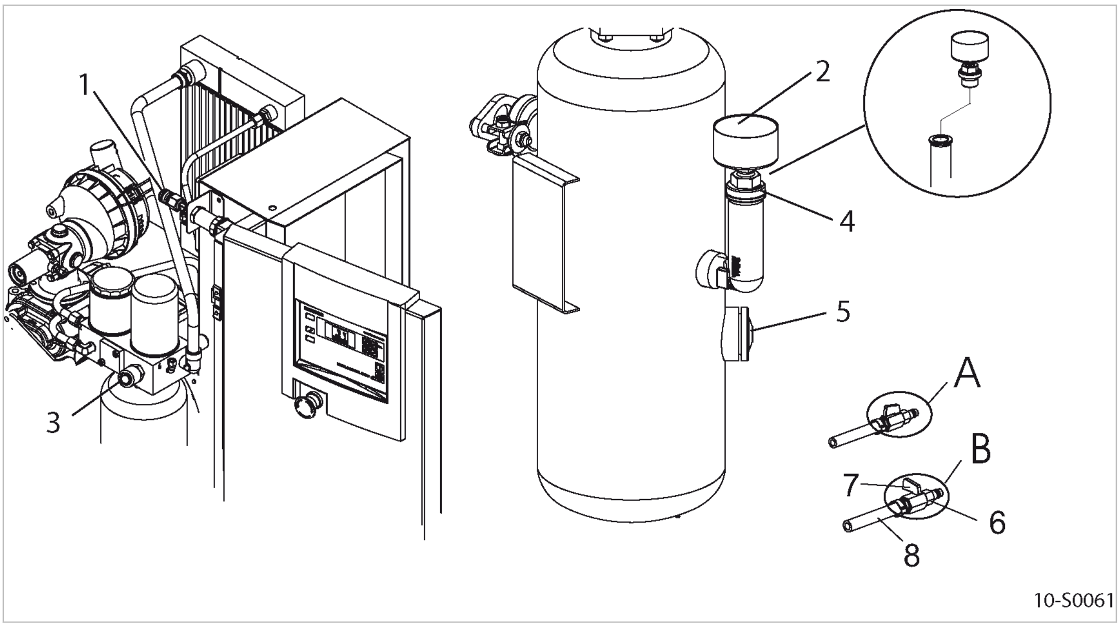


Fig. 33 Replenishing the cooling oil

1	Hose coupling (compressed air cooler venting)	6	Male hose fitting
2	Pressure gauge	7	Shut-off valve
3	Hose coupling (oil separator tank venting)	A	Shut-off valve open
4	Oil filler port with plug	B	Shut-off valve closed
5	Cooling oil level indicator	8	Maintenance hose

1. Vent the machine as described in section 10.14.1.
2. Fill with cooling oil and test run as described in section 10.14.2.

10.14.1 Venting the machine (de-pressurizing)

Venting takes place in three stages:

- Isolate the compressor from the air system.
- Vent the oil separator tank.
- Manually vent the compressed air cooler.

⚠ CAUTION

Escaping oil mist is damaging to health.

- *Do not direct the maintenance hose at persons while venting.*
- *Do not inhale the oil mist.*

Isolating the machine from the air system

- Close the user's shut-off valve between the machine and the air distribution network.

If no shut-off valve is provided by the user, the complete air distribution network must be vented.

Venting the oil separator tank

The oil circulation vents automatically as soon as the machine is stopped.

- Check that the oil separator tank pressure gauge reads 0 psig.

The pressure gauge does not read 0 psig after automatic venting?

- Make sure that the user's shut-off valve is closed or that the complete air system is vented.
- With the shut-off valve closed, insert the male hose fitting 6 into the hose coupling 3.
- Slowly open the shut-off valve 7 to release pressure.
- Disconnect the male hose fitting 6 and close the shut-off valve 7.
- If manual venting does **not** bring the oil separator tank pressure gauge to zero: Contact an authorized KAESER service representative.

Manually venting the compressed air cooler

After shutting down the compressor and venting the oil separator tank, the machine is still under pressure from the air system or the section from the shut-off valve to the minimum pressure/check valve.

1. With the shut-off valve closed, insert the male hose fitting 6 into the hose coupling 1.
2. Slowly open the shut-off valve 7 to release pressure.
3. Disconnect the male hose fitting 6 and close the shut-off valve 7.

10.14.2 Topping off with cooling oil and trial run

Replenishing the cooling oil

A sticker on the oil separator tank specifies the type of oil used.

1. **⚠ WARNING** *Compressed air!*
   Compressed air and devices under pressure can injure or cause death if the contained energy is released suddenly.
   - *De-pressurize all pressurized components and enclosures.*
2. **NOTICE** *The machine could be damaged by unsuitable oil!*
   - *Never mix different types of oil.*
   - *Never top off with a different type of oil than has already been used in the machine.*
3. Slowly unscrew the filler plug 4.

4. Top off to bring the oil to the correct level.
5. Replace the filler plug's sealing ring if necessary and screw the plug into the filler neck.

Starting the machine and carrying out a trial run

1. Close all access doors, replace and secure all removable panels.
2. Open the user's shut-off valve between the machine and the air distribution network.
3. After approx. 10 minutes of operation: Check the cooling oil level and top off if necessary.
4. Switch off the machine and check for leaks.

10.15 Changing the cooling oil

Always drain the oil completely from the following components:
- Airend
- Oil separator tank
- Oil cooler

Compressed air helps to expel the oil. This compressed air can be taken either from the compressor itself or from an external source.

An external source of compressed air is necessary in the following cases (examples):
- The machine is not operational.
- The machine is to be restarted after a long period of standstill.

The machine must be isolated from the compressed air network and completely vented before undertaking any work on the pressure system.

Material Cooling oil

Cooling oil receptacle

The maintenance hose with hose coupling and shut-off valve is stowed beneath the oil separator tank.

⚠ CAUTION

There is risk of burns from hot components and oil!

➤ *Wear long-sleeved garments and protective gloves.*

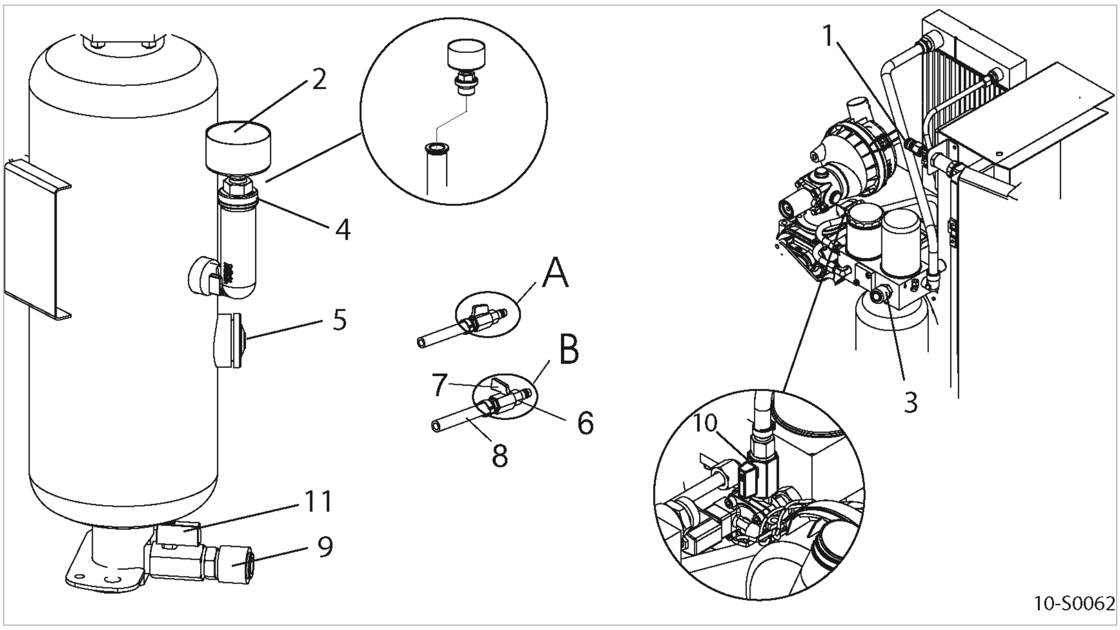


Fig. 34 Changing the cooling oil, oil separator tank

1	Hose coupling (air cooler venting)	A	Shut-off valve open
2	Pressure gauge	B	Shut-off valve closed
3	Hose coupling (oil separator tank venting)	8	Maintenance hose
4	Oil filler port with plug	9	Hose coupling (oil drain)
5	Cooling oil level indicator	10	Shut-off valve (venting line)
6	Male plug-in hose fitting	11	Shut-off valve (oil drain)
7	Shut-off valve		

Changing the cooling oil with internal pressure	Oil change with an external compressed air source
The machine has been running for, at least, 5 minutes in LOAD mode. The machine is fully vented, the pressure gauge on the oil separator tank reads 0 psig. 1. Close the shut-off valve [10] in the venting line. 2. Start the machine and watch the oil separator tank pressure gauge [2] until it reads 43–73 psi. 3. Switch off and lock out/tag out the power supply disconnecting device and verify the absence of any voltage. 4. Wait at least 2 minutes to allow the oil to flow back to the separator tank.	The power supply disconnecting device is switched off, lock out and tag out the device, the absence of any voltage has been verified. The machine is fully vented, the pressure gauge on the oil separator tank reads 0 psig. An external source of compressed air is available. 1. Close the shut-off valve [10] in the venting line. 2. With the shut-off valve closed, insert the male plug-in hose fitting[6] into the hose coupling [3]. 3. Connect the maintenance hose to the external air supply. 4. Open the shut-off valve [7] until the pressure gauge on the oil separator tank reads 43–73 psi. 5. Close the shut-off valve [7] and remove the male plug-in hose fitting from the hose coupling.

Draining the cooling oil from the airend

➤ Turn the upper pulley only in the direction of the arrow indicating the direction of rotation on the airend.

➤ Manually turn the upper pulley at least 5 times completely to extract remaining cooling oil from the airend into the oil separator tank.

Draining the cooling oil from the separator tank

Contact an authorized KAESER service representative if condensate is detected in the oil.
It is necessary to adapt the airend discharge temperature to individual ambient conditions.

Precondition: Switch off the power supply disconnecting device,
lock out and tag out the device,
verify the absence of any voltage.

1. Prepare a cooling oil receptacle.
2. With the shut-off valve closed, insert the male plug-in hose fitting[6] into the hose coupling [9].
3. Place the other end of the maintenance hose in the oil receptacle and secure it in place.
4. Open the shut-off valve [11].
5. Slowly open the shut-off valve [7] at the maintenance hose and allow oil and compressed air to drain completely.

   Pressure gauge on the oil separator tank indicates 0 psig.
6. Close the shut-off valve [11] and unplug the male plug-in hose fitting.

➤ Dispose of used oil in accordance with environment protection regulations.

Draining the oil from the cooler

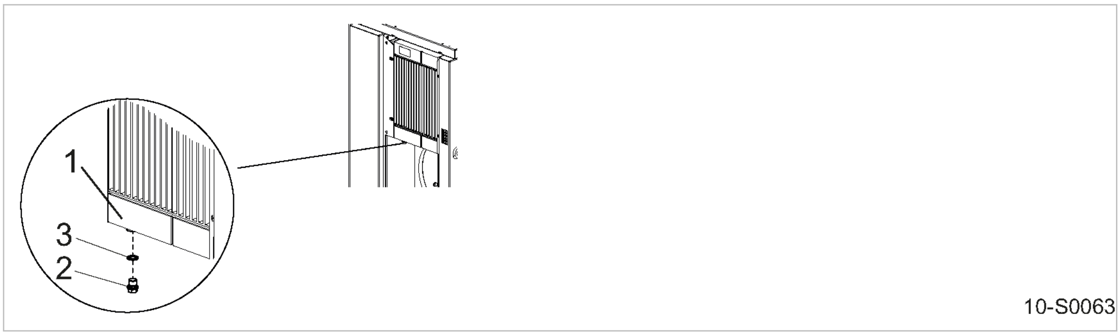


Fig. 35 Changing the cooling oil Oil cooler

1 Oil cooler
2 Screw plug (draining cooling oil)
3 Sealing ring

1. Prepare a cooling oil receptacle.
2. Slowly undo the screw plug 2 and completely drain the cooling oil.
3. Use new sealing ring 3 and tighten the screw plug 2.

➤ Dispose of used oil in accordance with applicable environmental provisions.

Filling with cooling oil

1. Slowly unscrew the filler plug 4 (see Fig. 34).
2. Fill with cooling oil.
3. Check the screw plug seal for damage and screw the filler neck with the screw plug back in.

Starting the machine and performing a test run

1. Close all access doors, replace and secure all removable panels.
2. Open the user's shut-off valve between the machine and the air distribution network.
3. Switch on the power supply disconnecting device and reset the maintenance interval counter.
4. Start the machine and check the cooling oil level again after about 10 minutes, topping off if necessary.
5. Switch off the machine and visually check for leaks.

10.16 Change the oil filter

The machine must be fully isolated from the compressed air network and completely depressurized before performing any work on the pressure system.

All the cooling oil has run out of the filter 1–2 minutes after shutting down. A cooling oil receptacle is not needed.

Material Spare Parts
Cleaning cloths

Precondition The power supply disconnecting device is switched off,
lockout and tagout the device,
verify the absence of any voltage.

The machine is fully vented, the pressure gauge on the oil separator tank reads 0 psig.

⚠ WARNING

There is risk of burns from hot components and oil!

➤ *Wear long-sleeved clothing and protective gloves.*

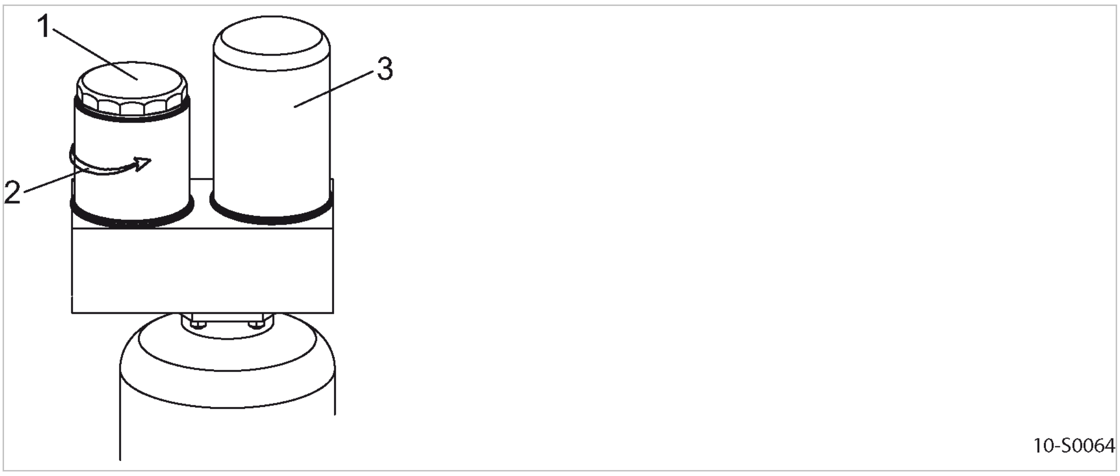


Fig. 36 Changing the oil filter

1 Oil filter
2 Direction to unscrew
3 Oil separator cartridge

Changing the oil filter

1. Unscrew the oil filter counter-clockwise and wipe off any drops of oil.
2. Lightly oil the new filter's gasket.
3. Turn the oil filter clockwise by hand to tighten.

➤ Dispose of parts and materiasl contaminated with cooling oil according to environment protection regulations.

Starting the machine and performing a trial run

1. Close all access doors, replace and secure all removable panels.
2. Open the user's shut-off valve between the machine and the compressed air network.
3. Switch on the power supply disconnecting device and reset the maintenance interval counter.
4. After approximately 10 minutes of operation: Check the cooling oil level and top off if necessa-ry.
5. Switch off the machine and visually check for leaks.

10.17 Changing the oil separator cartridge

The machine must be fully isolated from the compressed air network and completely depressurized before performing any work on the pressure system.

The life of the oil separator cartridge is influenced by:

- Contamination in the air drawn into the compressor,
- Adherence to the replacement intervals for:
  - Cooling oil
  - Oil filter
  - Air filter

All the cooling oil has run out of the separator cartridge 1–2 minutes after shutting down. A cooling oil receptacle is not needed.

Material
Spare parts
Cleaning cloths

Precondition
The power supply disconnecting device is switched off,
lockout and tagout the device,
verify the absence of any voltage.
The machine is fully vented, the pressure gauge on the oil separator tank reads 0 psig.

⚠ WARNING

There is risk of burns from hot components and oil!

➤ *Wear long-sleeved clothing and protective gloves.*

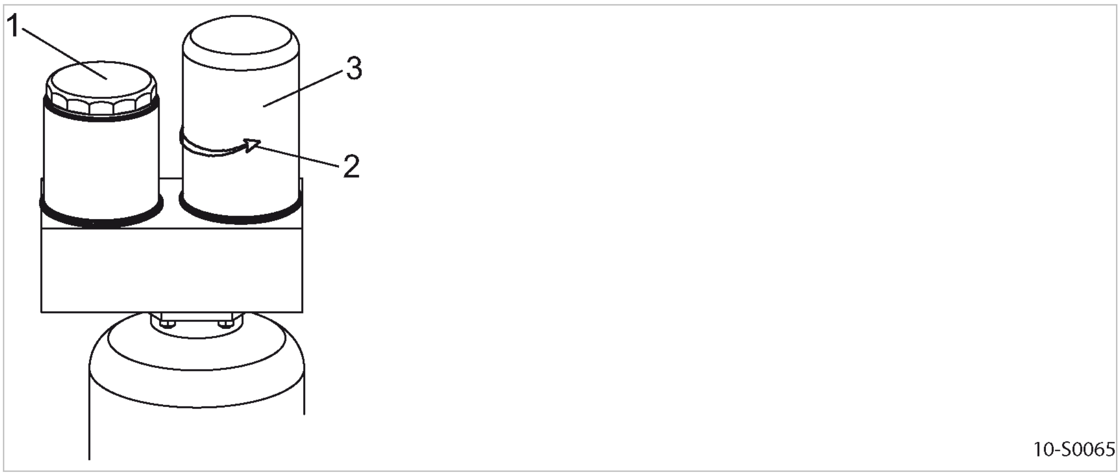


Fig. 37 Changing the oil separator cartridge

1 Oil filter
2 Direction to unscrew
3 Oil separator cartridge

Changing the oil separator cartridge

1. Unscrew the oil separator cartridge counter-clockwise and wipe off any drops of oil.
2. Lightly oil the new filter's gasket.
3. Turn the oil filter clockwise by hand to tighten.

➤ Dispose of parts and materials contaminated with cooling oil according to environment protection regulations.

Starting the machine and performing a trial run

1. Close all access doors, replace and secure all removable panels.
2. Open the user's shut-off valve between the machine and the compressed air network.
3. Switch on the power supply disconnecting device and reset the maintenance interval counter.
4. After approximately 10 minutes of operation: Switch off the machine and visually check for leaks.

10.18 Refrigerated dryer maintenance

The refrigeration circuit is a closed unit. Repairs may only be carried out by certified personnel.

Material
Compressed air for blowing out
Cleaning cloth
Vacuum cleaner

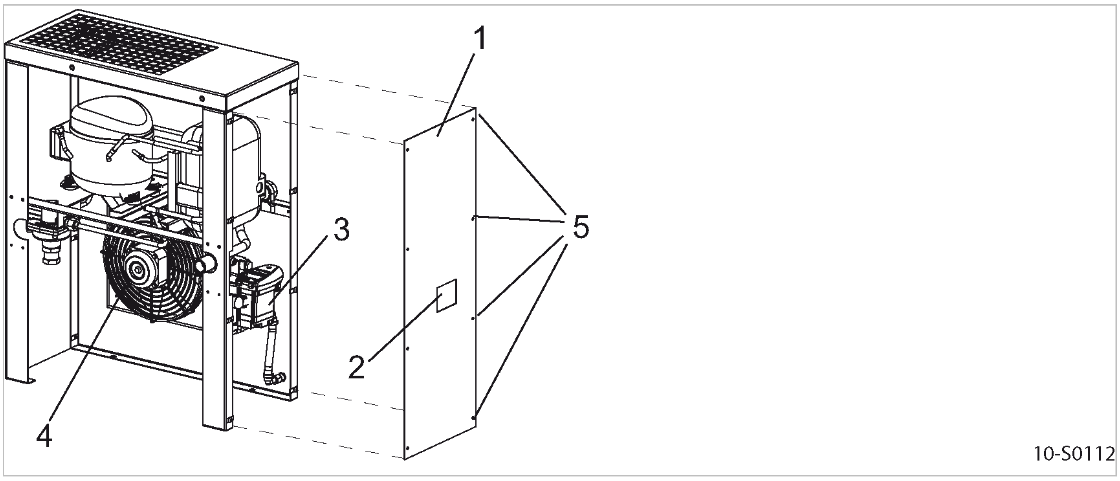


Fig. 38 Refrigerated dryer

1	Access panel	4	Refrigerant condenser
2	Sight glass: Condensate drain	5	Latch
3	Condensate drain		

10.18.1 Cleaning the refrigerant condenser

Do not use sharp objects to clean the refrigerant condenser. The refrigerant condenser could be damaged.

Avoid creating clouds of dust.

Precondition
The power supply disconnecting device is switched off,
the device is locked off,
the absence of any voltage has been verified.

1. Undo the latch 5 and remove the panel 1.

2. Use compressed air (<73 psig) to blow the condenser 4 through from outside to inside and then vacuum up the dirt.
3. Replace the panel again.

?

The refrigerant condenser can no longer be cleaned?

➤ Have stubborn clogging removed by an authorized KAESER service representative.

10.18.2 Maintaining the condensate drain

10.18.2.1 Checking the condensate drain

Precondition The machine is switched off.

The *Power* LED lights.

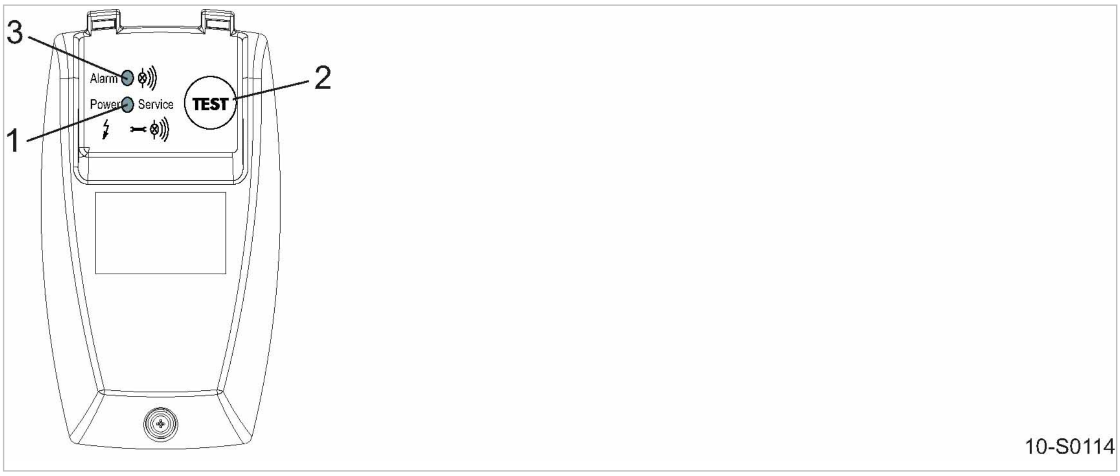


Fig. 39 Checking the condensate drain

1 *Power* LED
2 «TEST» key
3 *Alarm* LED

1. **⚠ CAUTION** *Danger of burns from hot components near the condensate drain!*
   ➤ *Work with caution.*
2. With one hand, lightly touch the condensate drain hose at the condensate drain.
3. With your other hand, push and hold the «TEST» key at the condensate drain for at least 2 seconds.

Result As soon as the condensate drain opens, you will feel a short burst at the condensate drain hose.

Service the condensate drain if you do **not** experience a burst during a manual check.

10.18.2.2 Changing the service unit

The condensate drain cannot be cleaned. The service unit must be changed if condensate does not drain.

Material Sealing tape for sealing the screw-in part

If required: O-ring 16x2 (5.1519.0)

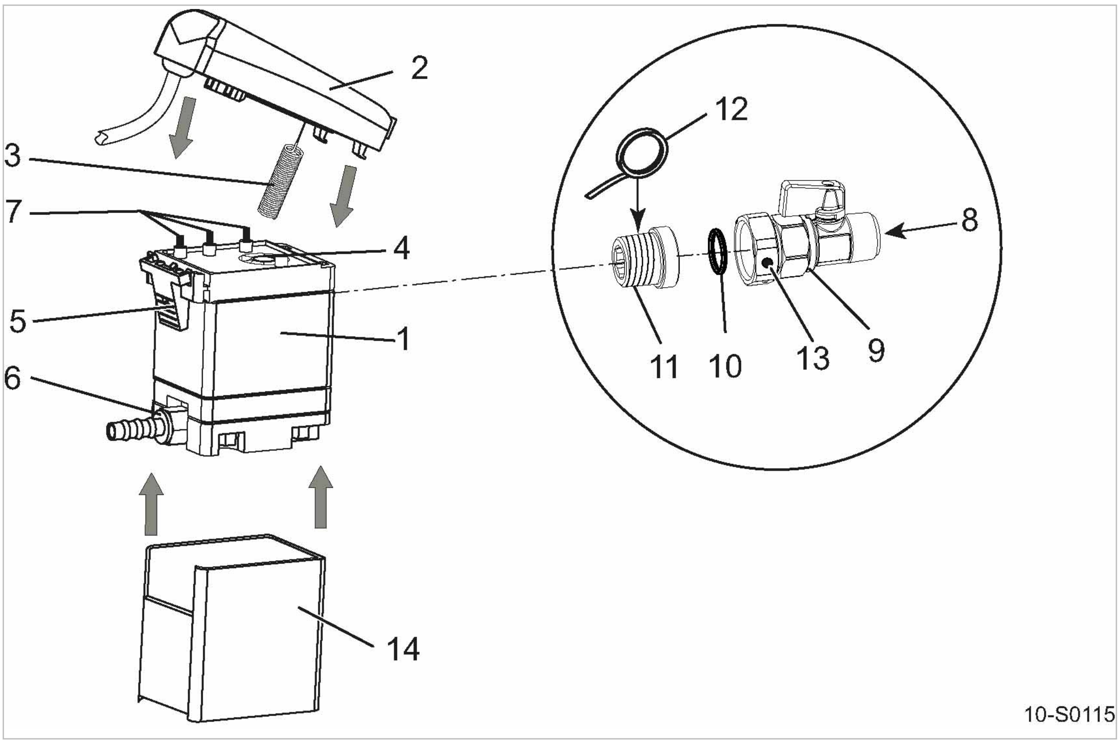


Fig. 40 Changing the service unit

1	Service unit	8	Condensate inlet
2	Control module	9	Shut-off valve
3	Sensor	10	O-ring
4	Sensor opening	11	Screw-in part
5	Snap fastener	12	Sealing tape
6	Condensate drain hose fitting	13	Union nut with vent holes
7	Contact springs	14	Insulation

Removing the service unit

1. **⚠ WARNING** *Serious injury or death can result from loosening or opening components under pressure!*
   ➤ *Fully vent all pressurized components and enclosures.*
2. Close the shut-off valve 9 upstream of the condensate drain.
3. Unscrew the fitting 6 at the condensate line.
4. Press the snap fastener 5 and carefully remove the control module 2 from the service unit 1.
5. Carefully loosen the union nut 13 at the shut-off valve 9 until remaining residual air has escaped through the venting holes.
6. Unscrew the screw-in part 11 from the service unit and place aside.
7. Remove the insulation 14 from the service unit.

Installing the service unit

Use only KAESER service units to ensure correct function of the condensate drain.

Precondition: Make sure that the top of the service unit and the contact springs are clean and dry.

1. Fit the insulation 14 to the service unit 1.

2. Carefully insert the sensor 3 of the control module 2 in the opening 4 at the service unit.
3. Place the snap fastener 5 of the control module into the service unit eyes.
4. Press the control module to the service unit until the snap fastener can be heard to click into place.
5. At the screw-in part 11, replace old sealing material with new sealing tape.
6. Install the screw-in part in the service unit.
7. If necessary, insert a new o-ring 10.
8. Tighten the union nut 13 at the shut-off valve 9.
9. Attach the condensate hose.
10. Open the shut-off valve upstream of the condensate drain.
11. Close the enclosure.

10.19 Air receiver maintenance

10.19.1 Draining condensate

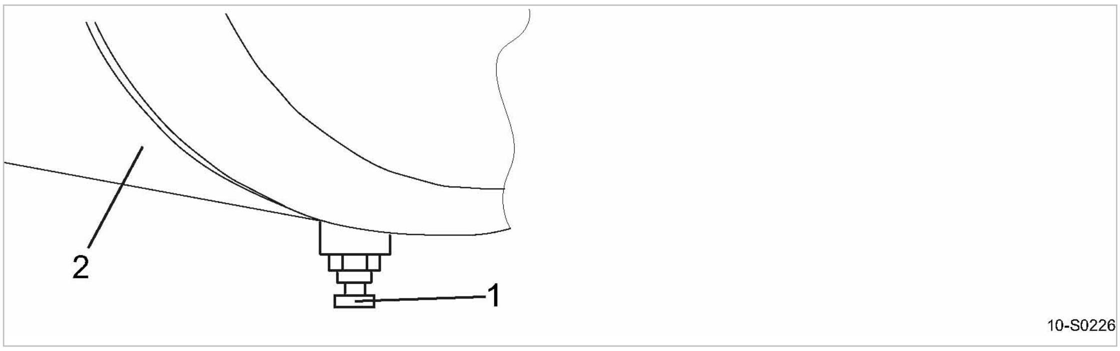


Fig. 41 Draining condensate

1 Shut-off valve
2 Air receiver

1. Open the shut-off valve on the air receiver drain and drain the condensate into a suitable container.
2. Close the shut-off valve.

➤ Dispose of the condensate in accordance with local environmental regulations.

10.19.2 Check the safety relief valve

Precondition The machine is switched off.
Always wear eye and ear protection.

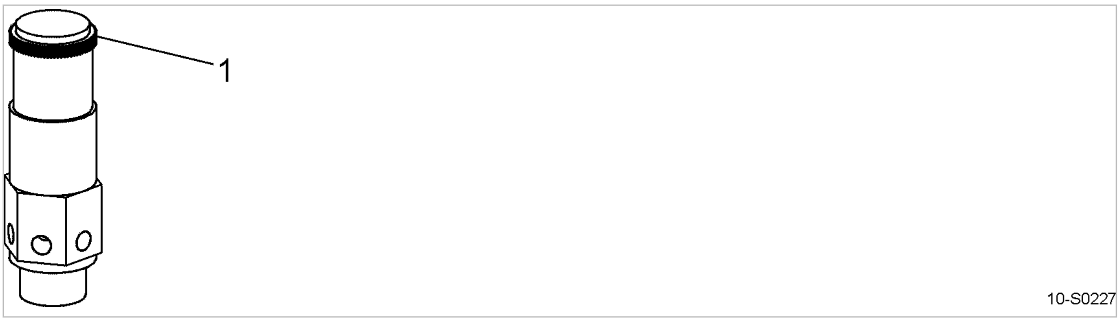


Fig. 42 Check the safety relief valve

1 Pull ring

1. Close the user's shut-off valve between the machine and the air distribution network.
2. Switch the machine on and wait until the air receiver is filled.

   The compressor switches to IDLE as soon as the air receiver is filled.
3. Shut down the machine and remove the panel on the air receiver.
4. Turn the ring counterclockwise until the safety relief valve opens.
5. Turn the ring back completely.
6. Reinstall the panel on the air receiver.
7. Open the user's shut-off valve between the machine and the air distribution network.

The safety relief valve fails to open:

➤ De-commission the machine until the defective safety relief valve is replaced.

10.20 Documenting maintenance and service work

Equipment number:

➢ Enter maintenance and service work carried out in the checklist below.

Date	Maintenance task carried out	Operating hours	Signature

Tab. 49 Logged maintenance tasks

11 Spares, Operating Materials, Service

11.1 Note the nameplate

The nameplate contains all information to identify your machine. This information is essential to us in order to provide you with optimal service.

➤ Please give the information from the nameplate with every inquiry and order for spare parts.

11.2 Ordering consumable parts and operating fluids/materials

KAESER consumable parts and operating materials are original KAESER products. They are specifically selected for use in KAESER machines.

Unsuitable or poor quality consumable parts and operating fluids/materials may damage the machine or impair its proper function.

Damage to the machine can also result in personal injury.

⚠ WARNING

There is risk of personal injury or damage to the machine resulting from the use of unsuitable spares or operating fluids/materials.

➤ *Use only original KAESER parts and operating fluids/materials.*

➤ *Have an authorized KAESER service representative carry out regular maintenance.*

Machine

Name	Number
Air filter element	1250
Filter mat (cooler)	1050
Filter mat (control cabinet)	1100
Oil filter	1200
Oil separator cartridge	1450
Cooling oil	1600
Drive belt	1801

Tab. 50 Consumable parts

11.3 KAESER AIR SERVICE

KAESER AIR SERVICE offers:

- authorized KAESER service representatives with KAESER factory training,
- increased operational reliability ensured by preventive maintenance,
- energy savings achieved by avoidance of pressure losses,
- optimum conditions for operation of the compressed air system,

- the security of genuine KAESER spare parts,
- increased legal certainty as all regulations are kept to.

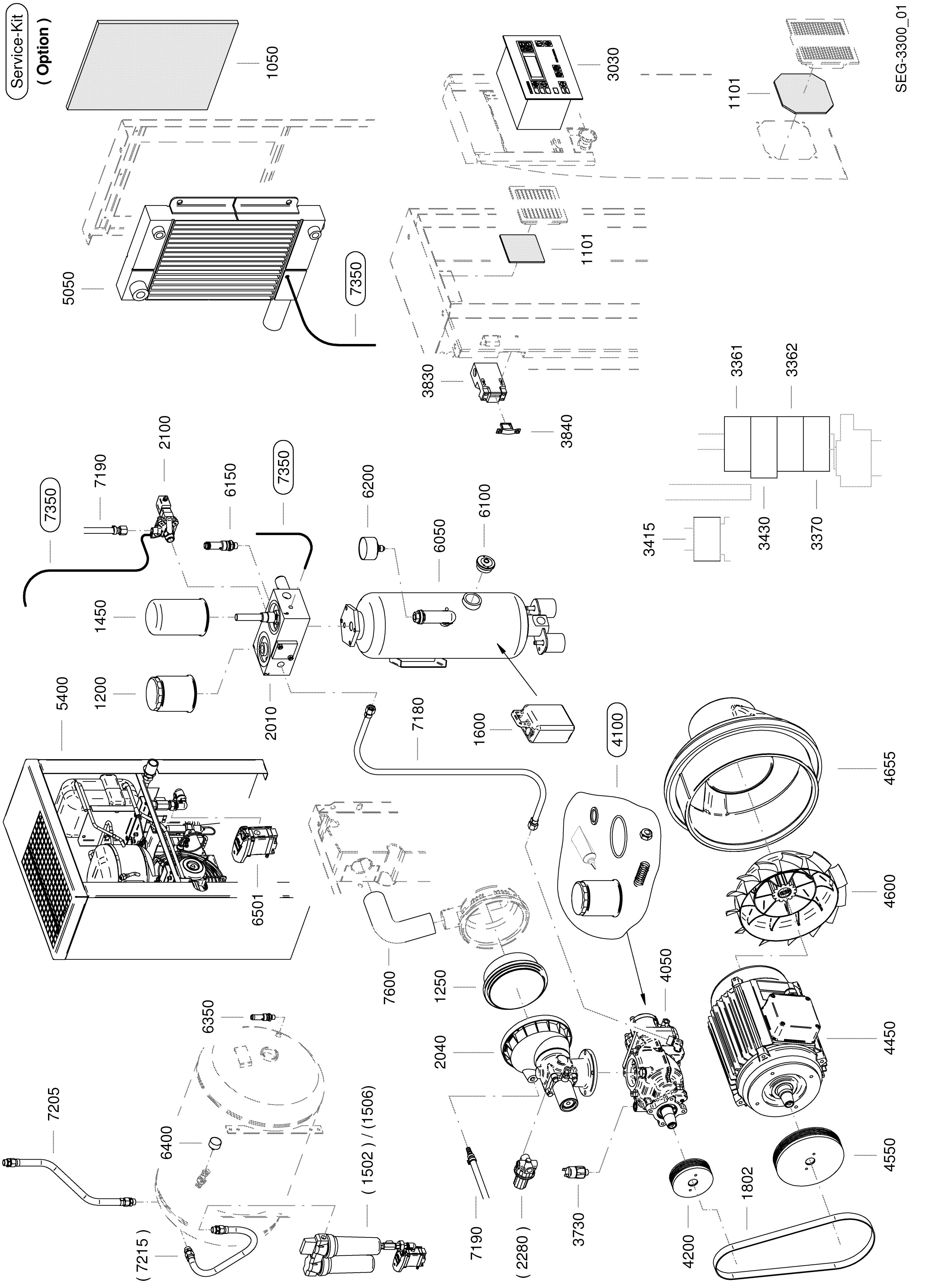
➢ Why not sign a KAESER AIR SERVICE maintenance agreement!

Result Your advantage:
lower costs and higher compressed air availability.

11.4 Replacement parts for service and repair

Use these parts lists to plan your material requirement according to operating conditions and to order the required spare parts.

➢ Make sure that any service or repair tasks not described in this manual are carried out by an authorized KAESER Service represenstative.


Service-Kit
(Option)
SEG-3300_01
1050
3030
1101
1101
5050
7350
3830
3840
3361
3362
3415
3430
3370
7350
7190
2100
6150
7350
6200
6100
6050
1450
1200
5400
2010
7180
1600
4100
4655
6501
4600
7600
1250
4050
6350
2040
4450
7205
6400
(1502) / (1506)
4550
1802
(7215)
7190
(2280)
3730
4200

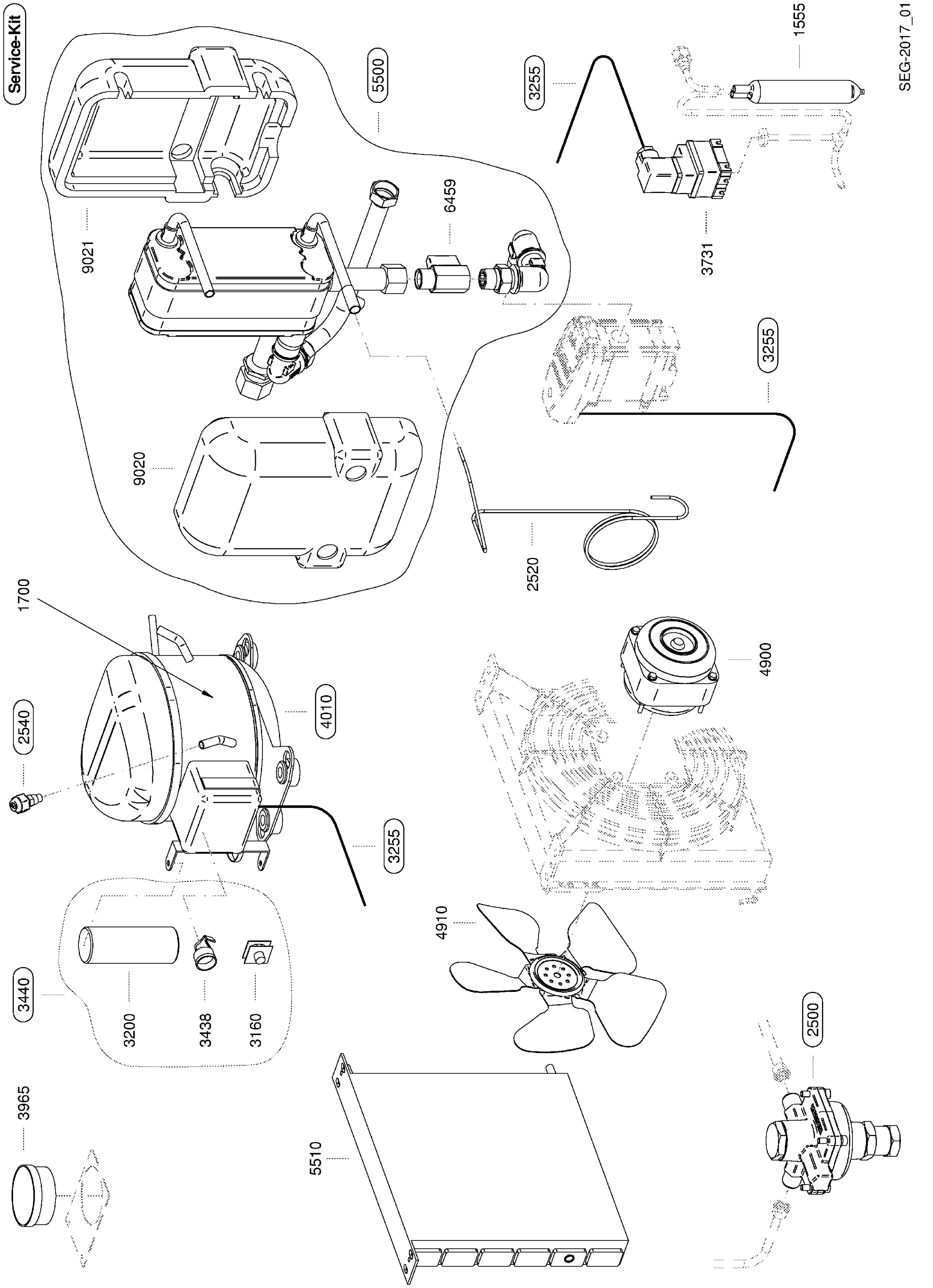
Legend	KAESER
Aircenter SX 3/4/5/7.5	SEL-2484_01USE

Item	Description	Option
1050	Filter mat, cooling air	
1101	Filter mat set, control cabinet	
1200	Oil filter	
1250	Air filter element	
1450	Oil separator cartridge	
1502	Air filters	X
1506	Compr. air filter combination	X
1600	Sigma Fluid *)	
1802	Drive belt	
2010	Valve block	
2022	Maintenance kit, MP/CV	
2024	Overhaul kit, MP/CV	
2062	Maintenance kit, combi. valve	
2064	Overhaul kit, combination valve	
2040	Inlet valve	
2042	Maintenance kit, inlet valve	
2044	Overhaul kit, inlet valve	
2100	Venting/control valve	
2102	Maintenance kit, VC valve	
2104	Overhaul kit, VC valve	
2280	Proportional controller	X
3030	SIGMA CONTROLLER	
3361	Mains contactor	
3362	Delta contactor	
3370	Star contactor	
3415	Contactor, dryer	
3430	Overload protection cutout	
3730	Rotating direction breaker	
3732	Protective cap	
3830	Safety interlock switch	
3840	Actuator (interlock switch)	
4050	SIGMA airend	
4100	Airend installation kit	
4200	Crankshaft pulley	
4450	Drive motor	
4451	Bearing set, drive motor	
4550	Drive motor pulley	
4600	Drive motor blower wheel	
4655	Motor cooling air flow guide	
5050	Cooler	
5400	Refrigeration dryer	
6050	Oil separator tank	
6100	Oil level indicator	
6150	OST pressure relief valve	
6200	Oil sep. tank pressure gauge	
6350	Air tank safety valve	
6400	Air tank pressure gauge	
6501	Condensate drain, dryer	
7180	Hose line	
7190	Hose line	
7205	Hose line	
7215	Hose line	X
7350	Control line kit	
7600	Inlet hose	

Please quote the part number and serial number of the machine together with the item number and the description of the part when ordering.

Before and during all work, be sure to read and follow the safety and service instructions in the machine's service manual.

*) See cooling fluid recommendations


Service-Kit
5500
3255
1555
SEG-2017_01
9021
6459
3731
3255
9020
2520
1700
4900
2540
4010
3255
4910
3440
3200
3438
3160
2500
3965
5510

Legend	KAESER
Refrigeration dryer ABT 4/8	SEL-2485_01E

Item		Description	Option
1555	*	Filter dryer	
1700	*	Refrigerant	
2500	*	Hot gas bypass regulator	
2520	*	Capillary tube	
2540	*	Refrigerant filling port	
3160		Starter relay	
3200		Starting capacitor	
3255		Control cabinet cable set	
3438		Circuit breaker, refr. compr.	
3440		Switch set	
3731	*	Safety pressure switch	
3965		Temperature gauge	
4010	*	Refrigerant compressor	
4900		Fan motor	
4910		Fan blade	
5500	*	Heat exchanger	
5510	*	Refrigerant condenser cpl.	
6459		Shut-off valve	
9020		Insulating jacket	
9021		Insulating jacket	

Please quote the part number and serial number of the machine together with the item number and the description of the part when ordering.

Before and during all work, be sure to read and follow the safety and service instructions in the machine's service manual.

* The replacement of the spare parts described requires an authorized and certified refrigerant technician.

12 Decommissioning, Storage and Transport

12.1 Decommissioning

Decommissioning is necessary, for example, under the following circumstances:

- The machine is (temporarily) not needed.
- The machine is to be moved to another location.
- The machine is to be scrapped.

➤ The following tasks are to be performed by authorized personnel only.

12.1.1 Temporary decommissioning

Precondition The machine can be started at regular intervals.

➤ Run the machine once a week for at least 30 minutes under LOAD to ensure sufficient protection against corrosion.

12.1.2 Long-term decommissioning

Precondition Immediately prior to decommissioning, run the machine under LOAD for at least 30 minutes.

12.1.2.1 Draining condensate

If the machine is equipped with a condensate drain, drain condensate from the condensate drain.

Precondition The power supply disconnecting device is switched off.

1. Drain condensate from the condensate drain and dispose of according to applicable environmental protection regulations.
2. Remove the user-supplied condensate lines.

The condensate drains are not supplied with power when the machine is switched off?

➤ Detach and drain the condensate drains.

12.1.2.2 Isolating the machine from supply lines

Precondition The power supply isolating device is switched off,
it has been locked out and tagged out against unintentional reactivation and
the absence of voltage has been verified.

The machine is fully vented (depressurized).
The user's shut-off valve to the compressed air network is closed or the compressed air network has been fully vented.

1. Allow the machine to completely cool down.
2. Detach the power supply and connecting line to the compressed air network at the user side.
3. Properly close all open connecting ports.

12.2 Packing

A suitable wooden crate is required for overland transport to protect the machine from mechanical damage.

Other measures must be taken for the transport of machines by sea or air. Please contact an authorized KAESER service representative for more information.

Material
Desiccant
Plastic sheeting
Wooden crate

Precondition
The machine is decommissioned.
The machine is dry and cooled down.

1. Place sufficient desiccant silica gel or desiccant clay in the machine.
2. Wrap the machine fully in plastic sheeting.
3. Protect the machine in a suitable wooden crate against mechanical damages.

12.3 Storage

Moisture can lead to corrosion, particularly on the surfaces of the airend and in the oil separator tank.

Frozen moisture can damage components, valve diaphragms and gaskets.

The following measures also apply to machines not yet commissioned.

> Please consult with KAESER if you have questions to the appropriate storage and commissioning.

1. **NOTICE** *Moisture and frost can damage the machine!*
   ➤ *Prevent ingress of moisture and formation of condensation.*
   ➤ *Maintain a storage temperature of >32 °F.*
2. Store the machine in a dry, frost-proof room.

12.4 Transport

12.4.1 Safety

Weight and center of gravity determine the most suitable method of transportation. The center of gravity is shown in the drawing in chapter 13.3.

> ➤ Please consult with KAESER if you intend to transport the machine in freezing temperatures.

Precondition
Transport only by forklift truck or lifting gear and only by personnel trained in the safe transportation of loads.

➤ Make sure the danger area is clear of personnel.

12.4.2 Transport with a forklift truck

Precondition
The forks are fully under the machine.

Fig. 43 Transport with a forklift truck

1. Take note of the center of gravity.
2. Drive the forks completely under the machine or palette and lift carefully.

12.4.3 Transport with a crane

Only suitable and approved load-carrying and attachment devices ensure proper transport of the machine with a crane. Suitable crossbeams ensure sufficient distance of the attachment resources from the machine housing to prevent damage.

The machine is not equipped with fixing points.

Examples of unsuitable fixing points:

- Pipe sockets
- Flanges
- Attached components such as centrifugal separators, condensate drains or filters
- Rain protection covers

➤ Consult KAESER if you require suitable load-carrying and attachment devices or have questions regarding the correct use.

Precondition Load-carrying and attachment devices meet the local safety regulations.

The hoist, load-carrying and attachment devices or the lifted machine do not endanger personnel.

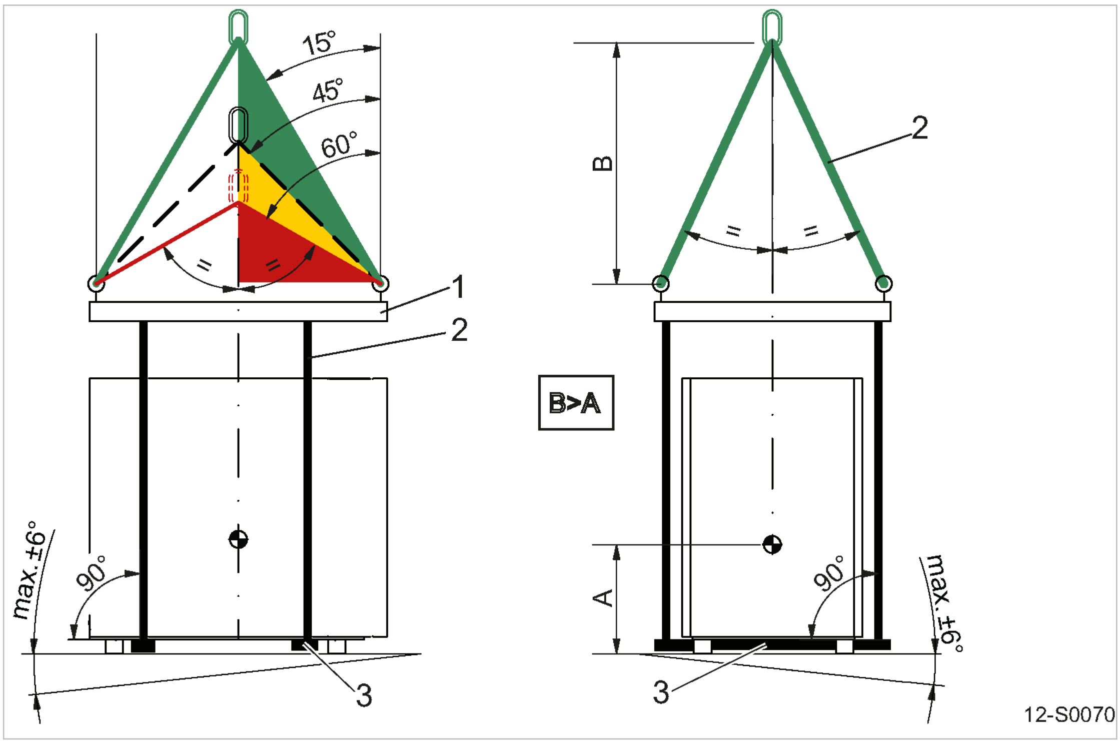


Fig. 44 Transport with a crane

1. Load carrying devices
2. Attachment resources
3. Crossbeam

1. **⚠ WARNING** *Risk of accident caused by incorrect use of load-carrying and attachment devices!*
   - ➤ *Comply with permissible load limits.*
   - ➤ *Comply with specific safety information of used load-carrying and attachment devices.*
2. Properly use load-carrying and attachment devices:
   - Ensure proper distribution of the fastening points relative to the center of gravity position (symmetrical load distribution).
   - Ensure equal slope angles of 15° to 45° for attachment devices with multiple strands.
     - Slope angles between 45° and 60° may be unsuitable.
     - Slope angles larger than 60° are prohibited.
   - Ensure the maximum incline of 6° of the machine to the horizontal.
   - Ensure sufficient distance of the attachment devices to the machine.
   - Ensure a positive stability height: Dimension B > Dimension A.
   - Do not attach the attachment devices to any machine component.
3. Carry out a lifting test:
   Slightly lift the machine to check whether machine remains in horizontal position and does not teeter.
4. Transport the machine only after a successful lifting test.

12.5 Disposal

When disposing of a machine, drain out all liquids and remove old filters.

Precondition: The machine is decommissioned.

1. Completely drain the cooling oil from the machine.
2. Remove used filters and the oil separator cartridge.
3. Hand the machine over to an authorized disposal expert.

➤ Parts contaminated with cooling oil must be disposed of in accordance with local environment protection regulations.

Compressors with refrigeration dryers

The refrigerant circuit still contains both refrigerant and oil.

➤ Refrigerant and oil must be drained and disposed of by authorized personnel.

12.5.1 Battery disposal in accordance with local environmental regulations.

Batteries contain substances that are harmful to living beings and the environment. For this reason, batteries must not be disposed of with unsorted residential waste. They must be disposed of in accordance with local environmental regulations. This procedure facilitates the handling and recycling of batteries.

The SIGMA CONTROL 2controller’s enclosure houses a battery.

Fig. 45 Battery labelling

1 Do not dispose of batteries with residential waste.
2 Battery contains lead (if applicable)

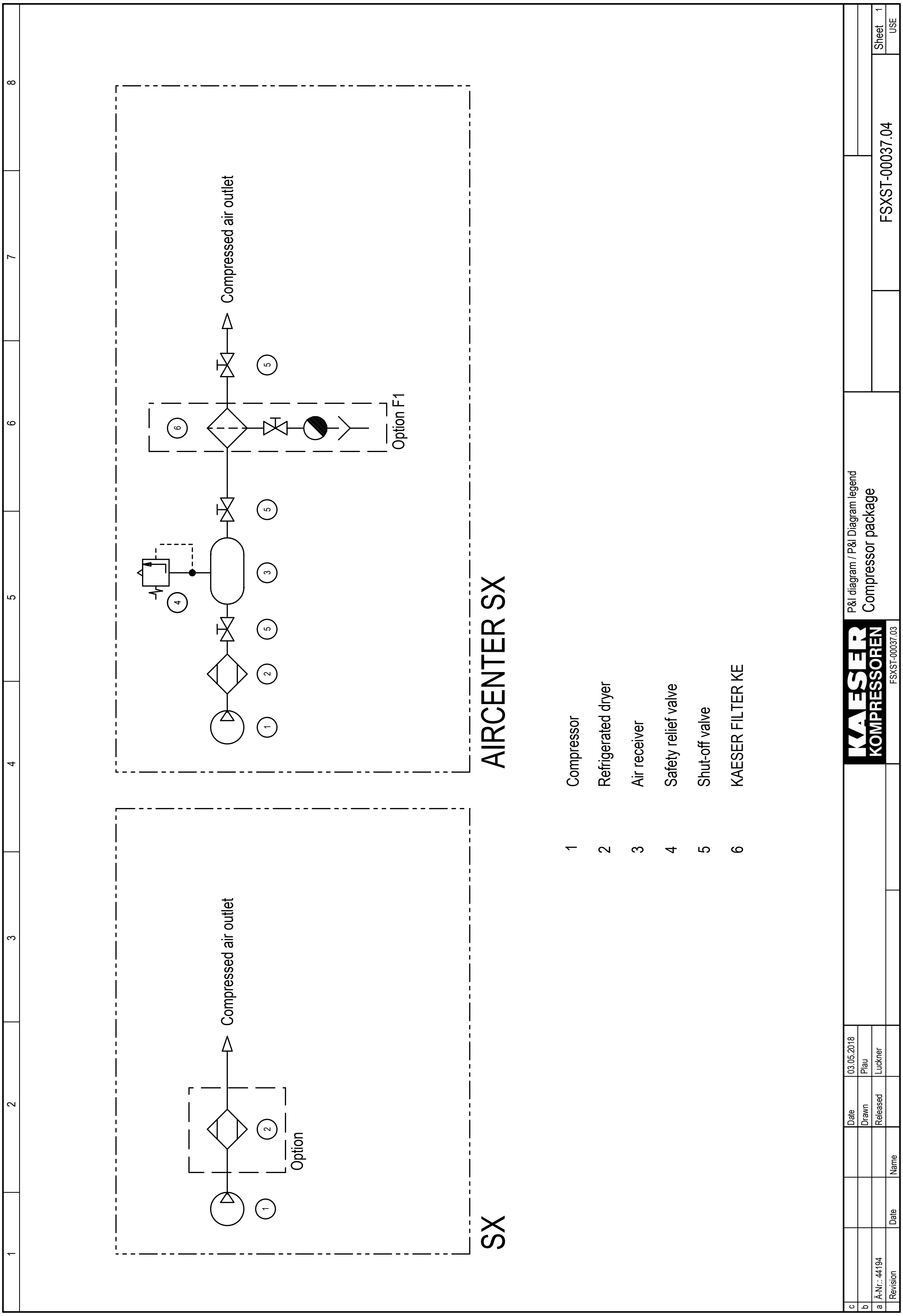
➤ Comply with national disposal regulations and dispose of batteries in an environmentally-friendly manner.

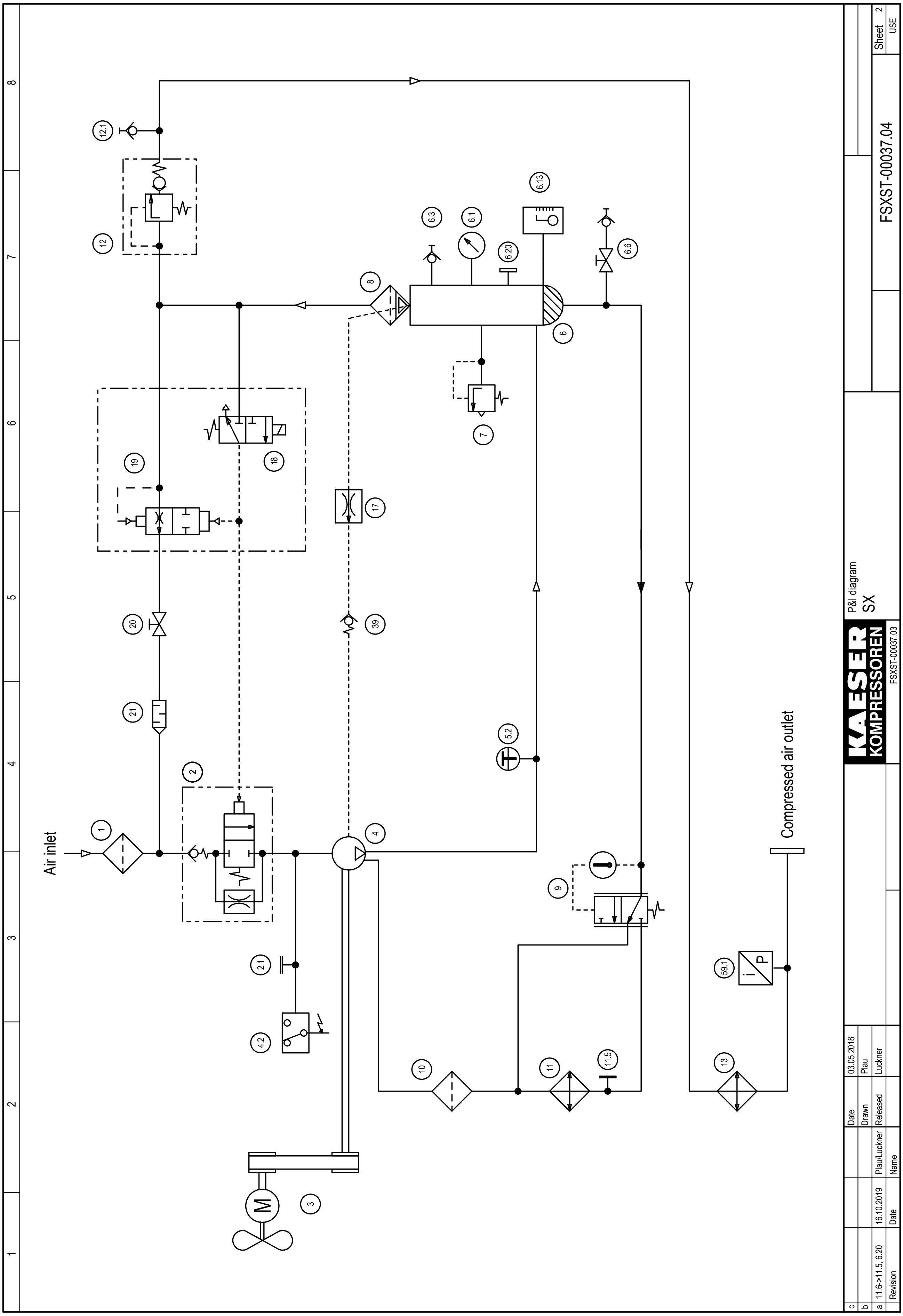
You actively contribute to the protection of our environment when you bring used batteries to the appropriate recycling system.

Further information: Refer to the SIGMA CONTROL 2user manual for details regarding battery removal and replacement.

13 Annex

13.1 Pipeline and instrument flow diagram (P+I diagram)


SX
Option
Compressed air outlet
AIRCENTER SX
Option F1
Compressed air outlet
1 Compressor
2 Refrigerated dryer
3 Air receiver
4 Safety relief valve
5 Shut-off valve
6 KAESER FILTER KE
P&I diagram / P&I Diagram legend
Compressor package
FSXST-00037.04
Sheet 1
USE
KAESER KOMPRESSOREN
FSXST-00037.03
Date 03.05.2018
Drawn Plau
Released Luckner
a Ä-Nr.: 44194
Revision
Date
Name


Air inlet
Compressed air outlet
1
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2.1
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4.2
5.2
6
6.1
6.3
6.6
6.13
6.20
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11.5
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12.1
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21
39
59.1
M
i
P
P&I diagram
SX
KAESER KOMPRESSOREN
FSXST-00037.03
FSXST-00037.04
Sheet 2
USE
Date 03.05.2018
Drawn Plau
Released Luckner
a 11.6->11.5, 6.20 16.10.2019 Plau/Luckner
Revision Date Name
b
c
1
2
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4
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6
7
8

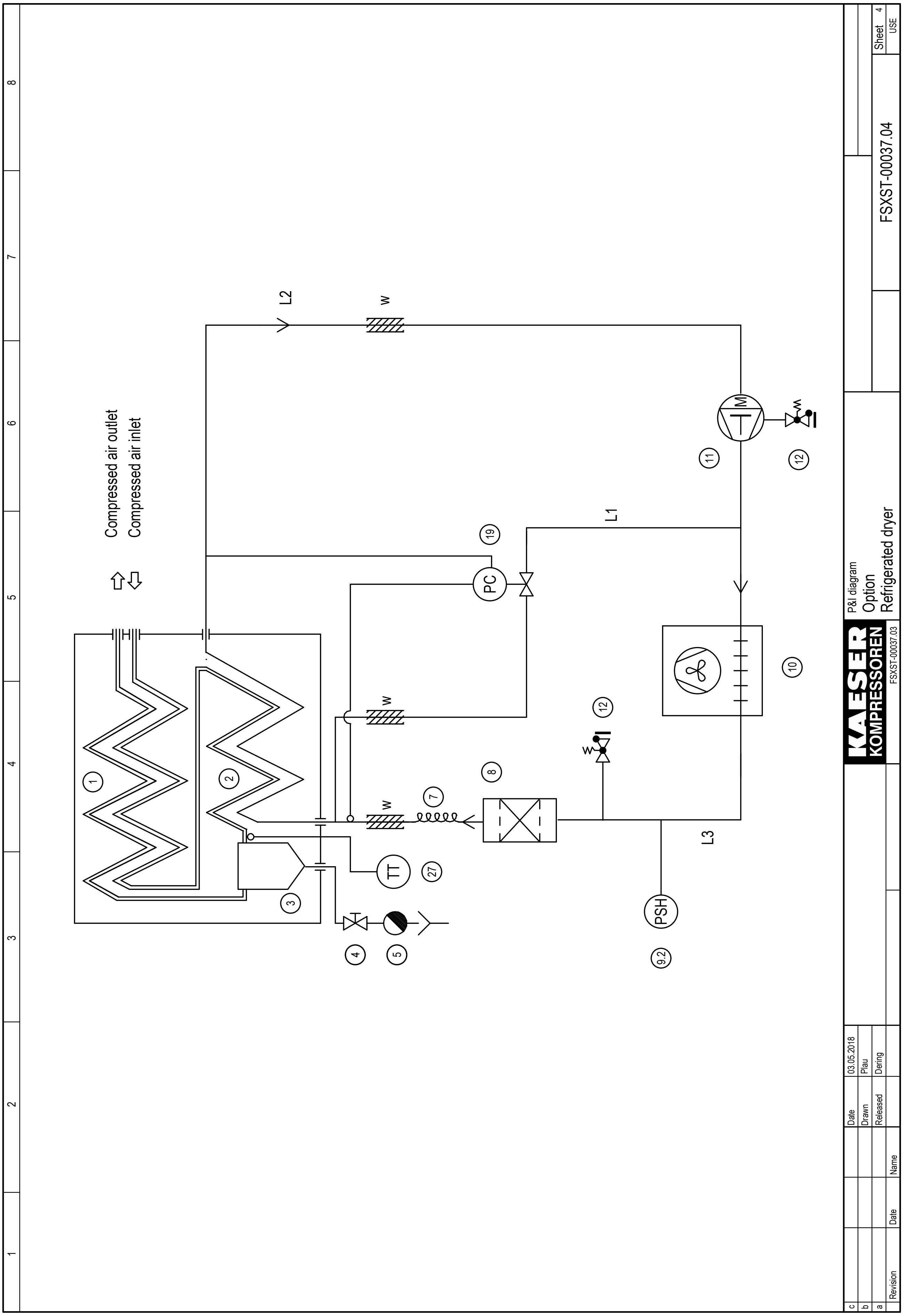
1	Air filter	9	Thermostatic valve
2	Inlet valve	10	Oil filter
2.1	Oil filler with screw plug	11	Oil cooler
3	Drive motor	11.5	Screw plug - Oil drain device
4	Rotary screw airend	12	Minimum pressure check valve
4.2	Pressure switch - Wrong direction of rotation	12.1	Hose coupling
5.2	Pt100 temperature sensor	13	Compressed air cooler
6	Oil separator tank	17	Nozzle
6.1	Pressure gauge	18/19	Combined control/venting valve
6.3	Hose coupling		18 Control valve
6.6	Shut-off valve with hose coupling - Oil drain device		19 Venting valve
6.13	Cooling oil level indicator	20	Shut-off valve - Venting line
6.20	Oil filler with screw plug	21	Silencer
7	Safety relief valve	39	Check valve
8	Oil separator cartridge	59.1	Pressure transducer - Network pressure

c				Date	03.05.2018
b				Drawn	Plau
a	11.6->11.5, 6.20	16.10.2019	Plau/Luckner	Released	Luckner
	Revision	Date	Name		

KAESER KOMPRESSOREN FSXST-00037.03

P&I Diagram legend
SX

FSXST-00037.04 Sheet 3 USE


Compressed air outlet
Compressed air inlet
L1
L2
L3
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PSH
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4
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7
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9.2
10
11
12
19
27
P&I diagram
Option
Refrigerated dryer
KAESER KOMPRESSOREN
FSXST-00037.03
FSXST-00037.04
Sheet 4
USE
Date 03.05.2018
Drawn Plau
Released Dering
Revision
Date
Name

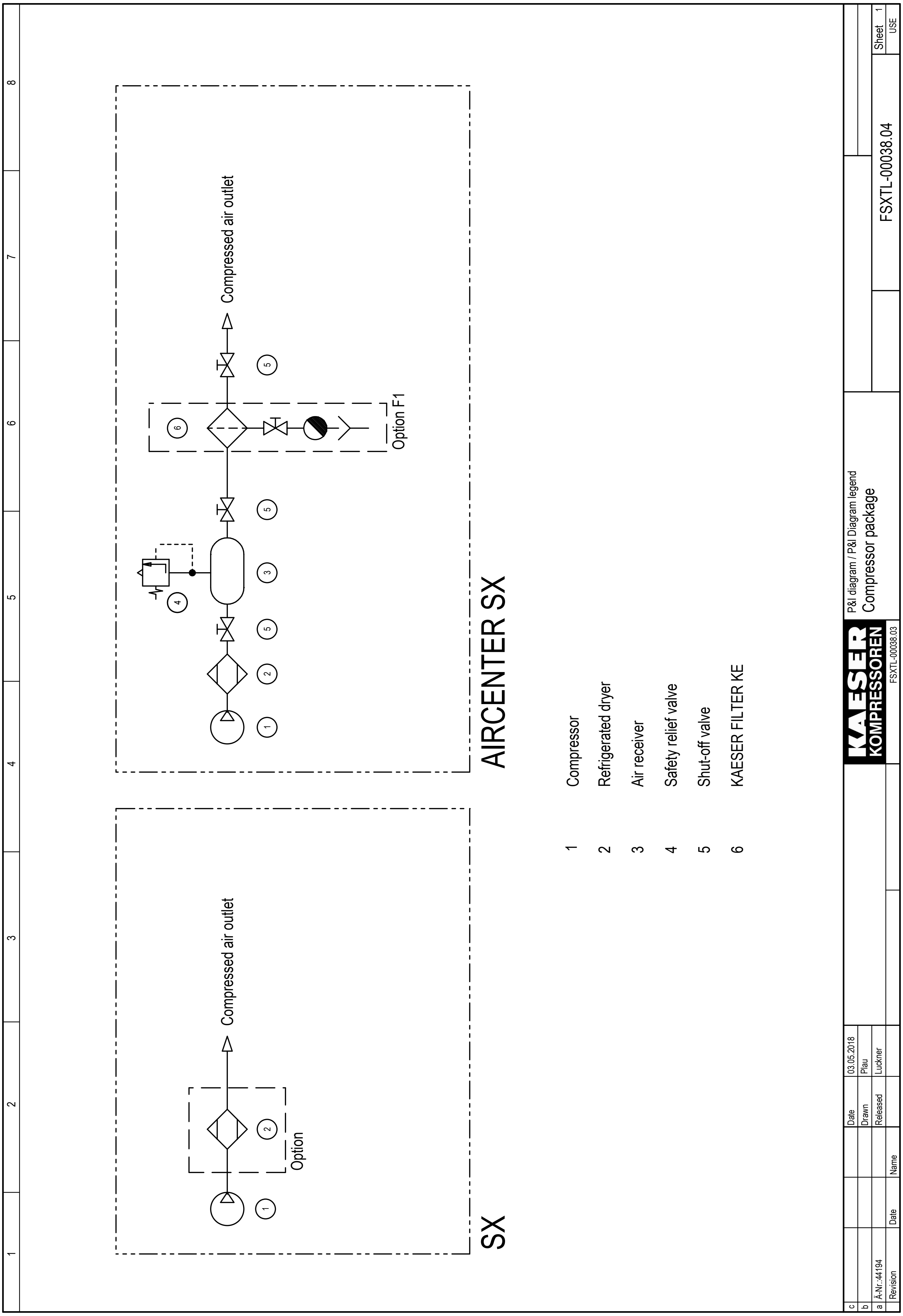
1	Air to air heat exchanger; heat insulated
2	Air to refrigerant heat exchanger (Evaporator); heat insulated
3	Condensate separator; heat insulated
4	Shut-off valve
5	Condensate drain
7	Capillary tube (Refrigerant injection)
8	Filter dryer
9.2	Pressure limiter
10	Refrigerant condenser
11	Refrigerant compressor
12	Service connection (Schrader valve)
19	Hot gas bypass regulator
27	Pt100 temperature sensor

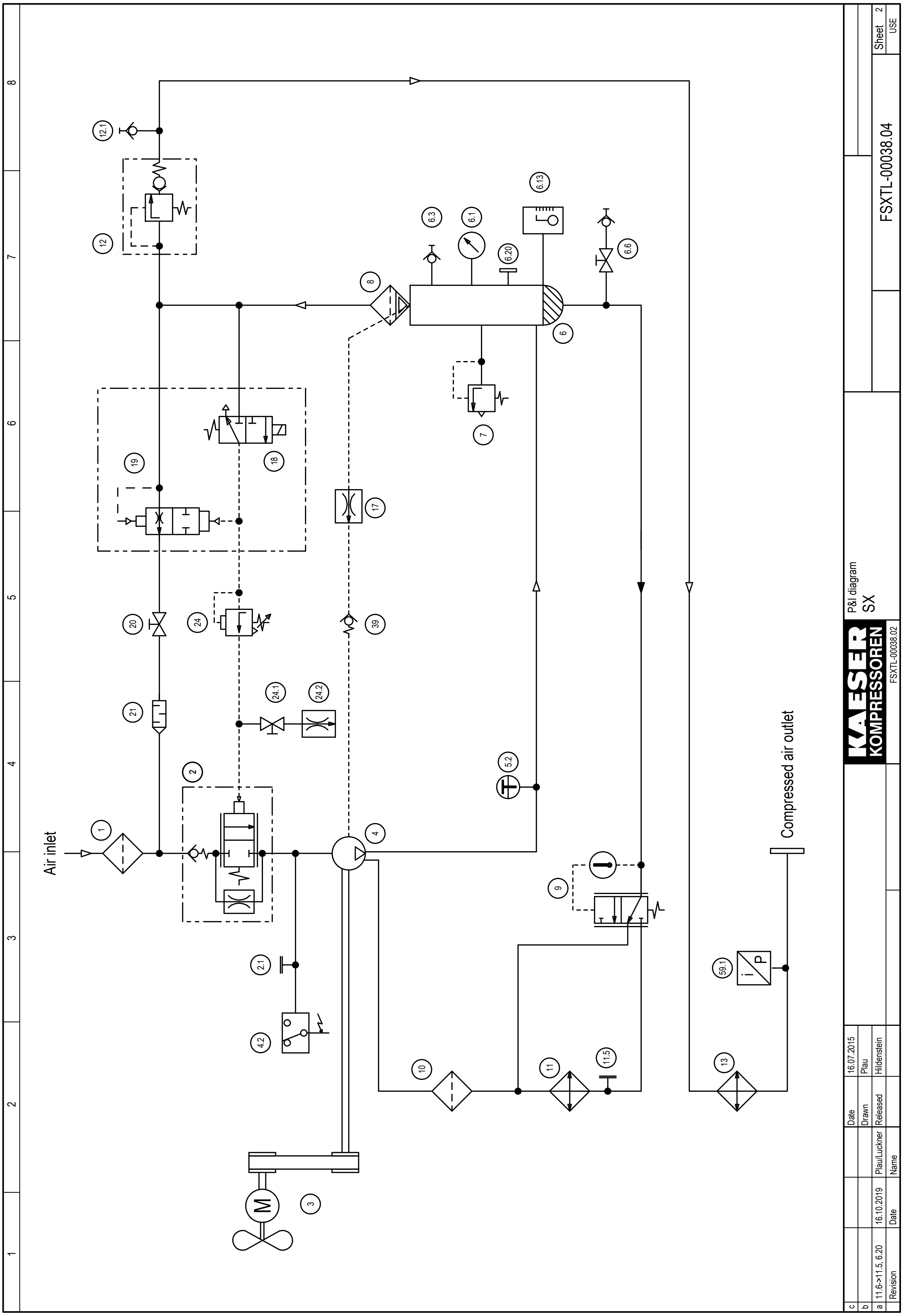
Piping:

L1	Bypass line
L2	CU-Pipe
L3	CU-Pipe
w	heat insulated

c			Date	03.05.2018	KAESER KOMPRESSOREN	P&I Diagram legend	
b			Drawn	Plau		Option	
a			Released	Dering		Refrigerated dryer	FSXST-00037.04 Sheet 5
Revision	Date	Name			FSXST-00037.03		USE

13.2 Option C1
Pipe and Instrument Flow Diagram (P+I diagram): MODULATING control mode


SX
Compressed air outlet
Option
AIRCENTER SX
Compressed air outlet
Option F1
1 Compressor
2 Refrigerated dryer
3 Air receiver
4 Safety relief valve
5 Shut-off valve
6 KAESER FILTER KE
P&I diagram / P&I Diagram legend
Compressor package
KAESER KOMPRESSOREN
FSXTL-00038.03
FSXTL-00038.04
Sheet 1
USE
Date 03.05.2018
Drawn Plau
Released Luckner
a Ä-Nr.:44194
Revision
Date
Name


Air inlet
Compressed air outlet
P&I diagram
SX
FSXTL-00038.04
FSXTL-00038.02
Sheet 2
USE
KAESER KOMPRESSOREN
Date 16.07.2015
Drawn Plau
Released Hildenstein
a 11.6->11.5, 6.20 16.10.2019 Plau/Luckner
Revision Date Name
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2.1
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4.2
5.2
6
6.1
6.3
6.6
6.13
6.20
7
8
9
10
11
11.5
12
12.1
13
17
18
19
20
21
24
24.1
24.2
39
59.1

1	Air filter	11	Oil cooler
2	Inlet valve	11.5	Screw plug - Oil drain device
2.1	Oil filler with screw plug	12	Minimum pressure check valve
3	Drive motor	12.1	Hose coupling
4	Rotary screw airend	13	Compressed air cooler
4.2	Pressure switch - Wrong direction of rotation	17	Nozzle
5.2	Pt100 temperature sensor	18/19	Combined control/venting valve
6	Oil separator tank	18	Control valve
6.1	Pressure gauge	19	Venting valve
6.3	Hose coupling	20	Shut-off valve - Venting line
6.6	Shut-off valve with hose coupling - Oil drain device	21	Silencer
6.13	Cooling oil level indicator	24	Proportional controller
6.20	Oil filler with screw plug	24.1	Shut-off valve
7	Safety relief valve	24.2	Nozzle
8	Oil separator cartridge	39	Check valve
9	Thermostatic valve	59.1	Pressure transducer - Network pressure
10	Oil filter		

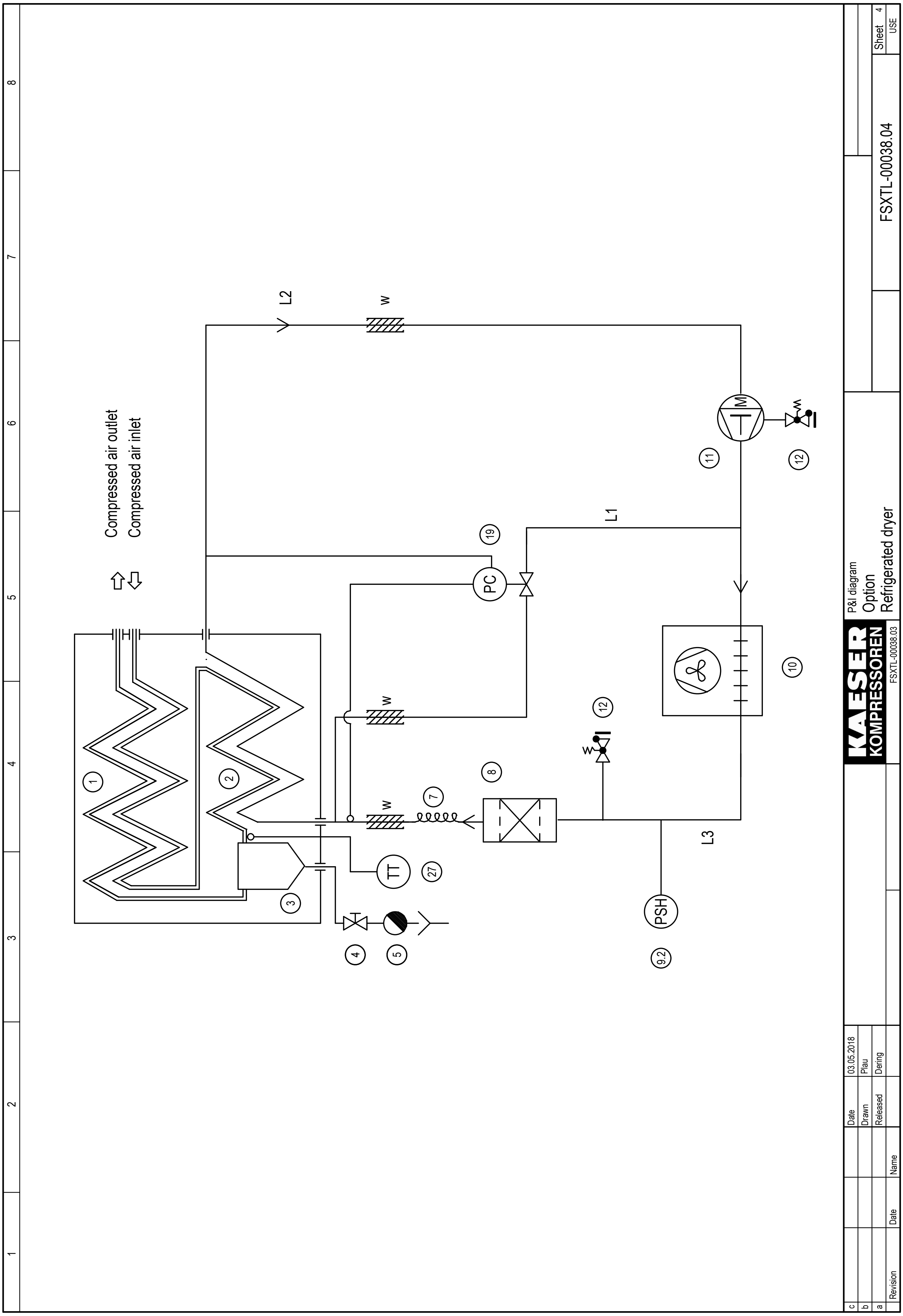
c					
b				Date	03.05.2018
a	11.6->11.5, 6.20	16.10.2019	Plau/Luckner	Drawn	Plau
Revision		Date	Name	Released	Luckner

KAESER KOMPRESSOREN
FSXTL-00038.03

P&I Diagram legend
SX

FSXTL-00038.04

Sheet 3
USE


Compressed air outlet
Compressed air inlet
L1
L2
L3
PC
TT
PSH
M
w
1
2
3
4
5
7
8
9.2
10
11
12
19
27
P&I diagram
Option
Refrigerated dryer
KAESER KOMPRESSOREN
FSXTL-00038.03
FSXTL-00038.04
Sheet 4
USE
Date 03.05.2018
Drawn Plau
Released Dering
Revision
Date
Name

1	Air to air heat exchanger; heat insulated
2	Air to refrigerant heat exchanger (Evaporator); heat insulated
3	Condensate separator; heat insulated
4	Shut-off valve
5	Condensate drain
7	Capillary tube (Refrigerant injection)
8	Filter dryer
9.2	Pressure limiter
10	Refrigerant condenser
11	Refrigerant compressor
12	Service connection (Schrader valve)
19	Hot gas bypass regulator
27	Pt100 temperature sensor

Piping:

L1	Bypass line
L2	CU-Pipe
L3	CU-Pipe
w	heat insulated

c			Date	FSXTL-00038.03	
b			Drawn	Plau	
a			Released	Dering	
Revision	Date	Name			

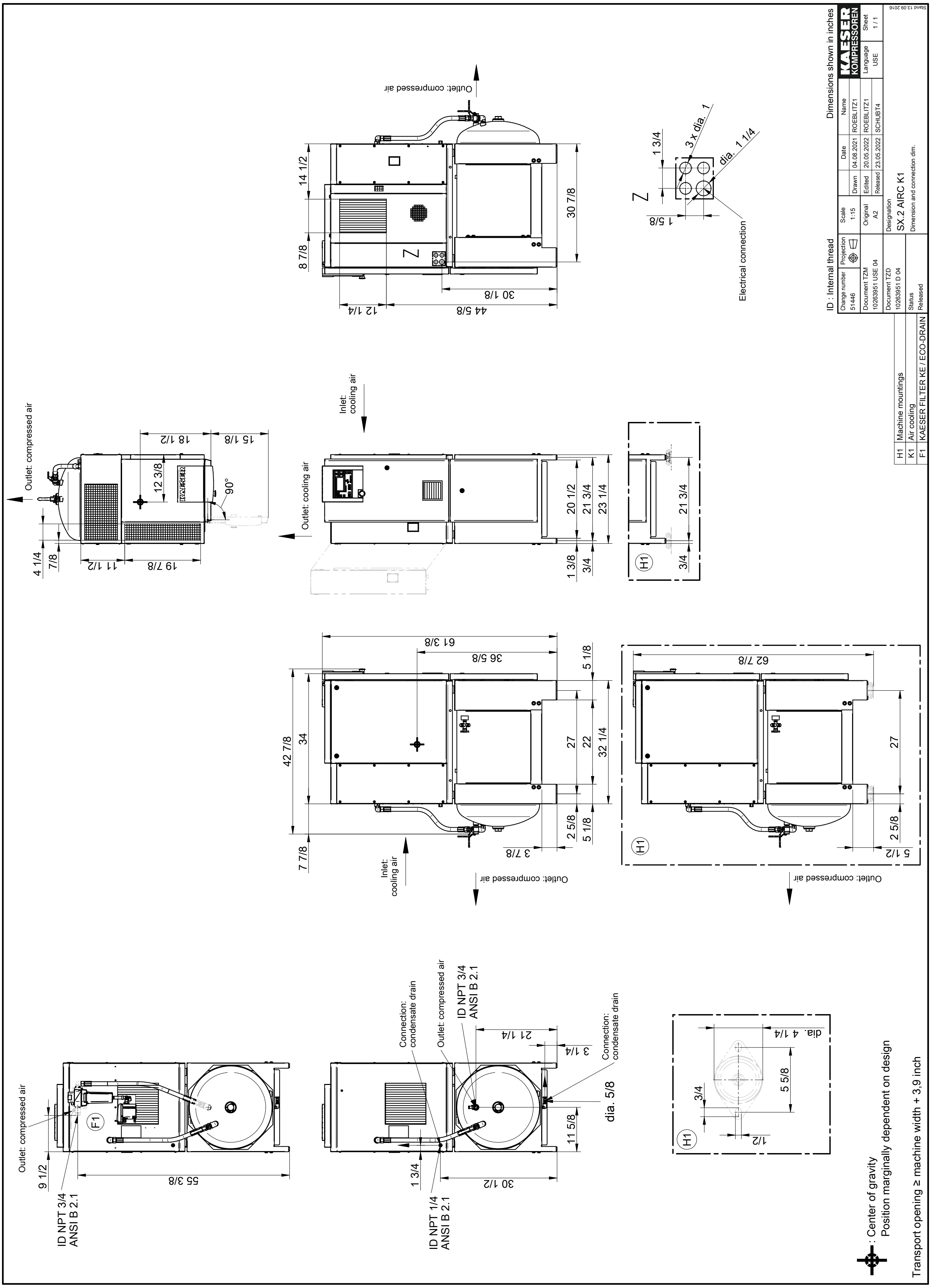
KAESER KOMPRESSOREN Plau

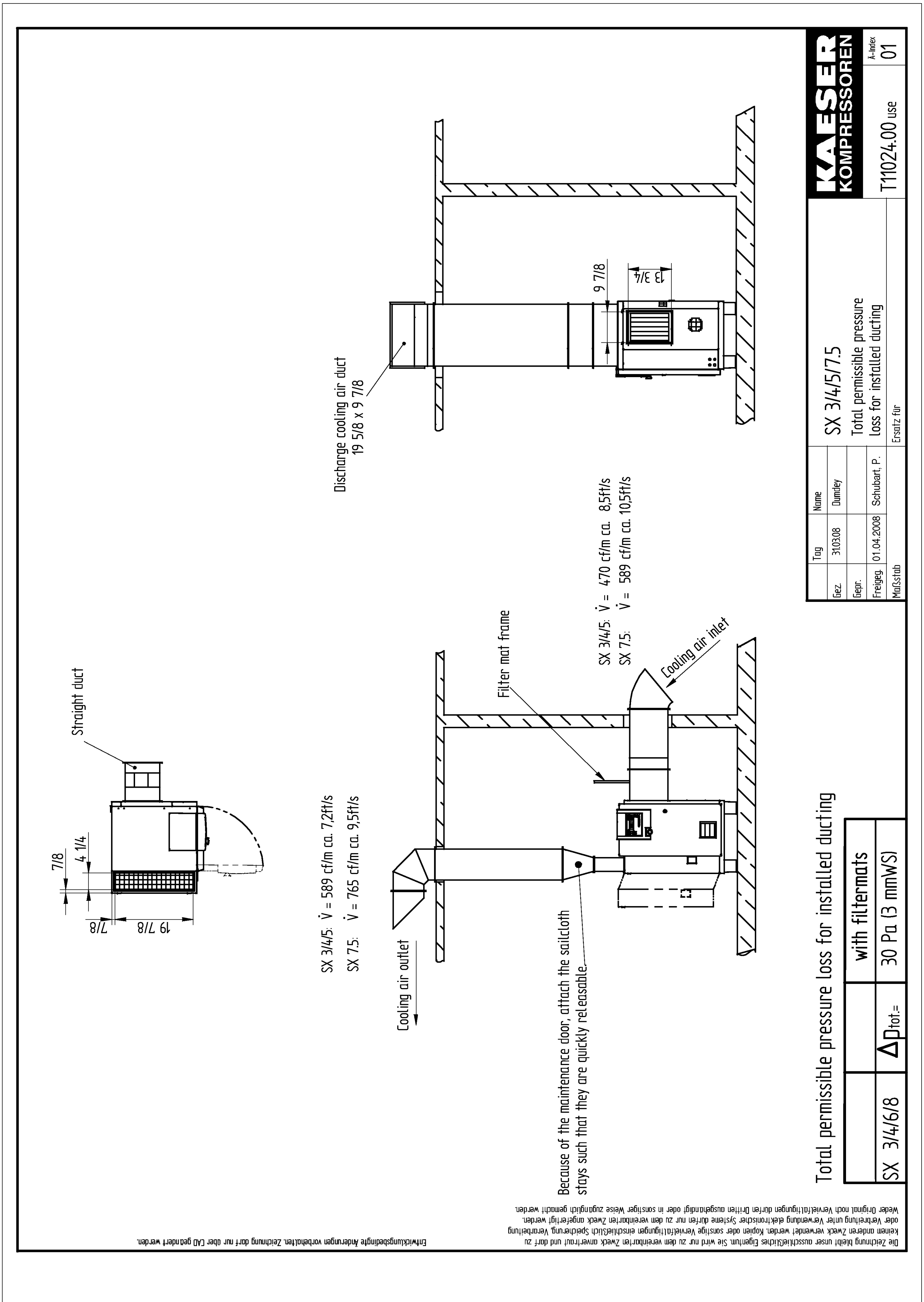
P&I Diagram legend
Option
Refrigerated dryer

FSXTL-00038.04

Sheet 5
USE

13.3 Dimensional drawing


Dimensions shown in inches
ID : Internal thread
Outlet: compressed air
Inlet: cooling air
Outlet: cooling air
Electrical connection
Connection: condensate drain
ID NPT 3/4 ANSI B 2.1
ID NPT 1/4 ANSI B 2.1
H1 Machine mountings
K1 Air cooling
F1 KAESER FILTER KE / ECO-DRAIN
Center of gravity
Position marginally dependent on design
Transport opening ≥ machine width + 3,9 inch
SX.2 AIRC K1
Dimension and connection dim.
Z
3 x dia. 1
dia. 1 1/4
dia. 5/8
dia. 4 1/4
90°


Total permissible pressure loss for installed ducting
SX 3/4/6/8
Δp tot.=
with filtermats
30 Pa (3 mmWS)
Straight duct
7/8
4 1/4
7/8
19 7/8
SX 3/4/5: V̇ = 589 cf/m ca. 7,2ft/s
SX 7.5: V̇ = 765 cf/m ca. 9,5ft/s
Cooling air outlet
Because of the maintenance door, attach the sailcloth stays such that they are quickly releasable.
Filter mat frame
SX 3/4/5: V̇ = 470 cf/m ca. 8,5ft/s
SX 7.5: V̇ = 589 cf/m ca. 10,5ft/s
Cooling air inlet
Discharge cooling air duct
19 5/8 x 9 7/8
9 7/8
13 3/4
Tag
Name
Gez.
31.03.08
Dumdey
Gepr.
Freigeg.
01.04.2008
Schubart, P.
Maßstab
SX 3/4/5/7.5
Total permissible pressure loss for installed ducting
Ersatz für
KAESER KOMPRESSOREN
T11024.00 use
Ä-Index
01

13.4 Electrical Diagram

1	2	3	4	5	6	7	8

ATTENTION !!!
The unit is non-functional when delivered.
Before connecting the electrical supply,
please follow the advice contained
in the 'Installation' chapter and
the wiring diagram of the service manual.

Wiring Diagram

compressor series SX T

208V±10% 3ph 60Hz
230V±10% 3ph 60Hz
460V±10% 3ph 60Hz

Tri-Voltage

Power supply:
WYE system with center point solidly grounded

manufacturer: KAESER COMPRESSORS
96450 COBURG
GERMANY

ATTENTION !!!
The document gives collective information on
power supply voltages and frequencies for all machines.
The voltage and frequency and local conditions under
which any particular machine may be used
are given on the nameplate of the machine
and in the accompanying service manual.

c				Date	16.09.2019 USE		
b				Drawn	Seubert		
a	47414	16.09.19	Bü/Se	Released	Büchner		
A	Change	Date	Name				

KAESER KOMPRESSOREN

cover page		=	
compressor series SX T		+	
	SC2 MCSIO	DSX.T-U3063.01	page 1 / 1 Sht.

Lfd. Nr. No.	Benennung Name		Zeichnungsnummer (Kunde) Drawing No. (customer)	Zeichnungsnummer (Hersteller) Drawing No. (manufacturer)	Blatt Page	Anlagenkennzeichen Unit designation
1	cover page			DSX.T-U3063.01	1	
2	list of contents			ZSX.T-U3063.01	1	
3	general instructions			USX.T-U3063.01	1	
4	electrical equipment identification			USX.T-U3063.01	2	
5	electrical component parts list			USX.T-U3063.01	3	
6	electrical component parts list			USX.T-U3063.01	4	
7	wiring diagram	power unit		SSX.T-U3063.01	1	
8	wiring diagram	control		SSX.T-U3063.01	2	
9	wiring diagram	power unit/control air dryer		SSX.T-U3063.01	3	
10	wiring diagram	control voltage tapping		SSX.T-U3063.01	4	
11	wiring diagram	condensate drain		SSX.T-U3063.01	5	
12	wiring diagram	power supply unit		SSX.T-U3063.01	6	
13	wiring diagram	IO-module/configuration		SSX.T-U3063.01	7	
14	wiring diagram	sensors/actuators		SSX.T-U3063.01	8	
15	wiring diagram	volt-free contacts		SSX.T-U3063.01	9	
16	wiring diagram	inputs/outputs		SSX.T-U3063.01	10	
17	wiring diagram	transformer diagrams		SSX.T-U3063.01	11	
18	wiring diagram	supply air dryer		SSX.T-U3063.01	12	
19	wiring diagram	Handling: Terminals / Feed line connection		SSX.T-U3063.01	13	
20	terminal connection	terminal strip -X0,-X1		KSX.T-U3063.01	1	
21	terminal connection	terminal strip -X11		KSX.T-U3063.01	2	
22	terminal connection	terminal strip -X31,-1X31,-2X31,-3X31		KSX.T-U3063.01	3	
23	lay-out	Switchboard		ASX.T-U3063.01	1	

c				Date	16.09.2019
b				Drawn	Seubert
a				Released	Büchner
B	Change	Date	Name		

KAESER KOMPRESSOREN

list of contents
compressor series SX T

=
+

SC2 MCSIO | ZSX.T-U3063.01 | page 1 | 1 Sht.

1	2	3	4	5	6	7	8

general instructions

ATTENTION !!!
Install supplies, grounding and shock protection
to local safety regulations.
Do not make or break
live plug-in connectors.

control cabinet wiring for non-designated conductors

primary circuits ungrounded:	black, UL-Style 1015, CSA-TEW
primary circuits grounded:	grey, UL-Style 1015, CSA-TEW
control voltage AC 115V ungrounded:	red, 18AWG UL-Style 1015, CSA-TEW
control voltage AC 115V grounded:	white, 18AWG UL-Style 1015, CSA-TEW
control voltage DC ungrounded:	blue, 18AWG UL-Style 1015, CSA-TEW
control voltage DC grounded:	white/blue, 18AWG UL-Style 1015, CSA-TEW
external voltage:	orange, 16AWG UL-Style 1015, CSA-TEW
measuring circuits:	violet, 18AWG UL-Style 1015, CSA-TEW
ground conductor:	green/yellow, UL-Style 1015, CSA-TEW

option F1 = KAESER FILTER KE

c				Date	16.09.2019	KAESER KOMPRESSOREN	general instructions		=
b				Drawn	Seubert		compressor series SX T		+
a				Released	Büchner			SC2 MCSIO	USX.T-U3063.01 page 1
C	Change	Date	Name						4 Sht.

electrical equipment identification

general components

-B25	overload relay, compressor motor
-1FU,-2FU	primary control fuse
-3FU	secondary control fuse
-4FU,-5FU	primary control fuse, air dryer
-6FU	secondary control fuse, air dryer
-M1	compressor motor
-M30	Refrigerant compressor, air dryer
-M31	vent motor, air dryer
-Q1	main contactor
-Q2	delta contactor
-Q3	wye contactor
-Q30	motor starter, air dryer
-S1	EMERGENCY STOP pushbutton
-S5	door safety interlock switch
-T11	control transformer
-T21	power unit
-T30	transformer, air dryer

control

-K20	*SC2 MCSIO MCSIO*
-X1	Ethernet
-X3	RS485-FC (USS)
-X5	SD card slot
-X6	ground connection
-X7	power supply unit, digital inputs
-X8	digital inputs, digital outputs
-X9	analog inputs, 4-20mA, Pt100
-X10	Relay outputs

terminal strips

-X0	terminal strip, power supply
-X1	terminal strip motor
-X11	terminal strip, control
-X31	terminal strip, air dryer
-1X31,-2X31	connector plug air dryer
-3X31	cable joint, air dryer

sensors/actuators

-B1	pressure transducer, air main pressure
-B2	safety air pressure switch-direction of rotation
-B30	safety air pressure switch (Pressure limiter), air dryer
-B39	temperature probe dew point temperature air dryer
-B40	temperature probe, airend discharge temperature
-K1	control valve
-K34	condensate drain, air dryer
-K13	condensate drain, microfilter/option F1

Date	16.09.2019	KAESER KOMPRESSOREN	electrical equipment identification	=	
Drawn	Seubert		compressor series SX T	+	
Released	Büchner				
C Change / Date / Name			SC2 MCSIO	USX.T-U3063.01	page 2 / 4 Sht.

electrical component parts list

model		SX 3 T	SX 4 T	SX 5 T	SX 7.5 T
machine power supply		208 V ±10 %, 60 Hz	208 V ±10 %, 60 Hz	208 V ±10 %, 60 Hz	208 V ±10 %, 60 Hz
		230 V ±10 %, 60 Hz	230 V ±10 %, 60 Hz	230 V ±10 %, 60 Hz	230 V ±10 %, 60 Hz
motor	-M1	3 hp	4 hp	5 hp	7,5 hp
		diagram 2, Sht. 1	diagram 2, Sht. 1	diagram 2, Sht. 1	diagram 2, Sht. 1
supply terminals	-X0:U1/V1/W1	7.3149.01960	7.3149.01960	7.3149.01960	7.3149.01960
	Wieland	WKFN10D1/2/35	WKFN10D1/2/35	WKFN10D1/2/35	WKFN10D1/2/35
	Stripped length	15 mm	15 mm	15 mm	15 mm
	Handling	fig. 2, Sht. 13	fig. 2, Sht. 13	fig. 2, Sht. 13	fig. 2, Sht. 13
	-X0:GRD	7.3149.01980	7.3149.01980	7.3149.01980	7.3149.01980
	Wieland	WKFN10D1/2/SL/35	WKFN10D1/2/SL/35	WKFN10D1/2/SL/35	WKFN10D1/2/SL/35
	Stripped length	15 mm	15 mm	15 mm	15 mm
	Handling	fig. 2, Sht. 13	fig. 2, Sht. 13	fig. 2, Sht. 13	fig. 2, Sht. 13
supply	connection	fig. 10, Sht. 13	fig. 10, Sht. 13	fig. 10, Sht. 13	fig. 10, Sht. 13
terminal strip	-X0	7.6836.00260 Wieland	7.6836.00260 Wieland	7.6836.00260 Wieland	7.6836.00260 Wieland
terminal strip	-X11/X31	7.6836.00700 Wieland	7.6836.00700 Wieland	7.6836.00700 Wieland	7.6836.00700 Wieland
	Handling	fig. 1, Sht. 13	fig. 1, Sht. 13	fig. 1, Sht. 13	fig. 1, Sht. 13
contactor	-Q1	7.8740.00350	7.8740.00360	7.8740.00360	7.8740.00360
		3RT2024-1AK60	3RT2025-1AK60	3RT2025-1AK60	3RT2025-1AK60
auxiliary switch		7.8740.05010	7.8740.05010	7.8740.05010	7.8740.05010
		3RH2911-1HA11	3RH2911-1HA11	3RH2911-1HA11	3RH2911-1HA11
interference suppressor		7.8740.05140	7.8740.05140	7.8740.05140	7.8740.05140
	Siemens	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00
contactor	-Q2	7.8740.00350	7.8740.00360	7.8740.00360	7.8740.00360
		3RT2024-1AK60	3RT2025-1AK60	3RT2025-1AK60	3RT2025-1AK60
interference suppressor		7.8740.05140	7.8740.05140	7.8740.05140	7.8740.05140
	Siemens	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00
contactor	-Q3	7.8740.00340	7.8740.00340	7.8740.00340	7.8740.00350
		3RT2023-1AK60	3RT2023-1AK60	3RT2023-1AK60	3RT2024-1AK60
interference suppressor		7.8740.05140	7.8740.05140	7.8740.05140	7.8740.05140
	Siemens	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00
contactor	-Q30	7.8740.03010	7.8740.03010	7.8740.03010	7.8740.03010
	Siemens	3RT2016-1JB41	3RT2016-1JB41	3RT2016-1JB41	3RT2016-1JB41
overload relay	-B25	7.8741 A00060	7.8741.00060	7.8741.00060	7.8741 A00070
		3RB3026-1SB0 (3-12 A)	3RB3026-1SB0 (3-12 A)	3RB3026-1SB0 (3-12 A)	3RB3026-1QB0 (6-25 A)
		setting: 4,6 A (208 V)	setting: 6,2 A (208 V)	setting: 8,3 A (208 V)	setting: 10,9 A (208 V)
		setting: 4,3 A (230 V)	setting: 5,8 A (230 V)	setting: 8,3 A (230 V)	setting: 10,4 A (230 V)
		NEC 430.32(C) incremental	NEC 430.32(C) incremental	NEC 430.32(C) incremental	NEC 430.32(C) incremental
		setting: 5,2 A (208 V)	setting: 6,9 A (208 V)	setting: 9,3 A (208 V)	setting: 12,2 A (208 V)
		setting: 4,8 A (230 V)	setting: 6,5 A (230 V)	setting: 9,3 A (230 V)	setting: 11,6 A (230 V)
adapter		7.8741.05010	7.8741.05010	7.8741.05010	7.8741.05010
	Siemens	3RU2926-3AA01	3RU2926-3AA01	3RU2926-3AA01	3RU2926-3AA01
fuses	-1FU...-3FU	7.3316.1	7.3316.1	7.3316.1	7.3316.1
	Ferraz	ATQR 1 1/2 (1,5 A, 600 V)	ATQR 1 1/2 (1,5 A, 600 V)	ATQR 1 1/2 (1,5 A, 600 V)	ATQR 1 1/2 (1,5 A, 600 V)
fuses	-4FU/-5FU	7.3161.00370	7.3161.00370	7.3161.00370	7.3161.00370
	Ferraz	ATDR 4 (4 A, 600 V)	ATDR 4 (4 A, 600 V)	ATDR 4 (4 A, 600 V)	ATDR 4 (4 A, 600 V)
fuses	-6FU	7.3161.00350	7.3161.00350	7.3161.00350	7.3161.00350
	Ferraz	ATDR 3 (3 A, 600 V)	ATDR 3 (3 A, 600 V)	ATDR 3 (3 A, 600 V)	ATDR 3 (3 A, 600 V)
fuse socket	-1FU...-6FU	7.3320.00060	7.3320.00060	7.3320.00060	7.3320.00060
	Wöhner	AMBUS EASYSWITCH	AMBUS EASYSWITCH	AMBUS EASYSWITCH	AMBUS EASYSWITCH
transformer	-T11	7.2221.10030	7.2221.10030	7.2221.10030	7.2221.10030
	Block	B0406056 (160 VA)	B0406056 (160 VA)	B0406056 (160 VA)	B0406056 (160 VA)
transformer	-T30	7.2292.10110	7.2292.10110	7.2292.10110	7.2292.10110
	Block	B0706107 (2,6 A)	B0706107 (2,6 A)	B0706107 (2,6 A)	B0706107 (2,6 A)
power supply	-T21	7.7605P0	7.7605P0	7.7605P0	7.7605P0
	Prodrive	PSDC24/2.5	PSDC24/2.5	PSDC24/2.5	PSDC24/2.5
connection	-W11	10 AWG black	10 AWG black	10 AWG black	10 AWG black
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
connection	-W12/-W13	12 AWG black	12 AWG black	12 AWG black	12 AWG black
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
connection	-W14	12 AWG black	12 AWG black	12 AWG black	12 AWG black
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
cables	-W19.1	7G14 AWG	7G14 AWG	7G12 AWG	7G12 AWG
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
cables	-W19.2	6x14 AWG	6x14 AWG	6x12 AWG	6x12 AWG
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
compressor control	-K20	7.9700.0	7.9700.0	7.9700.0	7.9700.0
	Prodrive	SIGMA CONTROL 2 MCSIO	SIGMA CONTROL 2 MCSIO	SIGMA CONTROL 2 MCSIO	SIGMA CONTROL 2 MCSIO
EMERGENCY STOP pushbutton	-S1	7.3217.0 / QRUV	7.3217.0 / QRUV	7.3217.0 / QRUV	7.3217.0 / QRUV
auxiliary contact	Schlegel	7.3218.0 / MHTOO	7.3218.0 / MHTOO	7.3218.0 / MHTOO	7.3218.0 / MHTOO
control cabinet	KAESER	212907.0	212907.0	212907.0	212907.0
control panel	KAESER	221359.0	221359.0	221359.0	221359.0

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4 Sht.
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USX.T-U3063.01
SC2 MCSIO
electrical component parts list
compressor series SX T
KAESER KOMPRESSOREN

Date	Drawn	Released
16.09.2019	Seubert	Büchner

	Change	Date	Name
a	47414	16.09.19	Bü/Se
b			
c			

electrical component parts list

model		SX 3 T	SX 4 T	SX 5 T	SX 7.5 T
machine power supply		**460 V ±10 %, 60 Hz**	**460 V ±10 %, 60 Hz**	**460 V ±10 %, 60 Hz**	**460 V ±10 %, 60 Hz**
motor	-M1	3 hp	4 hp	5 hp	7,5 hp
		diagram 1, Sht. 1	diagram 1, Sht. 1	diagram 1, Sht. 1	diagram 1, Sht. 1
supply terminals	-X0:U1/V1/W1	7.3149.01960	7.3149.01960	7.3149.01960	7.3149.01960
	Wieland	WKFN10D1/2/35	WKFN10D1/2/35	WKFN10D1/2/35	WKFN10D1/2/35
	Stripped length	15 mm	15 mm	15 mm	15 mm
	Handling	fig. 2, Sht. 13	fig. 2, Sht. 13	fig. 2, Sht. 13	fig. 2, Sht. 13
	-X0:GRD	7.3149.01980	7.3149.01980	7.3149.01980	7.3149.01980
	Wieland	WKFN10D1/2/SL/35	WKFN10D1/2/SL/35	WKFN10D1/2/SL/35	WKFN10D1/2/SL/35
	Stripped length	15 mm	15 mm	15 mm	15 mm
	Handling	fig. 2, Sht. 13	fig. 2, Sht. 13	fig. 2, Sht. 13	fig. 2, Sht. 13
supply	connection	fig. 10, Sht. 13	fig. 10, Sht. 13	fig. 10, Sht. 13	fig. 10, Sht. 13
terminal strip	-X0	7.6836.00260 Wieland	7.6836.00260 Wieland	7.6836.00260 Wieland	7.6836.00260 Wieland
terminal strip	-X11/X31	7.6836.00700 Wieland	7.6836.00700 Wieland	7.6836.00700 Wieland	7.6836.00700 Wieland
	Handling	fig. 1, Sht. 13	fig. 1, Sht. 13	fig. 1, Sht. 13	fig. 1, Sht. 13
contactor	-Q1	7.8740.00350	7.8740.00360	7.8740.00360	7.8740.00360
		3RT2024-1AK60	3RT2025-1AK60	3RT2025-1AK60	3RT2025-1AK60
auxiliary switch		7.8740.05010	7.8740.05010	7.8740.05010	7.8740.05010
		3RH2911-1HA11	3RH2911-1HA11	3RH2911-1HA11	3RH2911-1HA11
interference suppressor		7.8740.05140	7.8740.05140	7.8740.05140	7.8740.05140
	Siemens	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00
contactor	-Q2	7.8740.00350	7.8740.00360	7.8740.00360	7.8740.00360
		3RT2024-1AK60	3RT2025-1AK60	3RT2025-1AK60	3RT2025-1AK60
interference suppressor		7.8740.05140	7.8740.05140	7.8740.05140	7.8740.05140
	Siemens	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00
contactor	-Q3	7.8740.00340	7.8740.00340	7.8740.00340	7.8740.00350
		3RT2023-1AK60	3RT2023-1AK60	3RT2023-1AK60	3RT2024-1AK60
interference suppressor		7.8740.05140	7.8740.05140	7.8740.05140	7.8740.05140
	Siemens	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00	3RT2926-1CC00
contactor	-Q30	7.8740.03010	7.8740.03010	7.8740.03010	7.8740.03010
	Siemens	3RT2016-1JB41	3RT2016-1JB41	3RT2016-1JB41	3RT2016-1JB41
overload relay	-B25	7.8741.00120	7.8741.00060	7.8741.00060	7.8741.00060
		3RB3026-1PB0 (1-4 A)	3RB3026-1SB0 (3-12 A)	3RB3026-1SB0 (3-12 A)	3RB3026-1SB0 (3-12 A)
		setting: 2,1 A	setting: 2,9 A	setting: 4,2 A	setting: 5,2 A
		NEC 430.32(C) incremental	NEC 430.32(C) incremental	NEC 430.32(C) incremental	NEC 430.32(C) incremental
		setting: 2,4 A	setting: 3,3 A	setting: 4,7 A	setting: 5,8 A
adapter		7.8741.05010	7.8741.05010	7.8741.05010	7.8741.05010
	Siemens	3RU2926-3AA01	3RU2926-3AA01	3RU2926-3AA01	3RU2926-3AA01
fuses	-1FU...-3FU	7.3316.1	7.3316.1	7.3316.1	7.3316.1
	Ferraz	ATQR 1 1/2 (1,5 A, 600 V)	ATQR 1 1/2 (1,5 A, 600 V)	ATQR 1 1/2 (1,5 A, 600 V)	ATQR 1 1/2 (1,5 A, 600 V)
fuses	-4FU/-5FU	7.3161.00370	7.3161.00370	7.3161.00370	7.3161.00370
	Ferraz	ATDR 4 (4 A, 600 V)	ATDR 4 (4 A, 600 V)	ATDR 4 (4 A, 600 V)	ATDR 4 (4 A, 600 V)
fuses	-6FU	7.3161.00350	7.3161.00350	7.3161.00350	7.3161.00350
	Ferraz	ATDR 3 (3 A, 600 V)	ATDR 3 (3 A, 600 V)	ATDR 3 (3 A, 600 V)	ATDR 3 (3 A, 600 V)
fuse socket	-1FU...-6FU	7.3320.00060	7.3320.00060	7.3320.00060	7.3320.00060
	Wöhner	AMBUS EASYSWITCH	AMBUS EASYSWITCH	AMBUS EASYSWITCH	AMBUS EASYSWITCH
transformer	-T11	7.2221.10030	7.2221.10030	7.2221.10030	7.2221.10030
	Block	B0406056 (160 VA)	B0406056 (160 VA)	B0406056 (160 VA)	B0406056 (160 VA)
transformer	-T30	7.2292.10110	7.2292.10110	7.2292.10110	7.2292.10110
	Block	B0706107 (2,6 A)	B0706107 (2,6 A)	B0706107 (2,6 A)	B0706107 (2,6 A)
power supply	-T21	7.7605P0	7.7605P0	7.7605P0	7.7605P0
	Prodrive	PSDC24/2.5	PSDC24/2.5	PSDC24/2.5	PSDC24/2.5
connection	-W11	10 AWG black	10 AWG black	10 AWG black	10 AWG black
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
connection	-W12	12 AWG black	12 AWG black	12 AWG black	12 AWG black
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
connection	-W13	12 AWG black	12 AWG black	12 AWG black	12 AWG black
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
connection	-W14	12 AWG black	12 AWG black	12 AWG black	12 AWG black
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
cables	-W19.1	7G14 AWG	7G14 AWG	7G12 AWG	7G12 AWG
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
cables	-W19.2	6x14 AWG	6x14 AWG	6x12 AWG	6x12 AWG
		600 V, 90°C	600 V, 90°C	600 V, 90°C	600 V, 90°C
compressor control	-K20	7.9700.0	7.9700.0	7.9700.0	7.9700.0
	Prodrive	SIGMA CONTROL 2 MCSIO	SIGMA CONTROL 2 MCSIO	SIGMA CONTROL 2 MCSIO	SIGMA CONTROL 2 MCSIO
EMERGENCY STOP pushbutton	-S1	7.3217.0 / QRUV	7.3217.0 / QRUV	7.3217.0 / QRUV	7.3217.0 / QRUV
auxiliary contact	Schlegel	7.3218.0 / MHTOO	7.3218.0 / MHTOO	7.3218.0 / MHTOO	7.3218.0 / MHTOO
control cabinet	KAESER	212907.0	212907.0	212907.0	212907.0
control panel	KAESER	221359.0	221359.0	221359.0	221359.0

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SC2 MCSIO
electrical component parts list
compressor series SX T
KAESER KOMPRESSOREN

Date	16.09.2019
Drawn	Seubert
Released	Büchner

Change	Date	Name
a 47414	16.09.19	Bü/Se
b		
c		

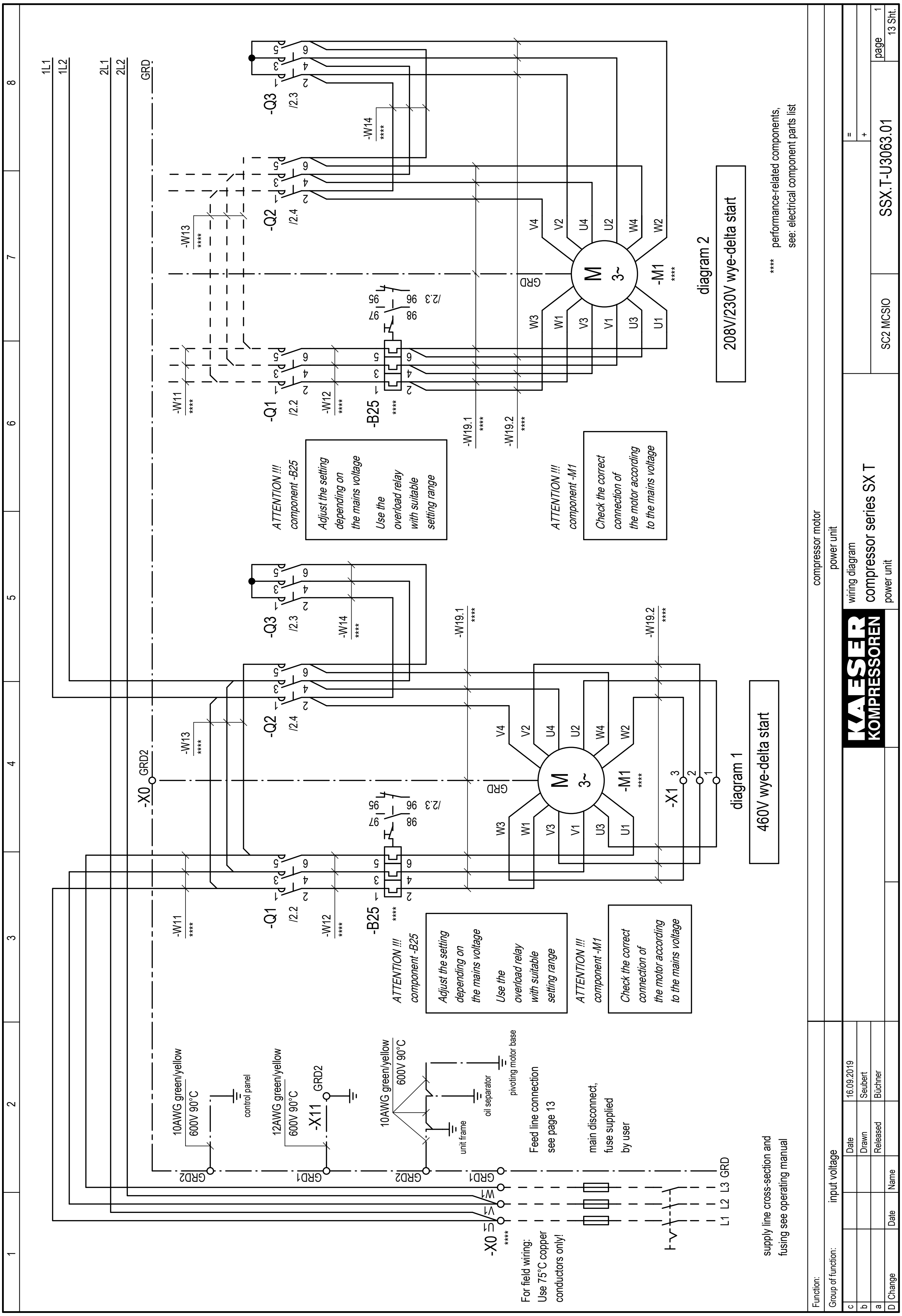
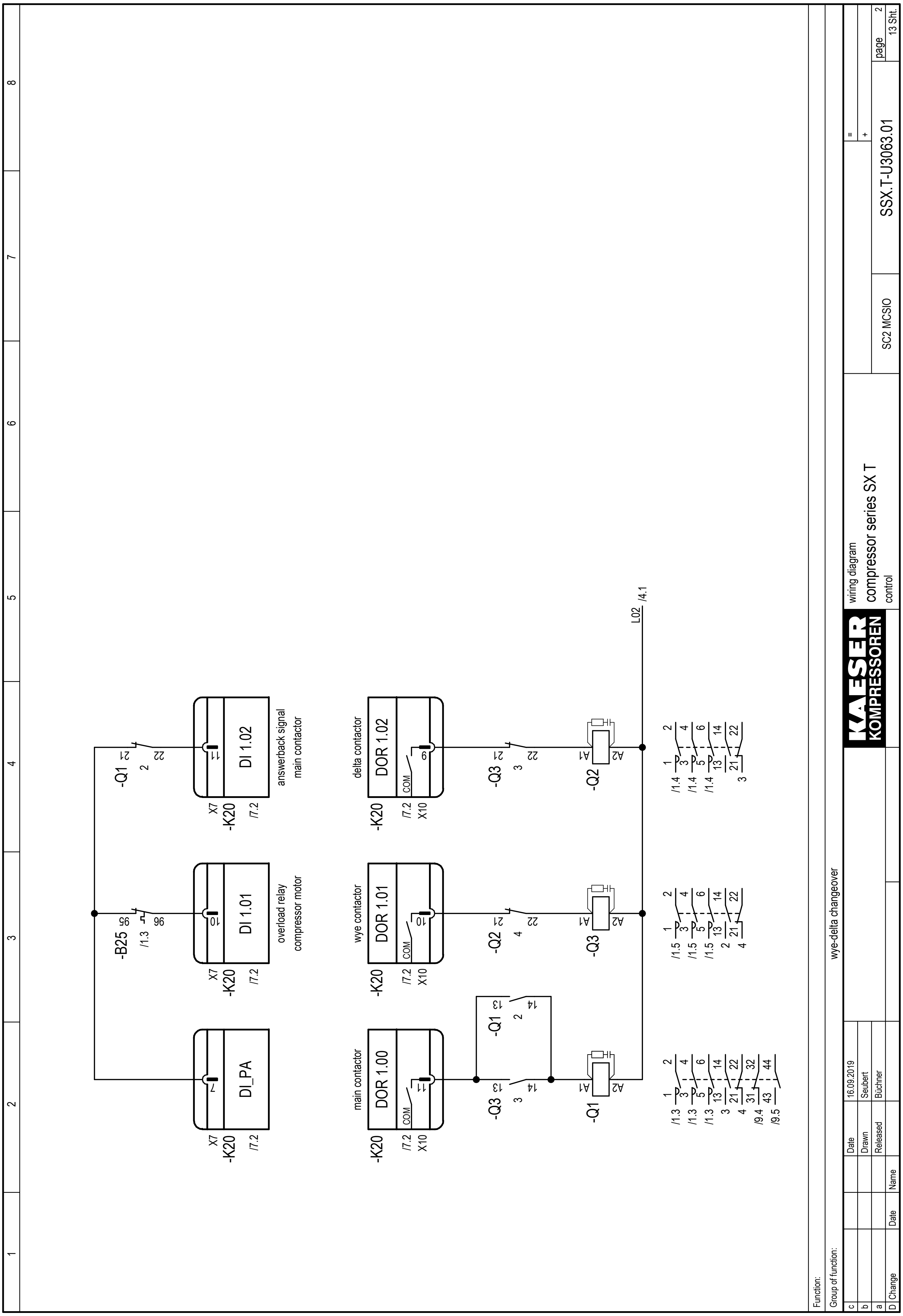
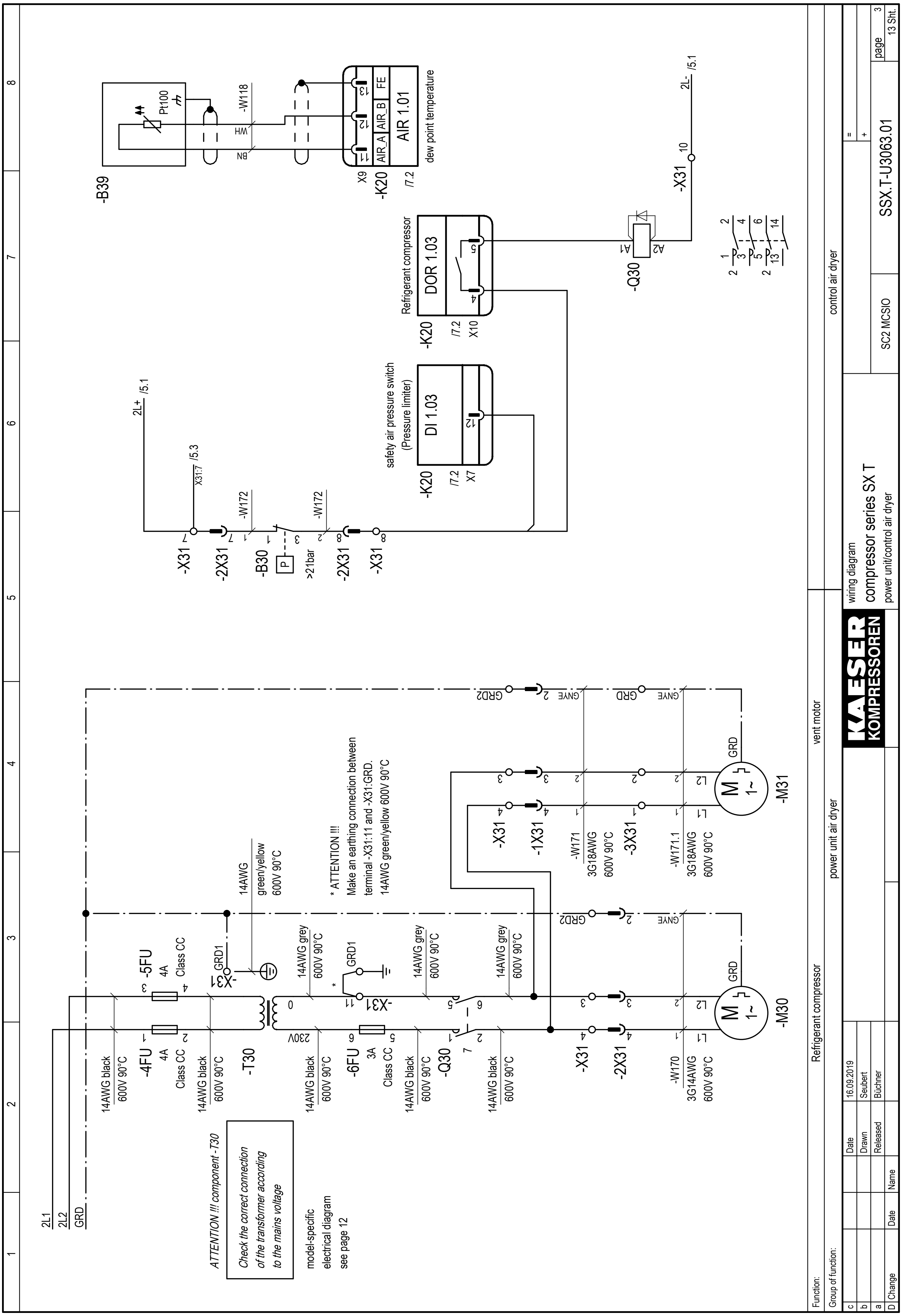
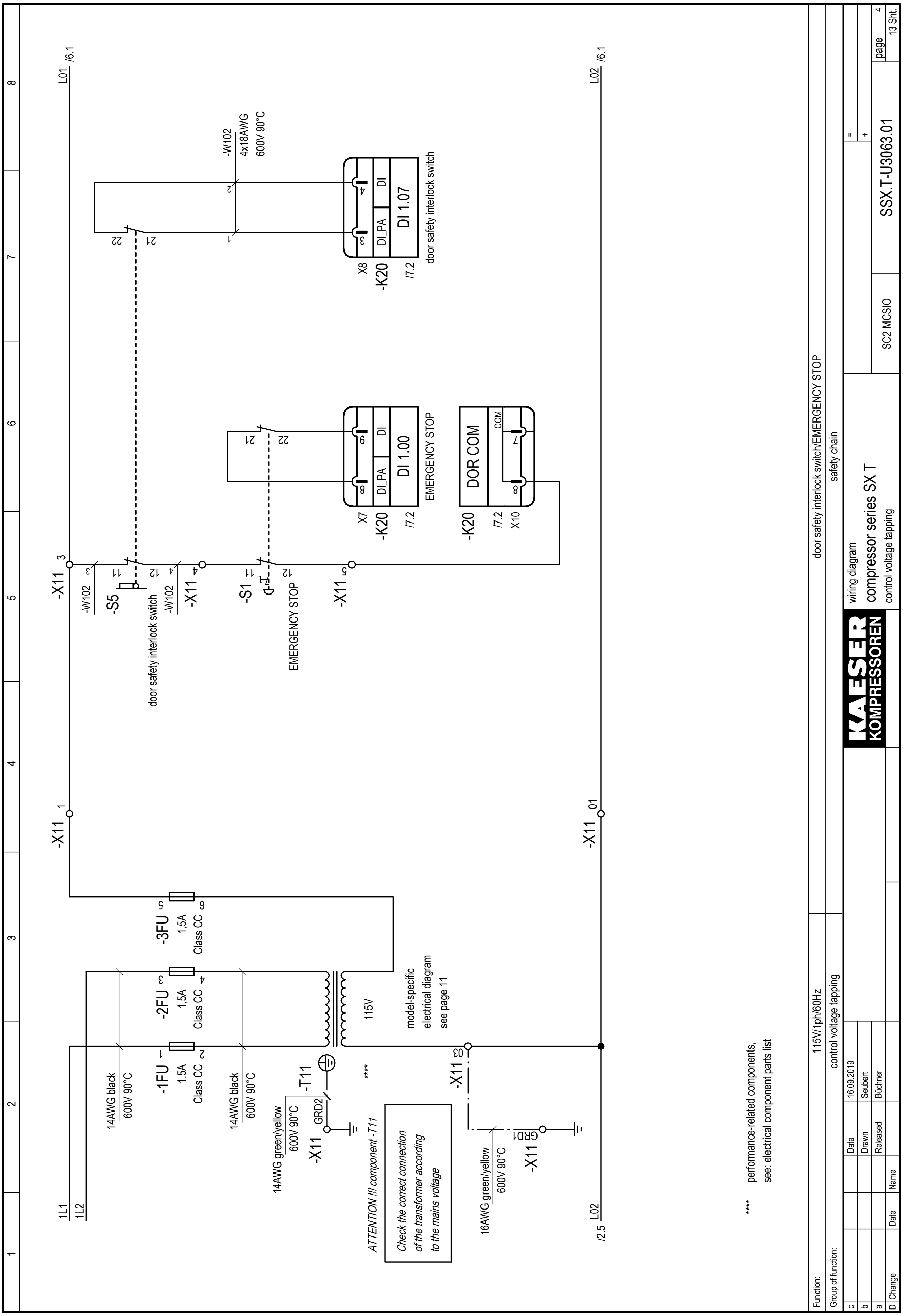


diagram 1
460V wye-delta start
diagram 2
208V/230V wye-delta start
ATTENTION !!!
component -B25
Adjust the setting depending on the mains voltage
Use the overload relay with suitable setting range
ATTENTION !!!
component -M1
Check the correct connection of the motor according to the mains voltage
For field wiring:
Use 75°C copper conductors only!
supply line cross-section and fusing see operating manual
main disconnect, fuse supplied by user
Feed line connection see page 13
10AWG green/yellow
600V 90°C
12AWG green/yellow
600V 90°C
control panel
oil separator
unit frame
pivoting motor base
**** performance-related components, see: electrical component parts list
compressor motor
power unit
input voltage
wiring diagram
compressor series SX T
power unit
SC2 MCSIO
SSX.T-U3063.01
16.09.2019
Seubert
Büchner
13 Sht.


wiring diagram
compressor series SX T
control
SSX.T-U3063.01
SC2 MCSIO
wye-delta changeover
Function:
Group of function:
Date
16.09.2019
Drawn
Seubert
Released
Büchner
Change
Name
page
2
13 Sht.
main contactor
DOR 1.00
wye contactor
DOR 1.01
delta contactor
DOR 1.02
DI_PA
DI 1.01
DI 1.02
overload relay
compressor motor
answerback signal
main contactor
-K20
-Q1
-Q2
-Q3
-B25
L02
/4.1


wiring diagram
compressor series SX T
power unit/control air dryer
SSX.T-U3063.01
SC2 MCSIO
page 3
13 Sht.
16.09.2019
Seubert
Büchner
Refrigerant compressor
power unit air dryer
vent motor
control air dryer
ATTENTION !!! component -T30
Check the correct connection of the transformer according to the mains voltage
model-specific electrical diagram see page 12
* ATTENTION !!!
Make an earthing connection between terminal -X31:11 and -X31:GRD.
14AWG green/yellow 600V 90°C
safety air pressure switch (Pressure limiter)
DI 1.03
Refrigerant compressor
DOR 1.03
AIR 1.01
dew point temperature
-B39
Pt100
-K20
-Q30
-T30
-M30
-M31
-B30
>21bar


door safety interlock switch
-S5
EMERGENCY STOP
-S1
-X11
-W102
4x18AWG
600V 90°C
-K20
DI 1.07
DI_PA
DI
door safety interlock switch
DI 1.00
EMERGENCY STOP
DOR COM
COM
-1FU
-2FU
-3FU
1,5A
Class CC
14AWG black
600V 90°C
14AWG green/yellow
16AWG green/yellow
-T11
GRD1
GRD2
115V
model-specific
electrical diagram
see page 11
ATTENTION !!! component -T11
Check the correct connection
of the transformer according
to the mains voltage
**** performance-related components,
see: electrical component parts list
Function:
115V/1ph/60Hz
door safety interlock switch/EMERGENCY STOP
Group of function:
control voltage tapping
safety chain
Date 16.09.2019
Drawn Seubert
Released Büchner
Change
Name
wiring diagram
compressor series SX T
control voltage tapping
SC2 MCSIO
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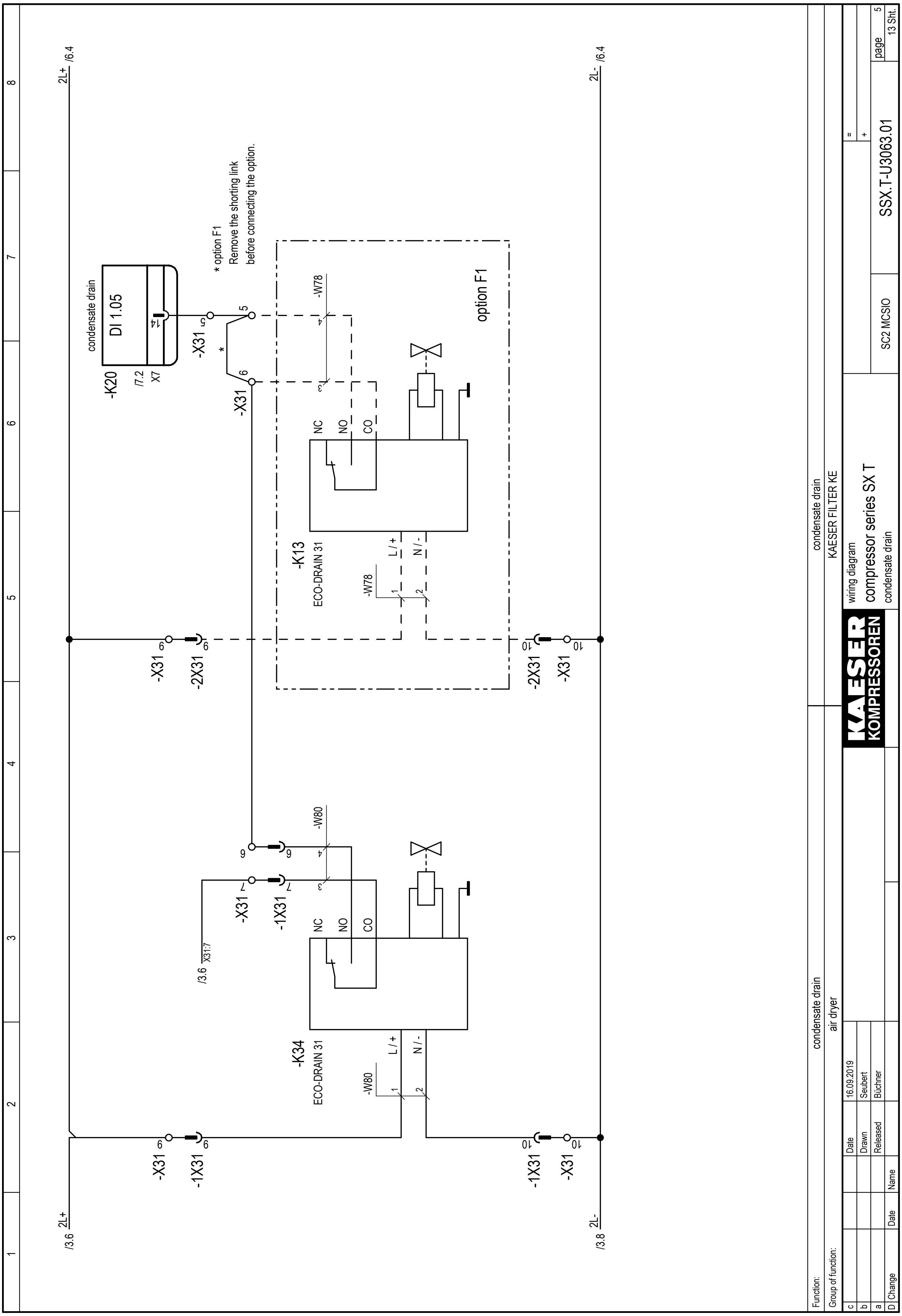
page 4
13 Sht.



condensate drain
DI 1.05
-K20
/7.2
X7
14
-X31
5
6
* option F1
Remove the shorting link
before connecting the option.
option F1
-W78
NC
NO
CO
-K13
ECO-DRAIN 31
L / +
N / -
-K34
ECO-DRAIN 31
-W80
-X31
-2X31
-1X31
X31:7
/3.6
2L+
2L-
/6.4
/3.8
Function:
Group of function:
condensate drain
KAESER FILTER KE
condensate drain
air dryer
wiring diagram
compressor series SX T
condensate drain
SC2 MCSIO
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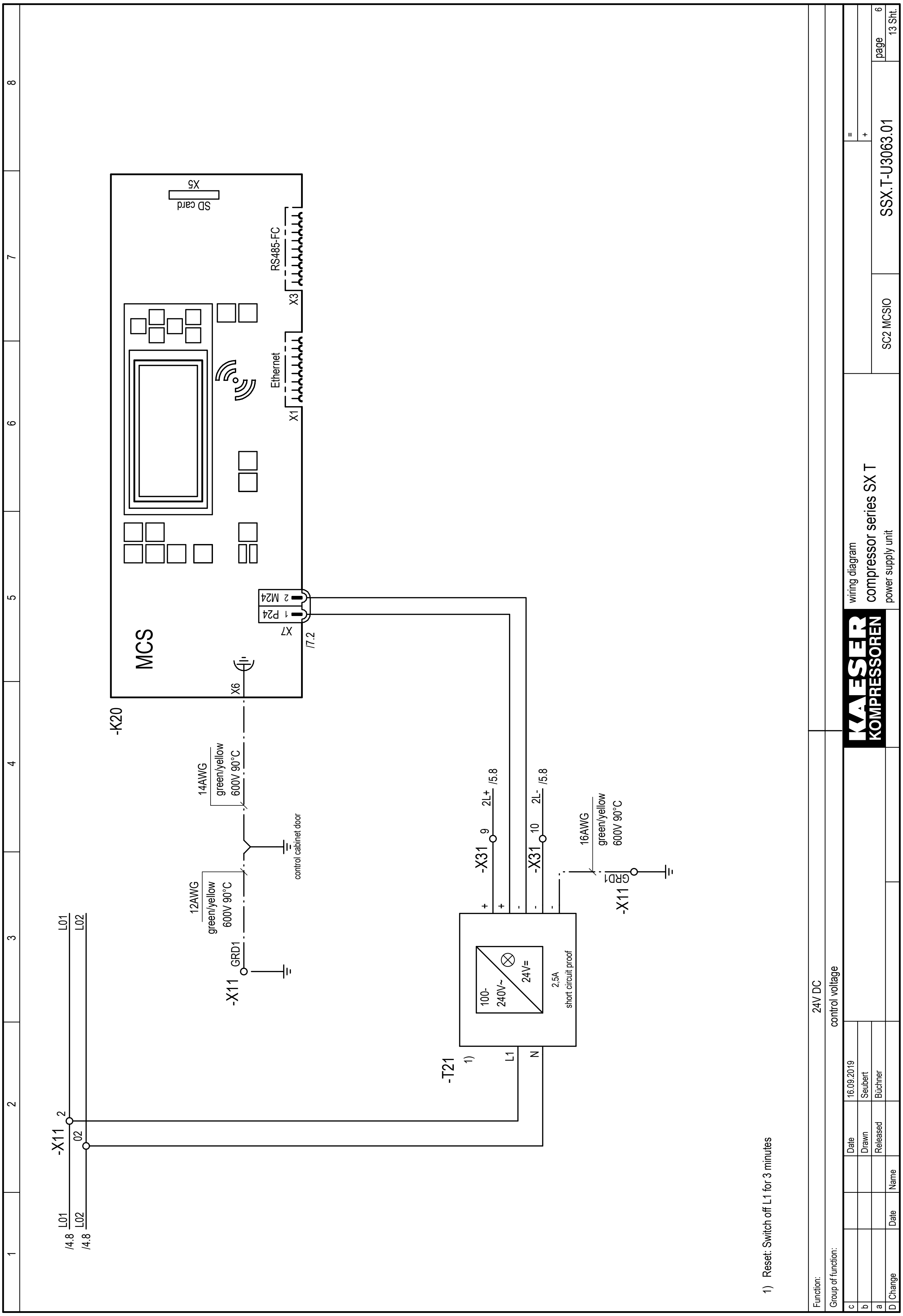
Date 16.09.2019
Drawn Seubert
Released Büchner
Change Date Name


MCS
-K20
SD card
X5
RS485-FC
X3
Ethernet
X1
X7
1 P24
2 M24
/7.2
X6
14AWG
green/yellow
600V 90°C
control cabinet door
12AWG
green/yellow
600V 90°C
-X11
GRD1
-X31
9
2L+
/5.8
-X31
10
2L-
/5.8
16AWG
green/yellow
600V 90°C
-X11
GRD1
100-
240V~
24V=
2,5A
short circuit proof
-T21
1)
L1
N
-X11
2
02
L01
L02
/4.8
/4.8
1) Reset: Switch off L1 for 3 minutes
Function:
24V DC
Group of function:
control voltage
Date
16.09.2019
Drawn
Seubert
Released
Büchner
Change
Date
Name
wiring diagram
compressor series SX T
power supply unit
SC2 MCSIO
SSX.T-U3063.01
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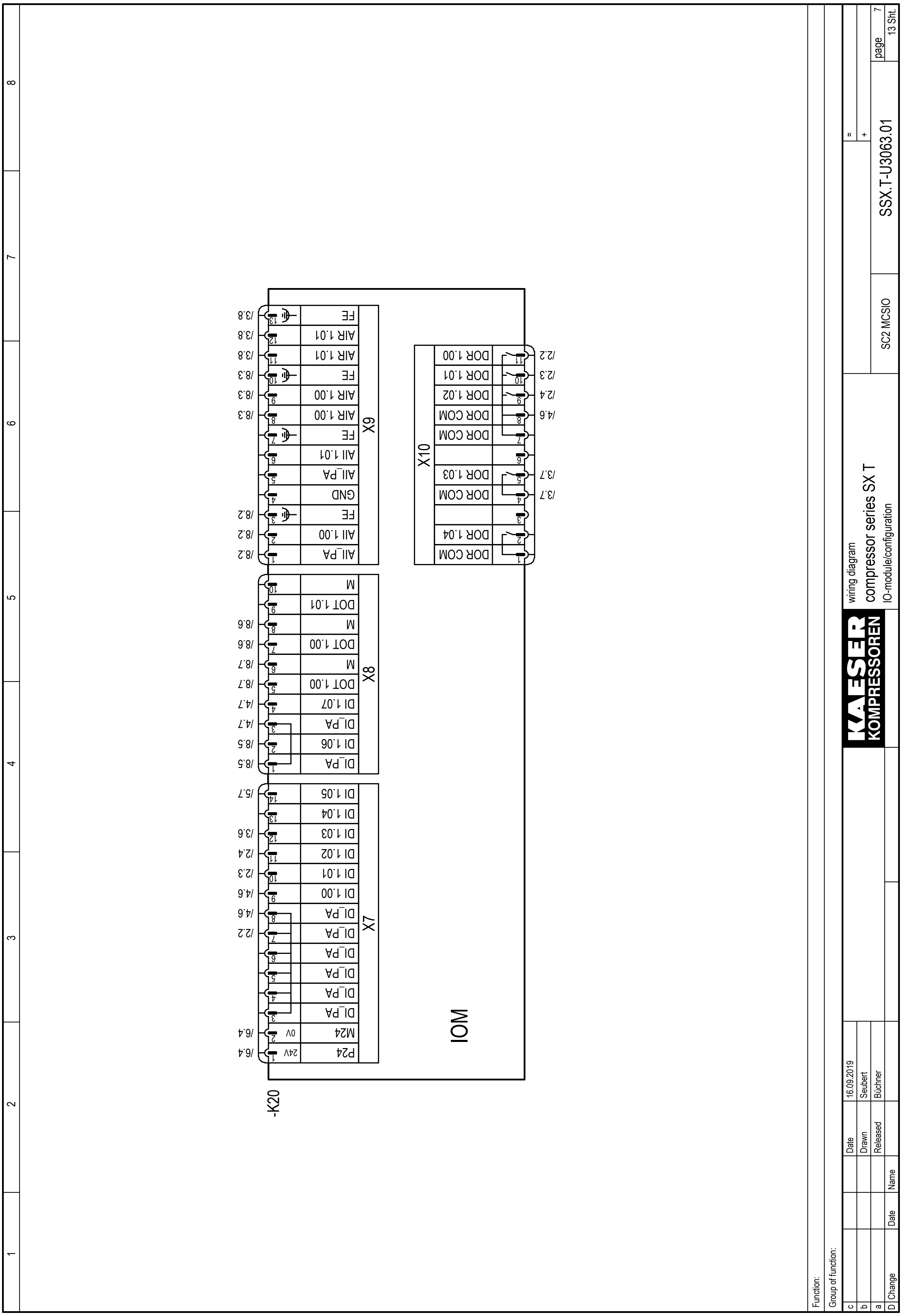

-K20
IOM
X7
X8
X9
X10
P24
M24
DI_PA
DI 1.00
DI 1.01
DI 1.02
DI 1.03
DI 1.04
DI 1.05
DI 1.06
DI 1.07
DOT 1.00
DOT 1.01
M
AII_PA
AII 1.00
AII 1.01
FE
GND
AIR 1.00
AIR 1.01
DOR COM
DOR 1.00
DOR 1.01
DOR 1.02
DOR 1.03
DOR 1.04
wiring diagram
compressor series SX T
IO-module/configuration
SC2 MCSIO
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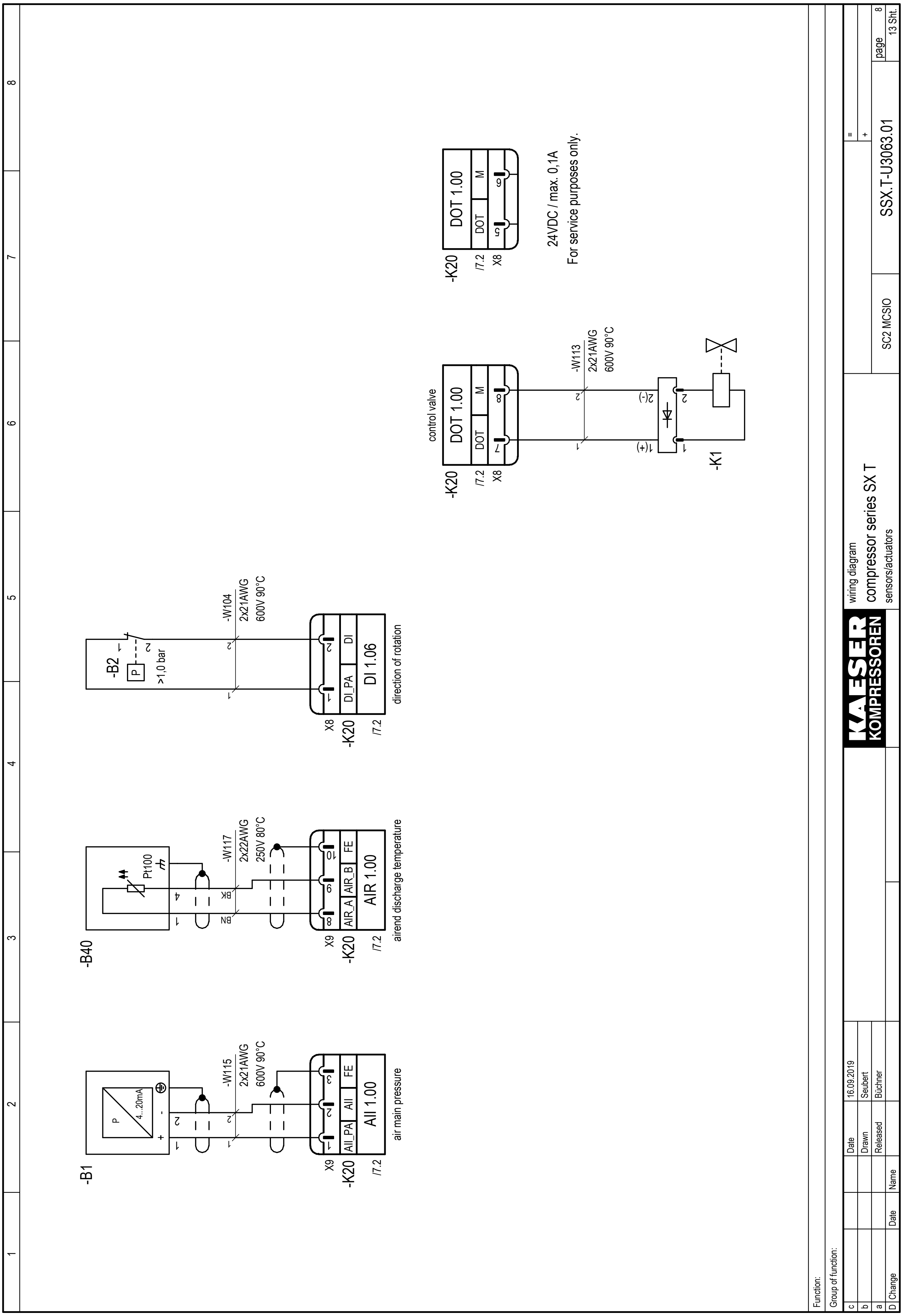
13 Sht.
Date 16.09.2019
Drawn Seubert
Released Büchner
Function:
Group of function:
Change
Date
Name



-B1
P
4...20mA
-W115
2x21AWG
600V 90°C
X9
-K20
/7.2
AII_PA
AII
FE
AII 1.00
air main pressure
-B40
Pt100
BN
BK
-W117
2x22AWG
250V 80°C
AIR_A
AIR_B
AIR 1.00
airend discharge temperature
-B2
P
>1,0 bar
-W104
2x21AWG
600V 90°C
X8
DI_PA
DI
DI 1.06
direction of rotation
control valve
DOT
M
DOT 1.00
-W113
2x21AWG
600V 90°C
1(+)
2(-)
-K1
24VDC / max. 0,1A
For service purposes only.
Function:
Group of function:
Change
Date
Name
Date 16.09.2019
Drawn Seubert
Released Büchner
KAESER KOMPRESSOREN
wiring diagram
compressor series SX T
sensors/actuators
SC2 MCSIO
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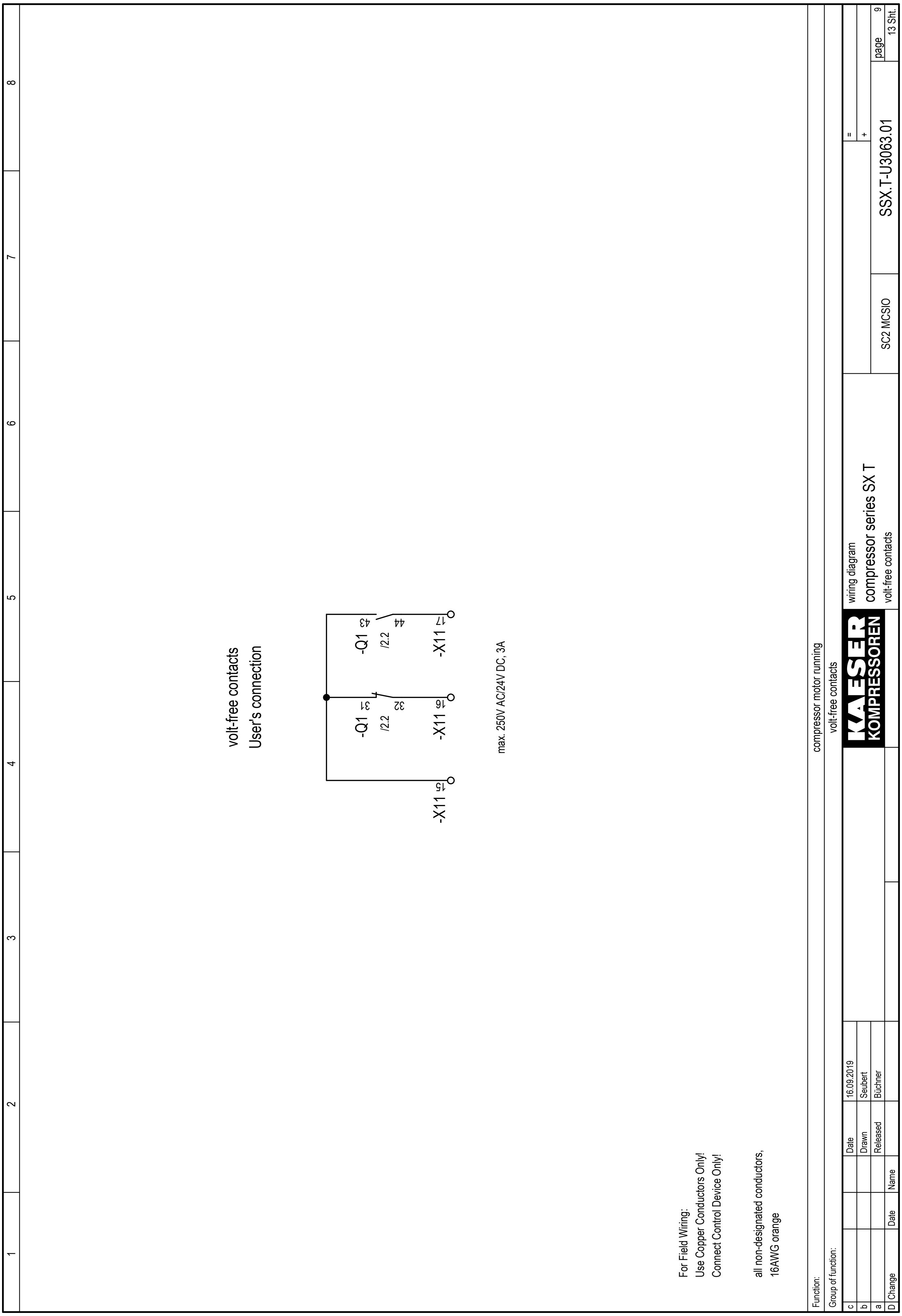
volt-free contacts
User's connection
-Q1
43
44
/2.2
-Q1
31
32
/2.2
-X11
15
-X11
16
-X11
17
max. 250V AC/24V DC, 3A
For Field Wiring:
Use Copper Conductors Only!
Connect Control Device Only!
all non-designated conductors,
16AWG orange
Function:
compressor motor running
Group of function:
volt-free contacts
Date
16.09.2019
Drawn
Seubert
Released
Büchner
Change
Date
Name
KAESER KOMPRESSOREN
wiring diagram
compressor series SX T
volt-free contacts
SC2 MCSIO
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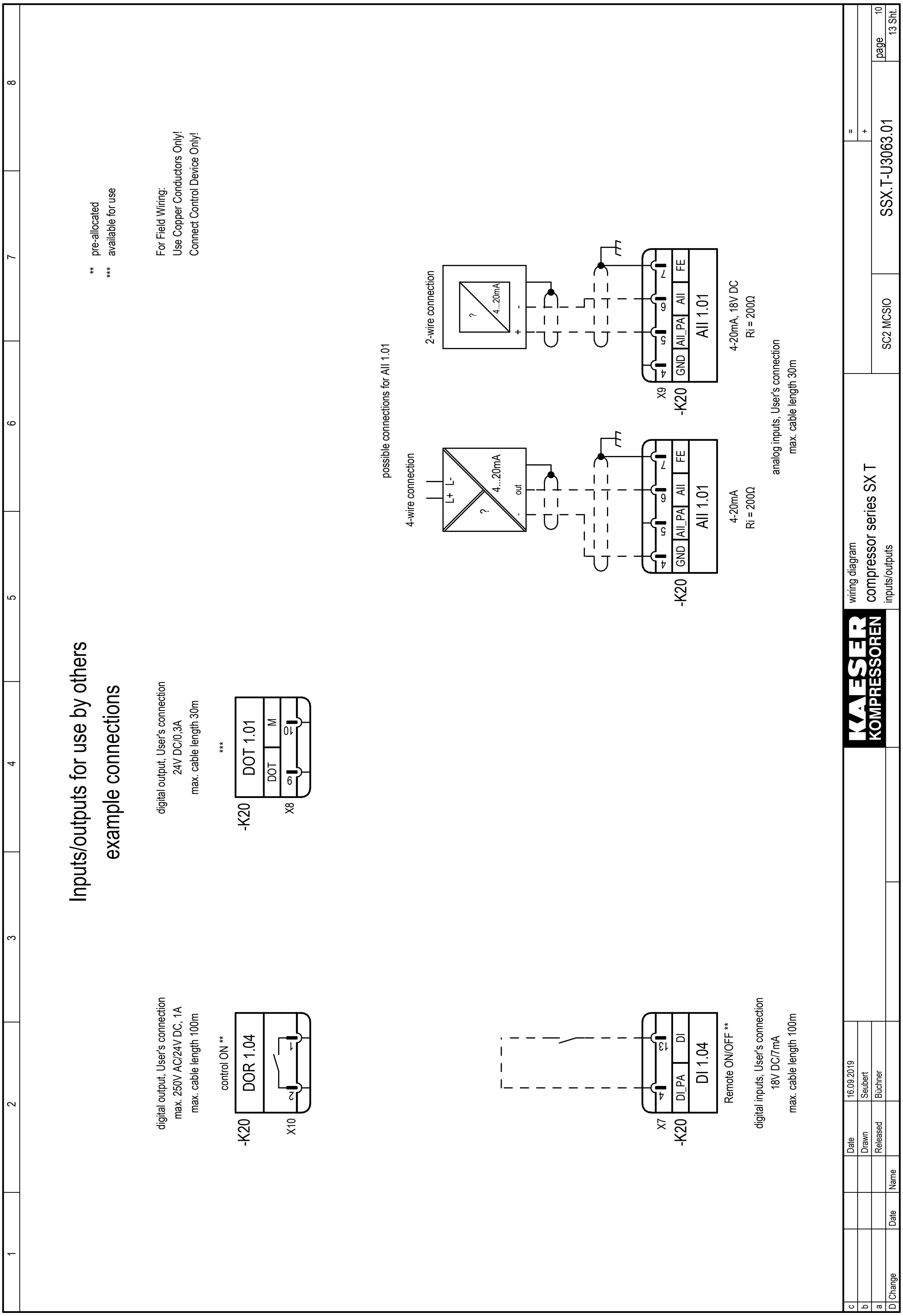
Inputs/outputs for use by others
example connections
** pre-allocated
*** available for use
For Field Wiring:
Use Copper Conductors Only!
Connect Control Device Only!
digital output, User's connection
max. 250V AC/24V DC, 1A
max. cable length 100m
control ON **
-K20
X10
DOR 1.04
digital output, User's connection
24V DC/0,3A
max. cable length 30m
***
DOT 1.01
DOT
M
X8
possible connections for AII 1.01
4-wire connection
2-wire connection
L+ L-
4...20mA
out
4-20mA
Ri = 200Ω
4-20mA, 18V DC
Ri = 200Ω
GND
AII_PA
AII
FE
AII 1.01
X9
analog inputs, User's connection
max. cable length 30m
Remote ON/OFF **
DI 1.04
DI_PA
DI
X7
digital inputs, User's connection
18V DC/7mA
max. cable length 100m
wiring diagram
compressor series SX T
inputs/outputs
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SSX.T-U3063.01
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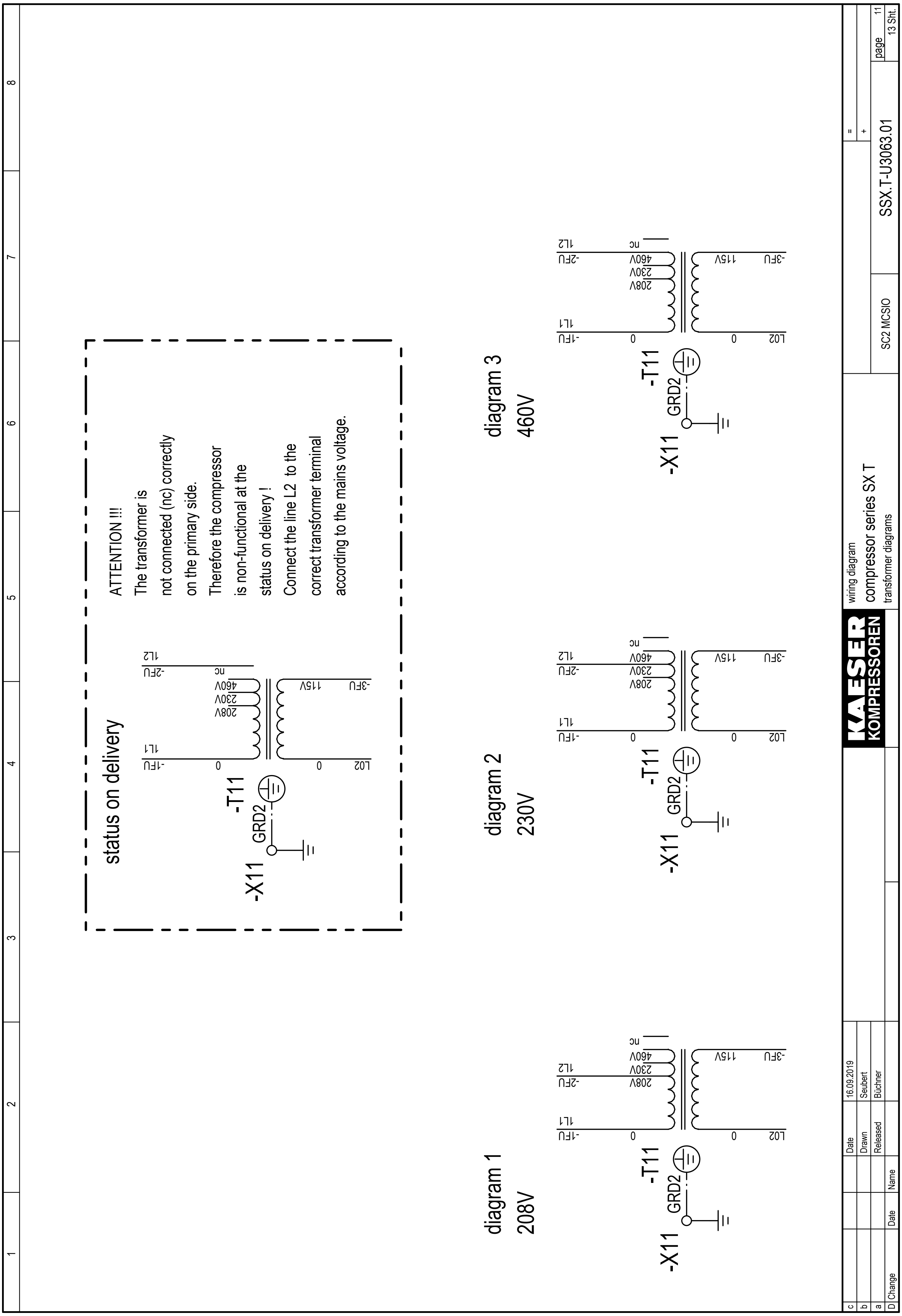
Date 16.09.2019
Drawn Seubert
Released Büchner
Change Date Name


status on delivery
-X11
GRD2
-T11
-1FU
1L1
-2FU
1L2
0
208V
230V
460V
nc
115V
-3FU
L02
ATTENTION !!!
The transformer is
not connected (nc) correctly
on the primary side.
Therefore the compressor
is non-functional at the
status on delivery !
Connect the line L2 to the
correct transformer terminal
according to the mains voltage.
diagram 1
208V
diagram 2
230V
diagram 3
460V
wiring diagram
compressor series SX T
transformer diagrams
SC2 MCSIO
SSX.T-U3063.01
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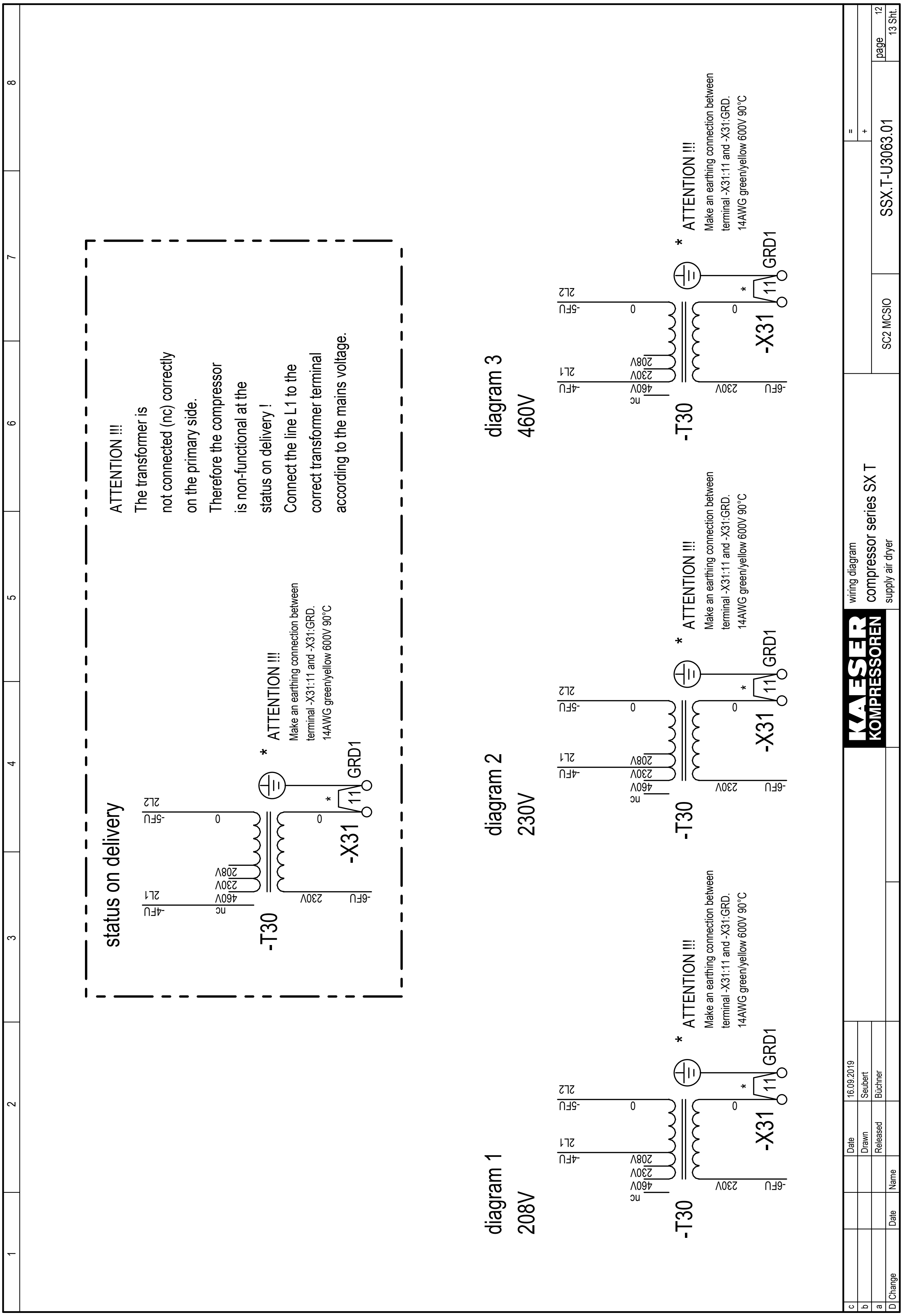
Date 16.09.2019
Drawn Seubert
Released Büchner
Change
Date
Name


status on delivery
ATTENTION !!!
The transformer is
not connected (nc) correctly
on the primary side.
Therefore the compressor
is non-functional at the
status on delivery !
Connect the line L1 to the
correct transformer terminal
according to the mains voltage.
-T30
-X31
GRD1
11
-4FU
2L1
-5FU
2L2
-6FU
nc
460V
230V
208V
0
* ATTENTION !!!
Make an earthing connection between
terminal -X31:11 and -X31:GRD.
14AWG green/yellow 600V 90°C
diagram 1
208V
diagram 2
230V
diagram 3
460V
wiring diagram
compressor series SX T
supply air dryer
SC2 MCSIO
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Released Büchner
Change
Date
Name

fig. 10: Feed line connection

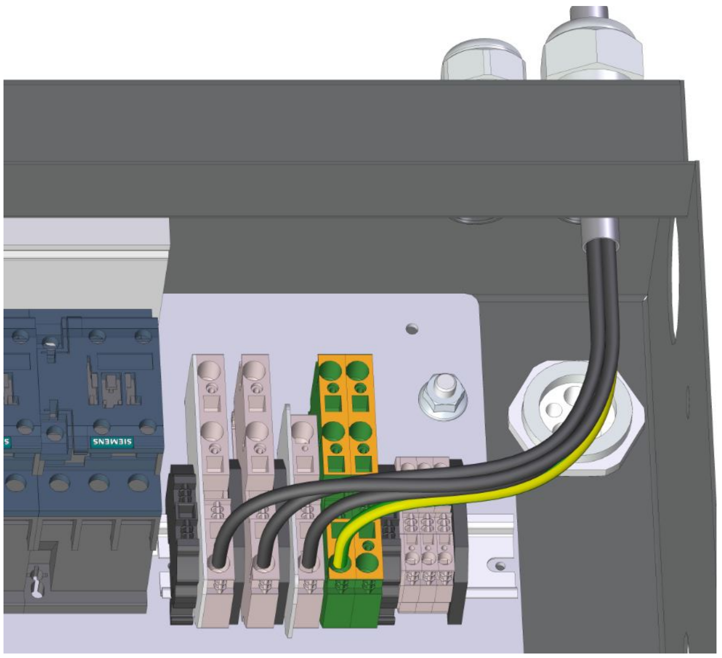


fig. 1: Handling: Control line terminal

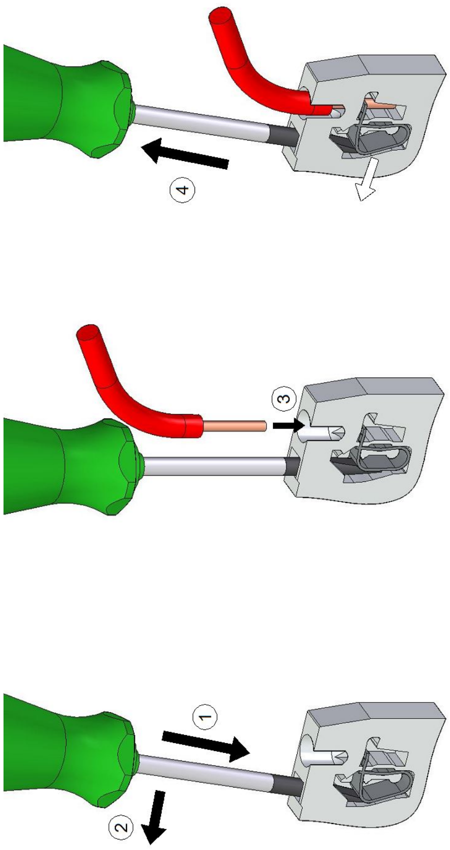
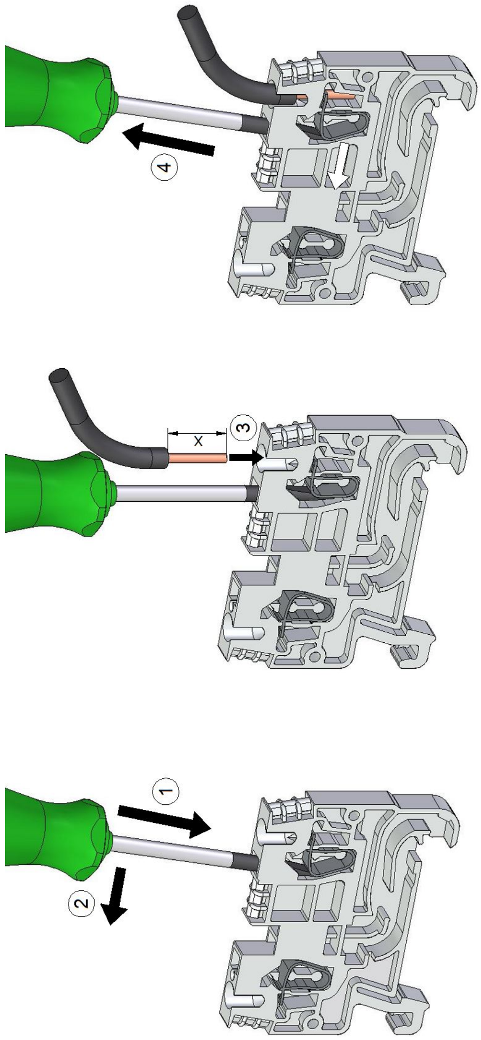


fig. 2: Handling: Supply terminal



Date	16.09.2019
Drawn	Seubert
Released	Büchner
Change / Date / Name	

KAESER KOMPRESSOREN

wiring diagram
compressor series SX T
Handling: Terminals / Feed line connection

SC2 MCSIO

SSX.T-U3063.01

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terminal strip: -X0
total 9 terminals

terminal-no.	cable-no. (inside)	destination inside: name of device	destination inside: connection	location	jumper wire	link	destination external: name of device	destination external: connection		-W19.1 ****	-W19.2 ****	-s.b.c. *
U1		-Q1	1	/1.1		●	supply					X
U1		-4FU	1			●						
V1		-Q1	3	/1.1		●	supply					X
V1		-5FU	3			●						
W1		-Q1	5	/1.2			supply					X
GRD1			.	/1.2		●	supply					X
GRD1		-X11	GRD2	/1.2		●						
GRD2		control panel		/1.4		●	-M1	GRD		X		
GRD2		unit frame		/1.2		●						

terminal strip: -X1
total 3 terminals

terminal-no.	-W19.2 **** (inside)	destination inside: name of device	destination inside: connection	location	jumper wire	link	destination external: name of device	destination external: connection		-W19.1 ****	-W19.2 ****	-s.b.c. *
1	X	-M1	U3	/1.4			-M1	U2	A)		X	
2	X	-M1	V3	/1.4			-M1	V2	A)		X	
3	X	-M1	W3	/1.4			-M1	W2	A)		X	

* supply line cross-section see operating manual

s.b.c. = supplied by customer

A) machine power supply 460V

**** performance-related components, see: electrical component parts list

c				Date	16.09.2019
b				Drawn	Seubert
a				Released	Büchner
H	Change	Date	Name		

KAESER KOMPRESSOREN

terminal connection
compressor series SX T
terminal strip -X0,-X1

SC2 MCSIO

KSX.T-U3063.01

=
+

page 1
3 Sht.

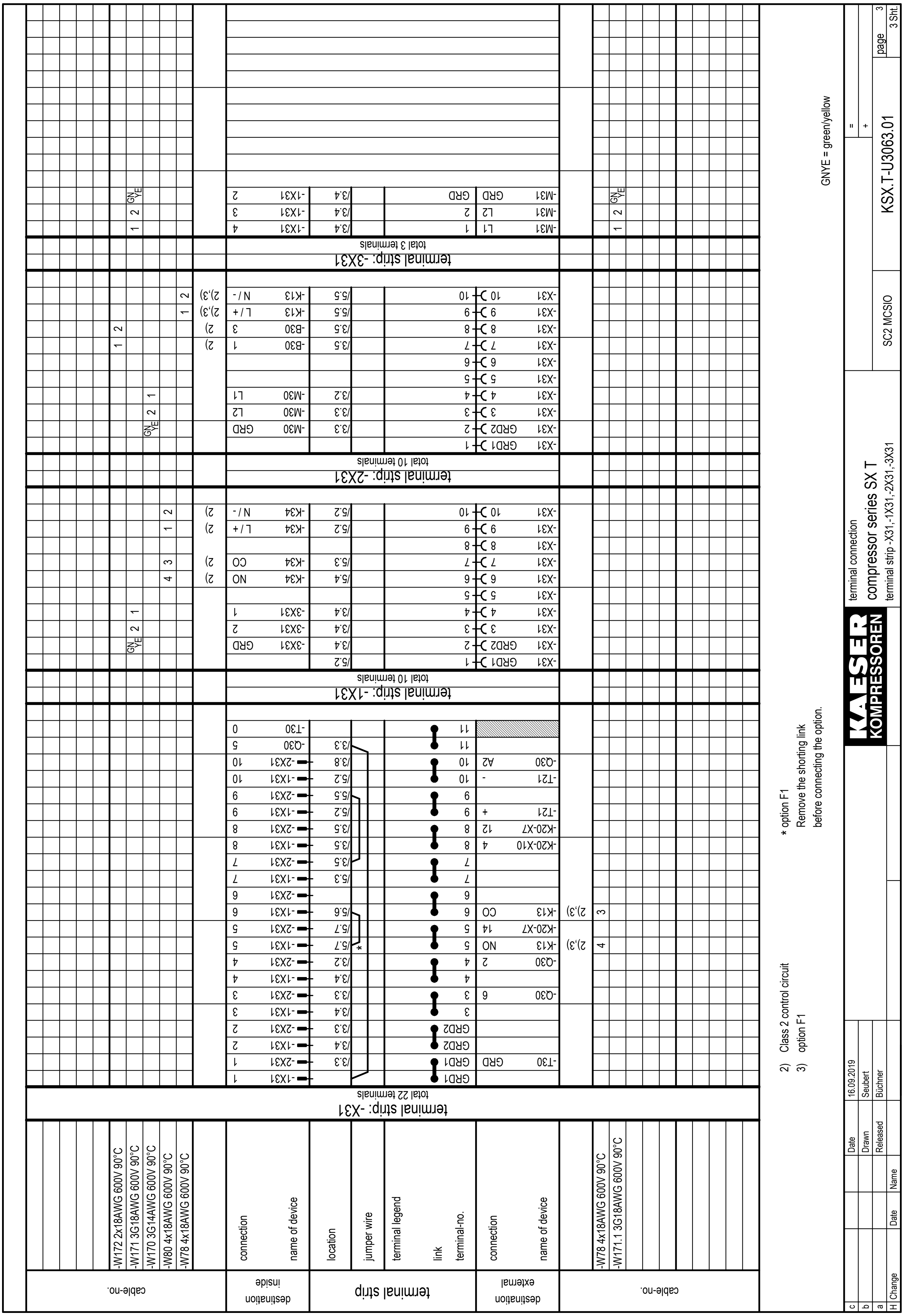
volt-free contacts, User's connection
1) compressor motor running

terminal strip: -X11
total 16 terminals

Cable-no.: -W102 4x18AWG 600V 90°C

cable-no. (-W102)	destination external: name of device	destination external: connection	location	jumper wire	link	terminal-no.	destination inside: connection	destination inside: name of device	cable-no.
	-3FU	5	/4.4			1			
	-Q1	A2	/4.4			01			
	-T21	L1	/6.2			2			
	-T21	N	/6.2			02			
3	-S5	11	/4.5			3			
	-T11	0	/4.2			03			
4	-S5	12	/4.5			4	11	-S1	
	-K20-X10	8	/4.5			5	12	-S1	
	control cabinet door		/4.2			GRD1		.	
	-T21	-	/6.3			GRD1			
	-T11	GRD	/4.2			GRD2			
	-X0	GRD1	/1.2			GRD2			
	-Q1	43	/9.4			15			1)
	-Q1	32	/9.4			16			1)
	-Q1	44	/9.5			17			1)
						18			

c				Date	16.09.2019		KAESER KOMPRESSOREN	terminal connection		=
b				Drawn	Seubert			compressor series SX T		+
a				Released	Büchner			terminal strip -X11	SC2 MCSIO	KSX.T-U3063.01
H	Change	Date	Name							page 2 / 3 Sht.


terminal strip: -X31
total 22 terminals
terminal strip: -1X31
total 10 terminals
terminal strip: -2X31
total 10 terminals
terminal strip: -3X31
total 3 terminals
2) Class 2 control circuit
3) option F1
* option F1
Remove the shorting link
before connecting the option.
GNYE = green/yellow
terminal connection
compressor series SX T
terminal strip -X31,-1X31,-2X31,-3X31
SC2 MCSIO
KSX.T-U3063.01
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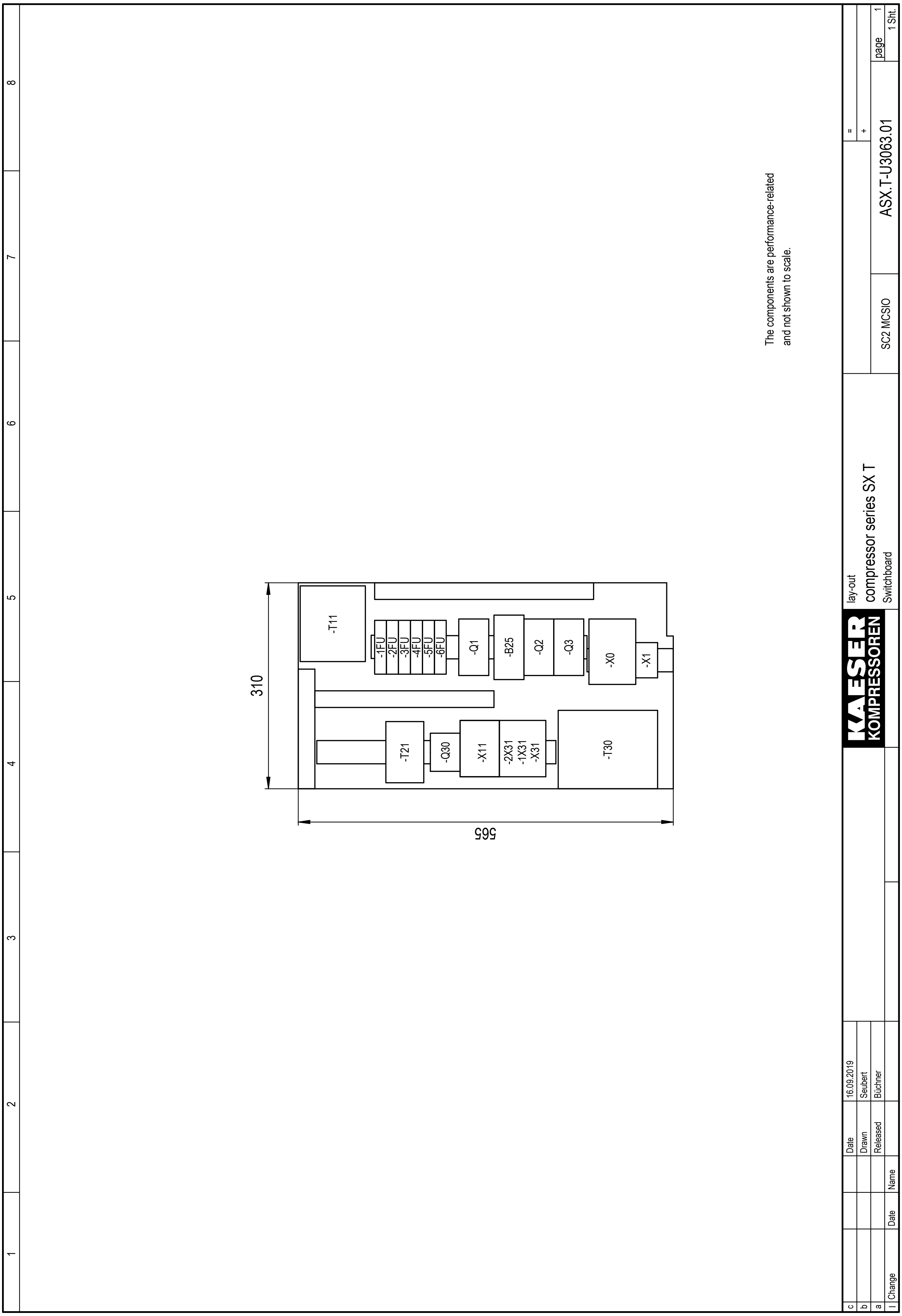
Date 16.09.2019
Drawn Seubert
Released Büchner
-W172 2x18AWG 600V 90°C
-W171 3G18AWG 600V 90°C
-W170 3G14AWG 600V 90°C
-W80 4x18AWG 600V 90°C
-W78 4x18AWG 600V 90°C
-W171.1 3G18AWG 600V 90°C
cable-no.
destination inside
connection
name of device
terminal strip
location
jumper wire
terminal legend
link
terminal-no.
destination external


1
2
3
4
5
6
7
8
The components are performance-related and not shown to scale.
-T11
-1FU
-2FU
-3FU
-4FU
-5FU
-6FU
-Q1
-B25
-Q2
-Q3
-X0
-X1
-T21
-Q30
-X11
-2X31
-1X31
-X31
-T30
310
565
c
b
a
I Change
Date
Name
Date
Drawn
Released
16.09.2019
Seubert
Büchner
KAESER KOMPRESSOREN
lay-out
compressor series SX T
Switchboard
=
+
SC2 MCSIO
ASX.T-U3063.01
page 1
1 Sht.

13.5 Service Manual for Compressed Air Filter

Operator Manual

Compressed air filter

KAESER FILTER F6 – F26

No.: 902429-TBA 01 USE

Read this manual before using this product.

Failure to follow the instructions and safety precautions in this manual can result in serious injury or death.

Manufacturer:

KAESER KOMPRESSOREN SE

96410 Coburg • PO Box 2143 • GERMANY • Tel. +49-(0)9561-6400 • Fax +49-(0)9561-640130

www.kaeser.com

/KKW/AFILT 2.21 en Z1 SBA-FILTER-TBA

20220207 085038

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List of Tables

1 Technical Data

1.1 Options

➤ Enter options here as a reference.

Option	Option code	Available?
KAESER FILTER KE Electronic condensate drain	F1	
KAESER FILTER KE Automatic condensate drain	F11	
Already available: ✓ Not available: —		

Tab. 1 Options

1.2 Model designation

Product	Filter size	Degree of filtration
E: Filter element F: Compressed air filter	6 9 16 22 26	KE: Coalescence filter Extra
Example:		
F	22	KE
My compressed air filter:		
My filter element:		

Tab. 2 Understanding the model designation

1.3 Type of application and use

Application

Degree of filtration	KE
Short description	Extra
Option	F1 / F11
Suitable fluids	Air Nitrogen

Degree of filtration	KE
Short description	Extra
Option	F1 / F11
Fluid properties	Non-corrosive Non-combustible Non-toxic Non-explosive Stable
Application	Used for higher compressed air quality
Fluid quality at the inlet	Free of condensate
Typical application near the compressor station	Downstream of compressed air dryers
Typical application near the consumers	Compressed air filter for higher air quality
Flow direction	From inside to outside

Tab. 3 Application

Pressure and temperature

Degree of filtration	KE
Short description	Extra
Option	F1 / F11
Permissible working pressure [psig] at the inlet	30 – 232
Permissible fluid temperature [°F] at the inlet	40 – 150
Permissible ambient temperature [°F]	40 – 120
Compression load	Static

Tab. 4 Pressure and temperature

1.4 Separation efficiency

The separation efficiency is highly dependent on the individual conditions in the compressed air network (composition of fluid, pressure and flow situation).

Consult an authorized KAESER service representative for advice.

1 Technical Data

1.4 Separation efficiency

Aerosol separation in accordance with ISO 12500-1

Degree of filtration	KE
Short description	Extra
Option	F1 / F11
Differential pressure[1] when new [psig]	<1.02
Initial differential pressure at saturation [psig]	<2.90
Residual aerosol content [mg/m³]	<0.01
[1] At maximum flow rate	

Tab. 5 Aerosol separation at 10 mg/m³ oil aerosol test concentration

2 Safety and Responsibility

2.1 Intended use

The compressed air filter is suitable for the following gaseous fluids:

- Air
- Nitrogen

The compressed air filter is designed exclusively for cleaning the fluids mentioned and for commercial use. Any other use shall be considered incorrect. The manufacturer shall not be liable for any damage that may result from incorrect use. The operator alone shall be liable for any risks incurred.

➤ Observe the information contained in this operating manual.

➤ Use the compressed air filter only within the performance limits and in accordance with the permissible operating conditions.

This compressed air filter is designed for stationary use only. Acceleration forces can damage the compressed air filter. This applies particularly to transportation in a depressurized state.

➤ The compressed air filter is to be used only in a stationary environment.

2.2 Improper use

Improper use can result in injuries or property damage.

➤ Do not operate the compressed air filter in compressed air networks with temperatures potentially exceeding 122°F. Such temperatures may occur, for example, with upstream heat-regenerated desiccant dryers.

➤ Do not use the compressed air filter in areas in which the specific requirements relating to explosion protection apply.

➤ Do not modify the compressed air filter or its components.

Unsuitable spare parts compromise the safety of the compressed air filter.

➤ Only use spare parts approved by KAESER for use in this compressed air filter.

➤ Use only genuine KAESER spare parts for components subject to pressure.

2.3 User’s responsibilities

2.3.1 Observe statutory and universally accepted regulations

Statutory regulations and recognized rules are, for example, the European directives implemented in national law, or the laws, safety and accident prevention regulations applicable in the country of use.

➤ Observe the relevant statutory and accepted regulations when installing, operating and maintaining the compressed air filter.

2.3.2 Qualified personnel

Suitable personnel are experts who, by virtue of their training, knowledge and experience, as well as their knowledge of relevant regulations, can assess the work to be undertaken and recognize the possible dangers involved.

Authorized personnel must possess the following qualifications:

- They must be of legal age.
- They must have read and understood the safety instructions and sections of the operating manual applicable to installation, operation and maintenance, and they must be capable of observing them.
- They must be fully familiar with the safety concepts and regulations of electrical and compressed air engineering.
- They must be able to recognize the possible dangers associated with electrical and compressed air devices and take appropriate measures to safeguard persons and property.
- They must have received adequate training in and authorization for the safe installation and maintenance of the compressed air filter.

➤ Ensure that the personnel entrusted with installation, operation and maintenance have the relevant qualifications and authorization required for the respective activity.

2.4 Safely handling potential sources of danger

This section provides information regarding the various types of danger that can arise in connection with the operation of the compressed air filter.

Wear suitable protective clothing for all work, e.g .:

- Approved work clothing, tight-fitting and flame-retardant
- Protective gloves
- Safety shoes
- Eye protection
- Ear protection

Forces of compression

A compressed fluid is stored energy. Uncontrolled release of this energy can cause serious injury or death.

➤ Before starting any work, check that all components and enclosures are completely depressurized.

➤ Do not carry out welding, heat treatment or mechanical modifications on pressure-bearing components, as this will adversely affect their compressive strength.

Compressed air quality

The composition of the fluid must be suitable for the specific application in order to exclude danger to life and limb.

➤ Use suitable compressed air treatment systems when using compressed air as breathing air and/or for the processing of foodstuffs.

Nitrogen

Nitrogen is a colorless, odorless and tasteless gas that can displace the oxygen out of the breathing air. Should the level of oxygen in the breathing air drop too low (<19.5 vol.%), sudden loss of consciousness can occur without warning. At high levels of nitrogen, only a few breaths can be fatal.

- Observe local safety regulations when handling gases that can displace oxygen from breathing air.
- Comply with the permissible limit values for harmful substances in breathing air as per OSHA 29CFR1910.134 / FDA 21CFR178.3570.
- Use suitable warning systems that monitor the oxygen content in the breathing air and which reliably warn personnel acoustically or visually in the event of a potentially dangerous situation.
- Before entering, sufficiently ventilate the machine room and ensure a permanent exchange of air.
- When it comes to entering rooms that could be subject to reduced levels of oxygen, do so only under the observation of a second person.

Heat

Surfaces can become hot during operation. Touching hot surfaces may lead to injury.
- Do not touch hot surfaces.
- If a surface temperature exceeding 122°F is anticipated, shield the surface or attach a suitable warning sign.

Transport and storage

In order to prevent accidents, weight and size require safety measures to be taken during transportation.
- Use suitable lifting hoists that comply with local safety regulations.
- Ensure that the compressed air filter is transported only by persons who, due to their training, are authorized to safely handle the transported goods.
- Only attach hoists to suitable load-bearing points.
- Ensure the danger zone is clear of personnel.
- Protect the compressed air filter from frost, direct sunlight, dust and rain.

Installation

- Use compressed air lines that are suitable and approved for the maximum working pressure.
- Make sure that pressure lines are installed tension-free.
- Do not induce any forces into the compressed air filter via the connections, whereby the compressive forces need to be balanced by bracing.
- Ensure accessibility so that all work on the compressed air filter can be carried out safely and without hindrance.
- Protect the compressed air filter from frost, direct sunlight, dust and rain.
- Do not install the compressed air filter in areas in which the specific requirements relating to explosion protection apply.

Operation and maintenance

Careless behavior in operation and maintenance can lead to accidents with serious health consequences.
- Do not use the compressed air filter or its components as a climbing aid.
- Have work performed by authorized personnel only.
- Check the absence of voltage on floating contacts.
- Use the compressed air filter only with a condensate drain suitable for the specific application.

- ➤ Check the following points regularly:
  visible damage
  safety devices
  components requiring monitoring
- ➤ Ensure that there is sufficient and suitable lighting so that displays can be read without glare and work can be carried out safely.
- ➤ Pay particular attention to cleanliness during maintenance work.
- ➤ Seal components and exposed openings with clean cloths, paper, or masking tape.

2.5 Environmental protection

Contaminated components can pose a risk to the environment.

- ➤ Do not allow cooling oil to escape to the environment or into the sewage system.
- ➤ Store and dispose of all contaminated components in accordance with applicable environmental protection regulations.
- ➤ Observe respective national regulations.
  This applies in particular to parts that are contaminated with cooling oil or grease.

2.6 Copyright

This operator manual is copyright protected. Queries regarding use or duplication of the documentation should be referred to KAESER. Correct use of information will be fully supported.

3 Design and Function

3.1 Design and function

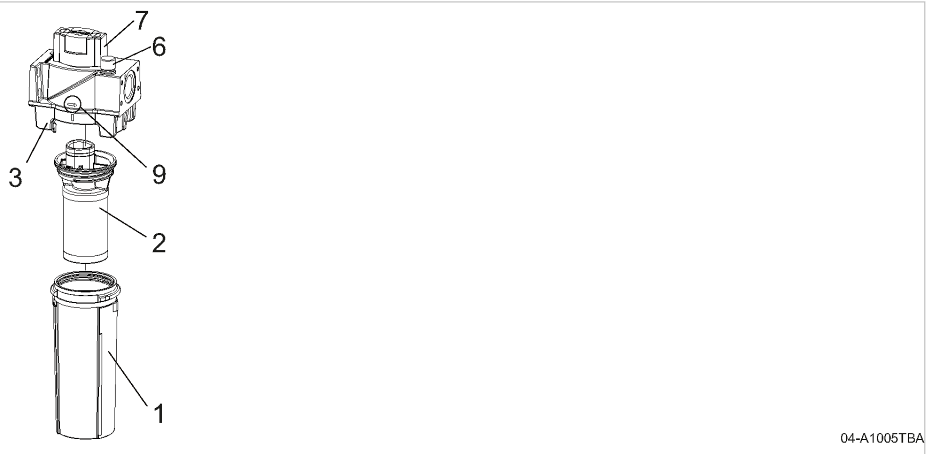


Fig. 1 Design

1	Filter bowl	6	Locking screw
2	Filter element	7	Differential pressure gauge
3	Filter head	9	Direction of flow

The filter element 2 is located in the filter bowl 1. The filter element separates unwanted elements from the fluid.

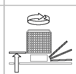
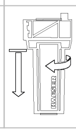
An arrow 9 indicates the direction of flow.

The locking screw 6 secures the filter housing against accidental opening. The compressed air filter is vented as soon as the locking screw is loosened.

The differential pressure gauge 7 provides information regarding the pressure differential between the fluid inlet and fluid outlet.

3.2 Labeling

Pictograms on the filter head inform you regarding the order in which the filter bowl is opened.

	Symbol	Meaning
1.		Ensure you have read and understood the operating manual and all safety instructions
2.		Loosen the locking screw by hand until you feel resistance. Wait until the compressed air filter is fully depressurized.
3.		Carefully unscrew the filter bowl

Tab. 6 Quick instructions

A maintenance sticker on the filter bowl shows the date for the next maintenance appointment.

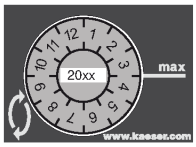


Fig. 2 Maintenance sticker

3.3 Condensate drain

KAESER FILTER feature different types of condensate drains to perfectly match your machine. Not all condensate drains described here are available for all machines.

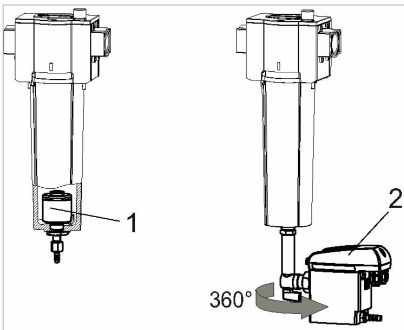


Fig. 3 Condensate drain

1 Automatic condensate drain (internal float)
2 Electronic condensate drain

Option F11 Automatic condensate drain

An automatic condensate drain with internal float is available for compressed air filters with degree of filtration KE. It opens automatically as soon as sufficient liquid has accumulated in the filter bowl.

Option F1 Electronic condensate drain

An electronic condensate drain is available for compressed air filters with degree of filtration KE. The electronic condensate drain opens automatically as soon as sufficient liquid has accumulated in the filter bowl.

In the variant of an electronic condensate drain with alarm contact, a floating contact will transmit a signal in the event of a fault.

The electronic condensate drain works more precisely, more reliably, causes lower pressure losses, and has a longer maintenance interval.

4 Installation and commissioning

4.1 Connecting the condensate drain

Make sure that the condensate can drain away without hindrance.

Only a compressed air filter with a maximum permissible gauge working pressure of 232 psig may be connected to the condensate collection pipe.

The illustration depicts an installation recommendation.
Condensate flows downwards into the condensate collection pipe. This prevents condensate from flowing back into the compressed air filter from the condensate collection pipe.

If condensate flows to multiple points in the condensate collection pipe, install a shut-off valve in each condensate line to ensure that the condensate lines can be shut-off individually.

Condensate line

Characteristic	Value
Max. length[1)] [ft]	50
Max. delivery head [ft]	16
Material (pressure-resistant, corrosion-proof)	Copper Stainless steel Plastic Hose line
[1)] For longer lengths, please contact an authorized KAESER service representative before installation	

Tab. 7 Condensate line

Condensate collection pipe

Characteristic	Value
Gradient [%]	≥3
Max. length[1)] [ft]	65
Material (pressure-resistant, corrosion-proof)	Copper Stainless steel Plastic Hose line
[1)] For longer lengths, please contact an authorized KAESER service representative before installation	

Tab. 8 Condensate collection pipe

Compressed air flow rate[1)] [cfm]	Line cross-section [“]
<350	3/4
350 – 700	1
[1)] Use compressed air flow rate as a guide for the expected condensate volume	

4 Installation and commissioning

4.1 Connecting the condensate drain

Compressed air flow rate[1)] [cfm]	Line cross-section ["]
701 – 1400	1 1/2
>1400	2

1) Use compressed air flow rate as a guide for the expected condensate volume

Tab. 9 Condensate collection pipe: Line cross-section

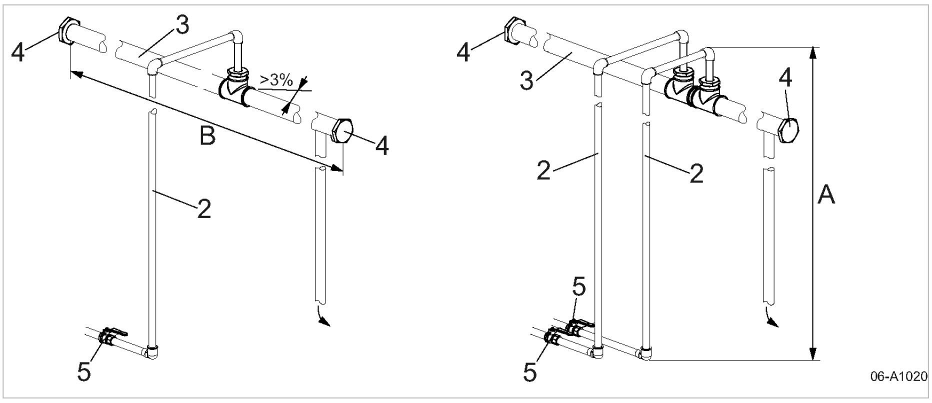


Fig. 4 Connecting the condensate drain

- 2 Condensate line
- 3 Condensate collection line
- 4 Screw plug
- 5 Shut-off valve
- A Delivery head
- B Length of condensate collection line

➤ Connect the condensate line to the hose connection.

➤ Collect the condensate in a suitable collection container and dispose of it in accordance with applicable environmental protection regulations.

5 Maintenance

5.1 Regular maintenance tasks

The table below provides an overview of the necessary maintenance tasks.

The intervals actually required depend on the conditions in which the compressed air filter is used.

➤ Determine economically viable intervals in consultation with an authorized KAESER service representative.

➤ Carry out maintenance tasks in a timely manner, in accordance with the operating conditions:

Interval	Maintenance task	See chapter
Weekly	Check function: ■ Electronic condensate drain	5.3.1
As per maintenance sticker At least annually	Replace the filter element	5.2
At least annually	Replace the float: ■ Automatic condensate drain:	5.4
At least every 2 years	Replace the service unit: ■ Electronic condensate drain	5.3.2
h = operating hours		

Tab. 10 Regular maintenance tasks

5.2 Replacing the filter element

Handle all components with care and install without the use of tools.

Material KAESER filter element, including silicone-free sealing grease and O-ring

⚠ WARNING

Risk of force being exerted on the body due to sudden release of pressure.

➤ *Fully vent all pressurized components and enclosures.*

5.2.1 Removing the filter element

The locking screw is secured against complete removal.

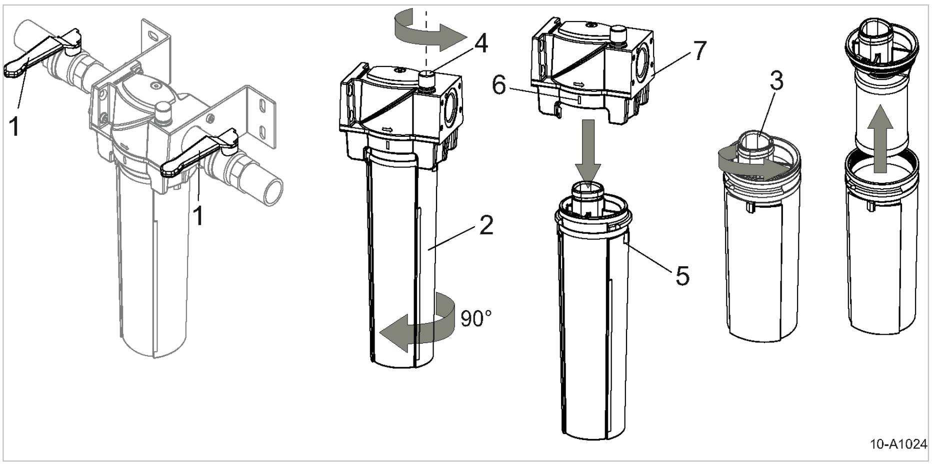


Fig. 5 Removing the filter element

1	Shut-off valve	5	Installation mark on filter bowl
2	Filter bowl	6	Installation mark on filter head
3	Filter element	7	Filter head
4	Locking screw		

1. Close the shut-off valve 1.
2. Loosen the locking screw 4 by hand until you feel resistance.

   If the compressed air filter is still pressurized, the remaining compressed air will escape.

If you can hear a continuous whistling sound:

The compressed air filter is pressurized.

➤ Shut off the compressed air filter from the compressed air network, or vent the entire network.

3. Gently shake the filter bowl 2 and then turn it 90° until the installation markings on the filter bowl 5 and filter head 6 are positioned opposite one another.
4. Remove the filter bowl, together with the screwed-in filter element, vertically downwards.
5. Unscrew the filter element 3 from the filter bowl by rotating approx. 1 1/2 times.
6. Drain and dispose of any condensate.
7. Check the filter bowl for corrosion.

If the filter bowl is noticeably corroded:

➤ Determine the cause (e.g. composition of the compressed air, prevailing operating conditions)

➤ Completely replace the compressed air filter.

Dispose of the contaminated filter element in accordance with environmental regulations.

5.2.2 Installing the filter element

➤ Do not touch the surface of the filter material.

Precondition Check that the inner surfaces of the filter head and filter bowl are clean.

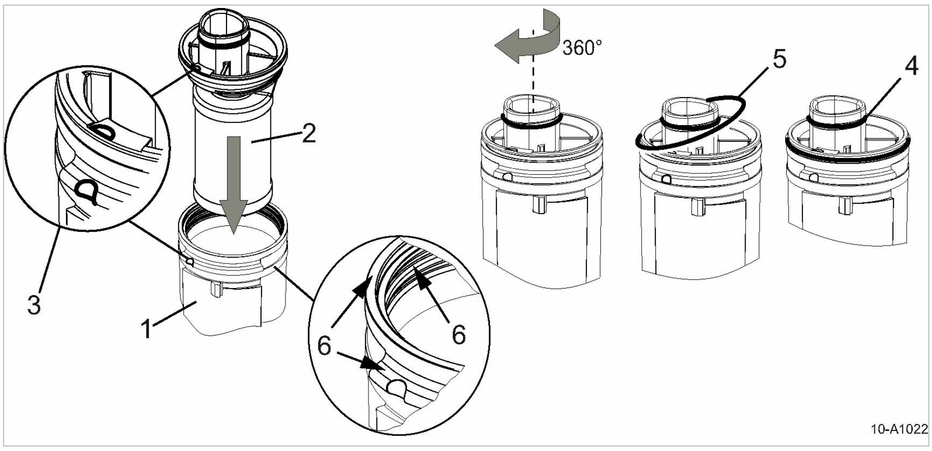


Fig. 6 Installing the filter element

1 Filter bowl	4 O-ring
2 Filter element	5 O-ring
3 Installation markings	6 Surface to be greased

1. Grease the thread, front face and bayonet catch on the filter bowl as per position 6.
2. Align the installation markings 3 with one another and then push the filter element 2 into the filter bowl 1.
3. Fasten the filter element into the filter bowl by rotating it once to the right.
4. Fully grease the O-ring 5 and insert between the filter element and the filter bowl.
5. Grease the O-ring 4.

5.2.3 Installing the filter bowl

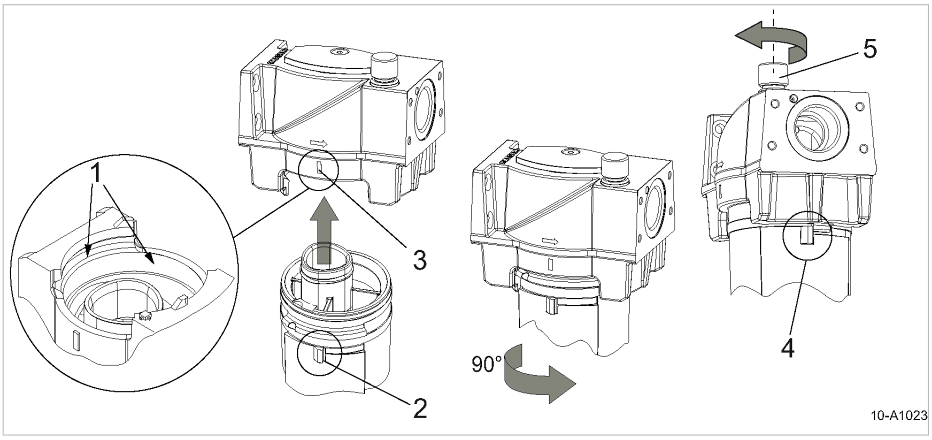


Fig. 7 Installing the filter bowl

1 Surface to be greased	4 Limit stop on filter head
2 Installation mark on filter bowl	5 Locking screw
3 Installation mark on filter head	

1. Grease the inner surface of the filter head as per position 1.
2. Align the installation markings on the filter bowl 2 and filter head 3 with one another and then push the filter bowl into the filter head.
3. Fasten the filter bowl in place by rotating it to the left as far as the limit stop 4.
4. Manually tighten the locking screw 5.

?

If the locking screw cannot be tightened:

The bayonet catch on the filter bowl is not fully closed.

➤ Rotate the filter bowl as far as the limit stop.

Fig. 8 Filling out the maintenance sticker

1 Maintenance sticker

5. Fill out the maintenance sticker with the year in which the next maintenance is due.
6. Align the month number on the maintenance sticker with the *max* marking and affix the sticker.

5.2.4 Pressurizing the compressed air filter

A high fluid flow velocity can damage the filter material.

1. Check that the locking screw has been tightened by hand.

2. Slowly open the shut-off valve at the fluid inlet.
3. Slowly open the shut-off valve at the fluid outlet.

5.3 Option F1 Maintaining the electronic condensate drain

5.3.1 Checking the condensate drain

Precondition The power supply disconnecting device is switched on.
Machine is pressurized.
The *Power* LED lights.

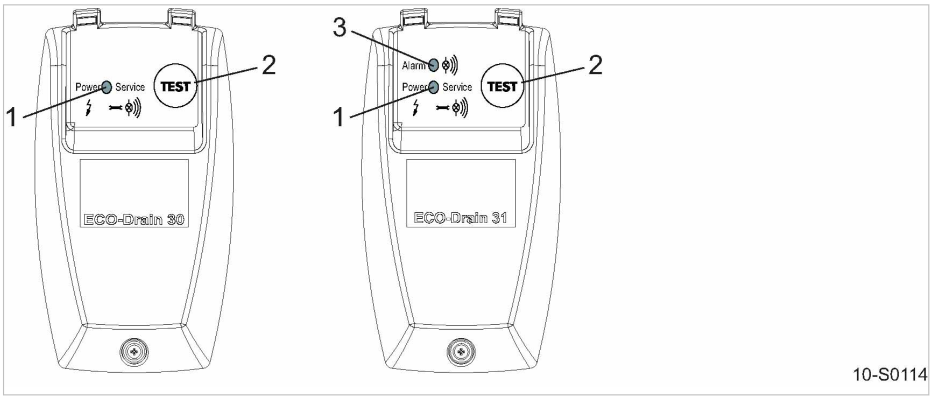


Fig. 9 Check condensate drain

1 *Power* LED
2 «TEST» key
3 *Alarm* LED

1. **CAUTION** *Danger of burns from hot components near the condensate drain!*
   ➢ *Work with caution.*
2. With one hand, lightly touch the condensate line at the condensate drain.
3. With your other hand, push and hold the «TEST» key at the condensate drain for at least 2 seconds.

Result As soon as the condensate drain opens, you will feel a short burst at the condensate line.
Replace the service unit if you do **not** experience a burst during manual test.

5.3.2 Changing the service unit

The condensate drain cannot be cleaned. The service unit must be changed if condensate does not drain.

Material Sealing tape for sealing the screw-in part
If required: O-ring 16x2 (5.1519.0)

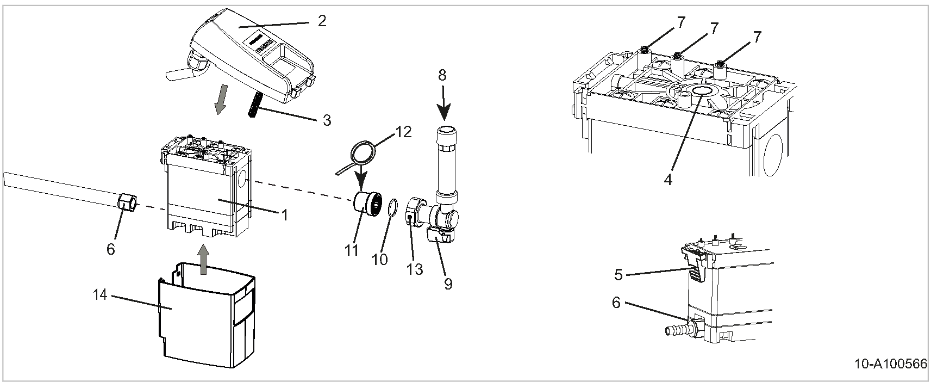


Fig. 10 Change the service unit

1	Service unit	8	Condensate inlet
2	Control unit	9	Shut-off valve
3	Sensor	10	O-ring
4	Sensor opening	11	Screw-in part
5	Snap fastener	12	Sealing tape
6	Condensate line screw connection	13	Clamping nut with vent hole
7	Contact springs	14	Casing

Removing the service unit

1. **⚠ WARNING** *Serious injury or death can result from loosening or opening components under pressure!*
   ➤ *Fully vent all pressurized components and enclosures.*
2. Close the shut-off valve 9 upstream of the condensate drain.
3. Unscrew the screw connection 6 at the condensate line.
4. Press the snap fastener 5 and carefully remove the control unit 2 from the service unit 1.
5. Carefully loosen the clamping nut 13 at the shut-off valve 9 until remaining residual air has escaped through the venting hole.
6. Unscrew the screw-in part 11 from the service unit and place aside.
7. Remove the casing 14 from the service unit.

Installing the service unit

Use only KAESER service units to ensure correct function of the condensate drain.

Precondition Make sure that the top of the service unit and the contact springs are clean and dry.

1. Fit the casing 14 to the service unit 1.
2. Carefully insert the sensor 3 of the control unit 2 in the opening 4 of the service unit.
3. Place the snap fastener 5 of the control unit into the service unit eyes.
4. Press the control unit against the service unit until the snap fastener can be heard clicking into place.
5. At the screw-in part 11, replace old sealing material with new sealing tape.
6. Install the screw-in part into the service unit and tighten to a maximum of 14.8 Nm.
7. If necessary, insert a new O-ring 10.

8. Tighten the clamping nut 13 at the shut-off valve 9.
9. Attach the condensate line.
10. Open the shut-off valve upstream of the condensate drain.

5.4 Option F11 Automatic condensate drain: Replacing the float

See chapter 5.2 for information on the removal and re-installation of the filter bowl.

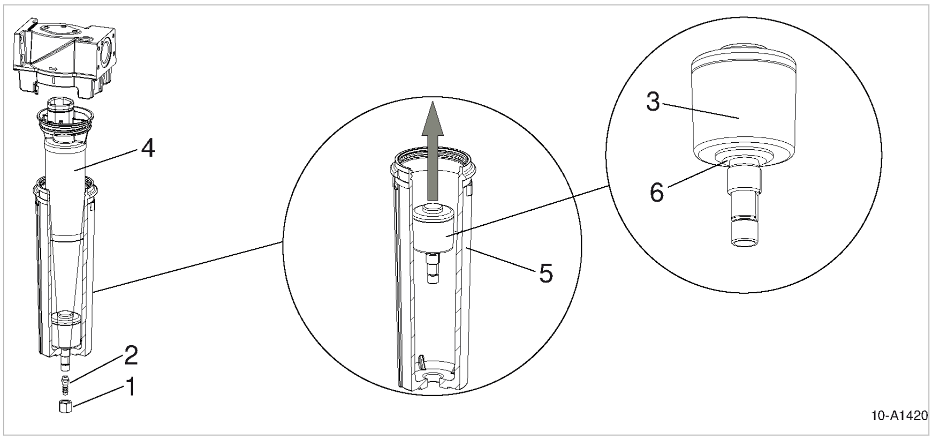


Fig. 11 Replacing the float

1	Clamping nut	4	Filter element
2	Male hose coupling	5	Filter bowl
3	Float	6	O-ring

1. **WARNING** *Risk of force being exerted on the body due to sudden release of pressure.*
   ➤ *Fully depressurize all pressurized components and enclosures.*
2. Dismantle the filter bowl as described in chapter 5.2.
3. Undo the clamping nut 1 and remove the male hose coupling 2.
4. Remove the filter bowl 5 and the filter element 4.
5. Turn the float 3 clockwise until it is fully removed from the filter bowl.
6. Check whether the O-ring 6 at the bottom of the new float is fully inserted in the groove.
7. Manually screw the float drain into the filter bowl and finally tighten with 3 lbf-ft.
8. Assemble the filter element and filter bowl as described in chapter 5.2.
9. Install the male hose coupling with the clamping nut.

6 Spares, Operating Materials, Service

6.1 Nameplate

The nameplate contains all information to identify your compressed air filter.

Please provide the information from the nameplate when making any enquiries and when ordering spare parts.

6.2 Ordering spare parts

KAESER replacement parts are original Kaeser components. They are designed for use in KAESER compressed air filters.

Spare parts of unsuitable or inferior quality can significantly impair the function of the compressed air filter and, consequently, the function of downstream components.

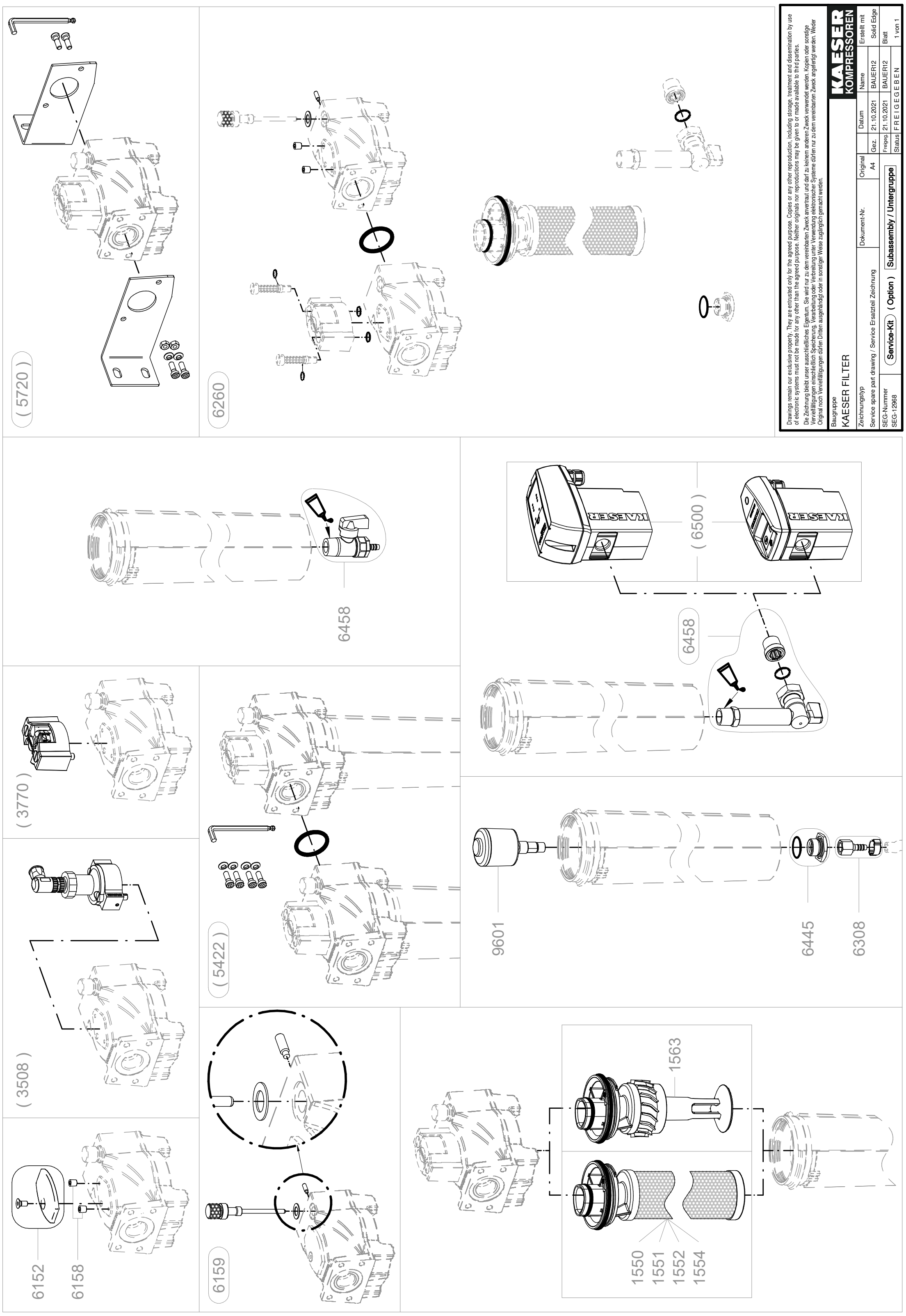
Personal injury may result from damage.

Do not attempt work other than that described in this operating manual.

KAESER KOMPRESSOREN

6 Spares, Operating Materials, Service

6.2 Ordering spare parts



Legend		
	KAESER FILTER	SEL-4551_01 E

Item	Description	Option
1550	Prefilter element	
1551	Microfilter element	
1552	Activat. carbon filter element	
1554	Particulate filter element	
1563	Centrifugal insert KC	
3508	Diff. pressure transducer	X
3770	Pressure diff. indicator	X
5422	Connecting kit	X
5720	Filter support	X
6152	Filter cover	
6158	Threaded plug	
6159	Locking screw	
6260	Gasket kit	
6308	Hose connection	
6445	Reduction piece	
6458	Stop valve	
6500	Condensate drain	X
9601	Maintenance kit, condens.drain	

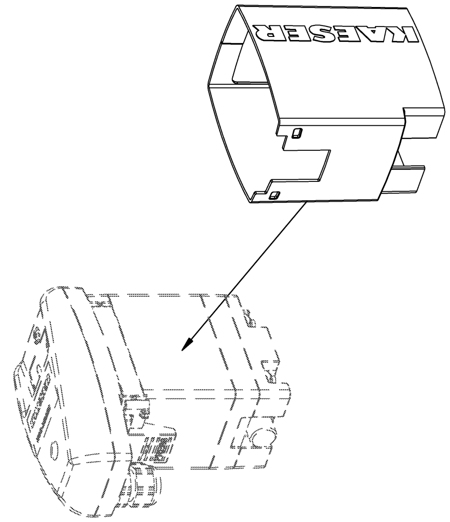
Please quote the part number and serial number of the machine together with the item number and the description of the part when ordering.

Before and during all work, be sure to read and follow the safety and service instructions in the machine's service manual!

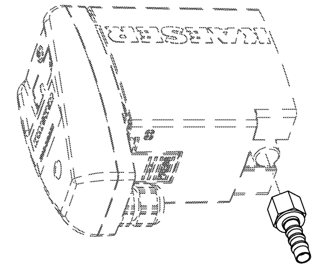
Service-Kit (Option)

Kondensatableiter / Condensate drain

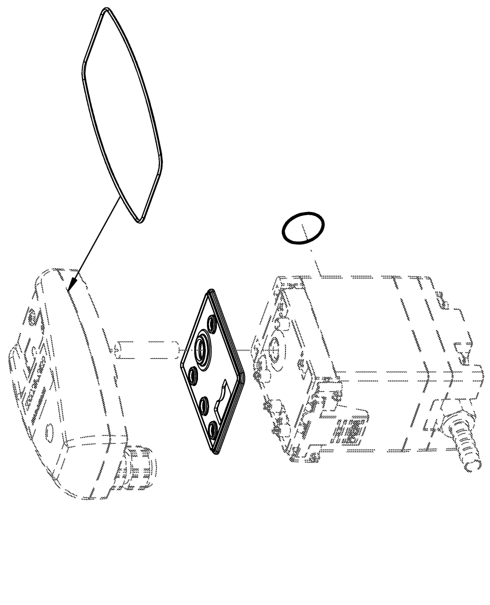
SEG-5405_01



9022



6307



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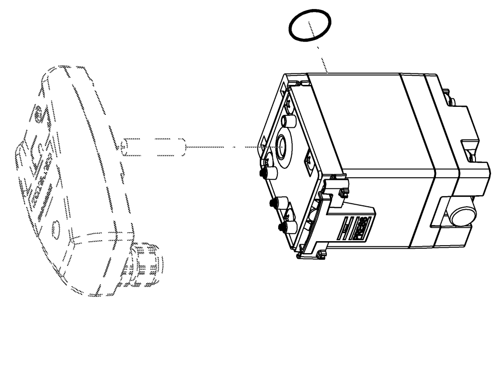

9602

Legend		KAESER
	Condensate drain	SEL-3886_01 E

Item	Description	Option
6307	Hose connection	
9022	Panelling	
9602	Condensate drain service-unit	
9603	Condensate drain gasket kit	

Please quote the part number and serial number of the machine together with the item number and the description of the part when ordering.

Before and during all work, be sure to read and follow the safety and service instructions in the machine's service manual!